Differential Equations

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Lecture 1: Direction fields, existence and uniqueness of solutions

A differential equation is a relation between a function and its derivatives. Scientific laws typically describe how a system changes from one instant to the next; the task of the theory is to turn this short-term information into long-term behavior. One models a system by a differential equation, extracts information about its solutions, and interprets that information back in the original setting. Newton's law $F = ma$ is the basic example: forces act on the position x only through its second derivative, so it is a second-order equation.

1.1 First-order equations and separation of variables

This unit studies first-order equations, involving only the first derivative:

$$y' = F(x, y).$$

Example: The equation $y' = 2x$ is solved by integrating: $y = x^2 + c$. An expression involving an arbitrary constant that describes all solutions is the general solution.

Example: The equation $y' = ky$ has general solution

$$\boxed{y = Ce^{kx}}.$$

This is the central example of the course. In fact, e^x can be defined as the solution of $y' = y$ with $y(0) = 1$.

Separation of variables applies when $F(x, y)$ factors as a function of x times a function of y: put all the y's on one side and the x's on the other, integrate both sides, combine the constants, and solve for y if possible.

Example: For $dy/dx = 2y + 1$, separate as $dy/(2y + 1) = dx$ and integrate:

$$\tfrac{1}{2}\ln|2y + 1| = x + c, \qquad |2y + 1| = e^{2c}e^{2x}, \qquad 2y + 1 = Ce^{2x},$$

so $y = Ce^{2x} - \frac{1}{2}$. The sign of e^{2c} is absorbed into C, and $C = 0$ restores the constant solution $y = -\frac{1}{2}$ lost by dividing by $2y + 1$. Answers are easy to check: $y' = 2Ce^{2x} = 2(y + \frac{1}{2}) = 2y + 1$.

Whether an equation is separable may only become visible after rearranging. For example, $y' + xy = x$ is separable, since $y' = x(1 - y)$ gives $dy/(1 - y) = x\,dx$.

Most equations arising in practice cannot be solved explicitly, and often the general properties of solutions matter more than formulas. Geometric methods address this.

1.2 Direction fields and integral curves

Definition 1.1 [Direction field]. The equation $y' = F(x, y)$ assigns the slope $F(x, y)$ to every point of the region where F is defined. This assignment is the direction field (or slope field). A solution is a function whose graph has the prescribed slope at every point it passes through; its graph is an integral curve.

Symmetries of the field are inherited by the solutions. For $y' = 2x$ the slope does not depend on y: the field is invariant under vertical translation, and so a vertical translate of a solution is again a solution (the integral curves are nested parabolas). For $y' = y$ the slope does not depend on x: the field is invariant under horizontal translation, and the integral curves are horizontal translates of one another (together with $y = 0$). To pick out one solution, give an initial condition: the value $y(a) = b$ at one point $x = a$.

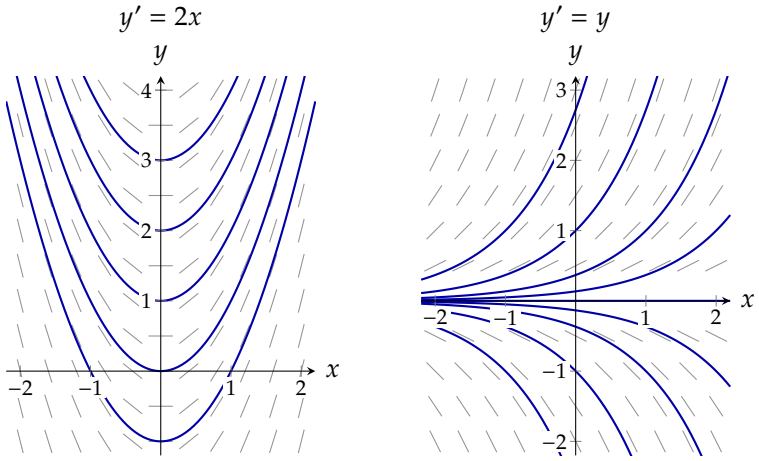


Figure 1: Direction fields and integral curves of $y' = 2x$ (the parabolas $y = x^2 + c$, $c = -1, \ldots, 3$) and of $y' = y$ (the curves $y = \pm e^{x-c}$, $c = -1, \ldots, 2$, and $y = 0$).

1.3 Isoclines

To sketch a direction field by hand, draw the curves on which the slope is constant.

Definition 1.2 [Isocline]. For a constant m, the isocline of slope m is the curve $F(x, y) = m$. Every integral curve crossing it does so with slope m.

Example: The equation $y' = y^2 - x$ has no solutions expressible in elementary functions, but its integral curves can still be described. Its isoclines are the parabolas

$$m = 0 : \; x = y^2, \qquad m = 1 : \; x = y^2 - 1, \qquad m = -1 : \; x = y^2 + 1,$$

and in general $x = y^2 - m$, all horizontal translates of one another (Figure 2).

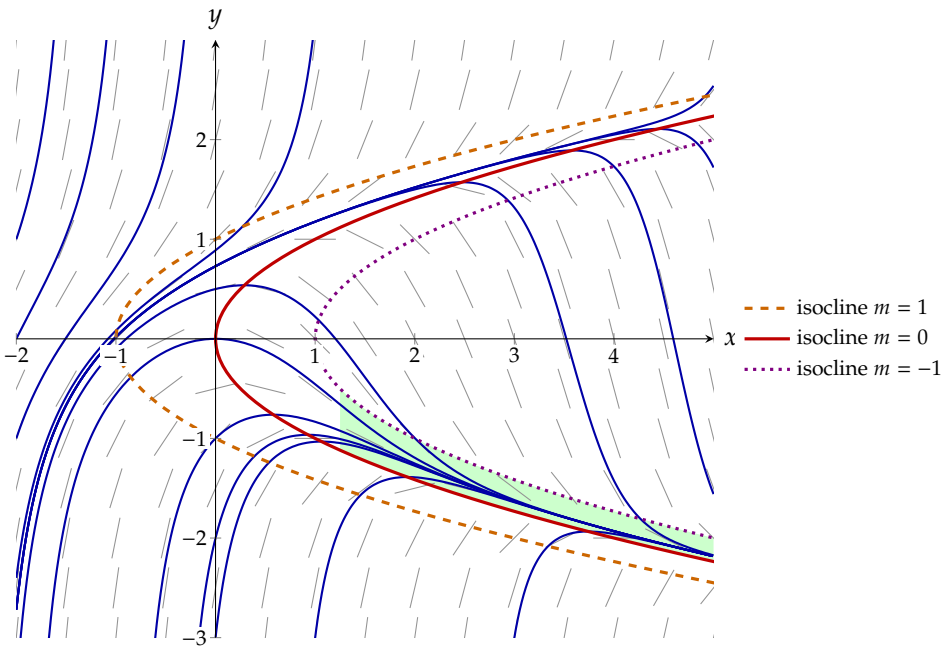


Figure 2: Direction field of $y' = y^2 - x$, the isoclines $m = 1, 0, -1$, and integral curves (computed numerically). The shaded funnel lies between the lower branches of the isoclines $m = 0$ and $m = -1$.

1.4 Existence and uniqueness

Theorem 1.3 [Existence and uniqueness**].** *Suppose F and $\partial F/\partial y$ are continuous on a region of the plane. Then for every point (a, b) of the region, the equation $y' = F(x, y)$ has exactly one solution with $y(a) = b$. This solution may only be defined for x near a.*

Geometrically, *every point of the region lies on exactly one integral curve*: integral curves never cross or touch. A plot that seems to show several solutions merging only shows curves that are very close.

1.5 Funnels

In Figure 2, many solutions of $y' = y^2 - x$ crowd together along the lower branch of the parabola $x = y^2$. The isoclines explain this. Consider the region between the lower branches of the isoclines $m = 0$ and $m = -1$,

$$-\sqrt{x} < y < -\sqrt{x - 1}, \qquad x > \tfrac{5}{4}.$$

Inside it, $x - 1 < y^2 < x$, so solutions have slope between -1 and 0.

Proposition 1.4 [A funnel**].** *A solution of $y' = y^2 - x$ that lies in this region at some $x_0 > \frac{5}{4}$ stays in it for all $x > x_0$. Every such solution satisfies $y(x) + \sqrt{x} \to 0$ as $x \to \infty$.*

Proof. A solution leaving the region at some x_1 would cross one of the two boundary curves, and at the crossing its slope would be at least (leaving upward) or at most (leaving downward) the slope of that curve. On the upper boundary $y = -\sqrt{x-1}$ the solution has slope -1, while the boundary has slope $-1/(2\sqrt{x-1}) > -1$ for $x > \frac{5}{4}$; so the solution cannot cross it upward. On the lower boundary $y = -\sqrt{x}$ the solution has slope 0, while the boundary has slope $-1/(2\sqrt{x}) < 0$; so the solution cannot cross it downward. Finally,

$$0 < \sqrt{x} - \sqrt{x-1} = \frac{1}{\sqrt{x} + \sqrt{x-1}} \to 0,$$

so the two boundaries, and every solution trapped between them, are asymptotic to $-\sqrt{x}$. □

Such a region is a FUNNEL. Once a solution enters it, its long-term behavior is known with increasing accuracy and does not depend on the initial condition. This insensitivity is a form of STABILITY. Solutions that never enter the funnel can behave quite differently.

Direction fields show the qualitative behavior of solutions, which is often what one wants to know. The following lectures add numerical and analytic (formula-based) methods.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 1, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 2: Numerical methods

Differential equations are studied by three kinds of methods: analytic (exact, symbolic) methods, qualitative methods (direction fields, isoclines), and numerical methods. Even an explicit solution leaves the question of computing its values. For instance, e is the value $y(1)$ of the solution of $y' = y$, $y(0) = 1$, and its decimal expansion $e = 2.718281828459045\ldots$ is found numerically.

2.1 The tangent line approximation

An initial value problem (IVP) is an equation $y' = F(x, y)$ together with an initial condition $y(a) = y_0$. The running example is

$$y' = y^2 - x, \qquad y(0) = -1.$$

This solution is trapped in the funnel of Lecture 1, so for large x it is close to $-\sqrt{x}$; for instance $y(100) \approx -10$. Its value at $x = 1$ is $y(1) \approx -0.83$. How is such a value computed?

If (a, y_0) lies on the graph of y and $m_0 = y'(a)$, the tangent line approximation is

$$y(a + h) \approx y_0 + m_0 h,$$

the height of the tangent line above $a + h$. For an IVP the slope is known without solving: here $m_0 = F(0, -1) = 1$, so a single hop of length 1 gives $y(1) \approx -1 + 1 = 0$. This is poor because the hop is long; the approximation is much better for small h.

2.2 Euler's method

Take many small hops, choosing each new direction from the direction field at the end of the previous hop. To estimate $y(b)$ for $b > a$, divide $[a, b]$ into steps of width h with $x_k = a + kh$.

Algorithm 2.1: Euler's method

Input: F, initial point $(x_0, y_0) = (a, y(a))$, step size h, number of steps n.
Output: Approximations $y_k \approx y(x_k)$, $k = 1, \ldots, n$.

```
for k ← 0 to n − 1 do
    m_k ← F(x_k, y_k)
    x_{k+1} ← x_k + h
    y_{k+1} ← y_k + m_k h
```

The segments from (x_k, y_k) to (x_{k+1}, y_{k+1}) are Euler struts; together they form the Euler polygon, an approximation to the graph of y. The points x_k, y_k depend on h, and only $y_0 = y(x_0)$ is exact: y_k

for $k \geq 1$ merely approximates $y(x_k)$.

Example: For $y' = y^2 - x$, $y(0) = -1$, with $h = \frac{1}{2}$:

k	x_k	y_k	$m_k = y_k^2 - x_k$	$m_k h$
0	0	-1	1	0.5
1	0.5	-0.5	-0.25	-0.125
2	1	-0.625		

Smaller steps give better estimates of $y(1)$:

h	1	0.5	0.25	0.125	actual
estimate of $y(1)$	0.00	-0.62	-0.75	-0.80	-0.83

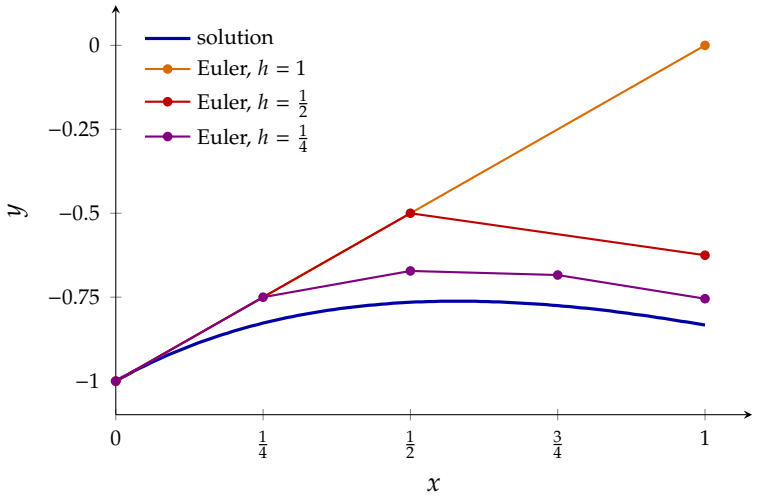


Figure 3: Euler polygons for $y' = y^2 - x$, $y(0) = -1$, with $h = 1, \frac{1}{2}, \frac{1}{4}$, and the solution (computed numerically). The solution is concave here, and every Euler estimate of $y(1)$ lies above it.

2.3 Sources of error

Euler's method is rarely exact.

1. *Round-off accumulation.* Each computation has a small rounding error, and these accumulate, which argues for few steps. Small steps, on the other hand, keep the polygon close to the solution.

2. *Variation of the direction field.* Usually more important: the polygon is not a solution, and the field at its vertices differs more and more from the field along the true solution. Where the field changes rapidly the approximation degrades quickly; the integral curves bend away from the struts.

Whether Euler overestimates or underestimates is predicted by convexity. If the solution is concave (bending down), it falls below its tangent lines and the strut ends up too high; if it is convex (bending up), the strut ends up too low. Concavity at the initial point can be read off the equation by differentiating it:

$$y' = y^2 - x \quad \Longrightarrow \quad y'' = 2yy' - 1, \qquad y''(0) = 2(-1)(1) - 1 = -3.$$

Proposition 2.1 [Sign of the first-step error]. *If $y''(x_0) < 0$, the Euler estimate is too high; if $y''(x_0) > 0$, it is too low (for small enough h).*

Proof. By Taylor's theorem, $y(x_0 + h) = y_0 + m_0 h + \frac{1}{2} y''(\xi) h^2$ for some ξ between x_0 and $x_0 + h$; for small h, $y''(\xi)$ has the sign of $y''(x_0)$. □

In the example, $y''(0) = -3 < 0$, consistent with the estimates in Figure 3 lying above the solution.

2.4 Higher-order methods

For small h the error of Euler's method is at most $C_1 h$ for some constant C_1: it is a first-order method (unrelated to first-order equations). Shrinking h reduces the error, but cleverer uses of the direction field do much better.

Definition 2.2 [Heun's method]. The improved Euler method (or Heun's method, or RK2) polls the slope field at the start of the step and at the end of the Euler strut, and uses their average:

$$m_k = F(x_k, y_k), \qquad n_k = F(x_k + h,\ y_k + m_k h), \qquad y_{k+1} = y_k + \frac{m_k + n_k}{2}\, h.$$

Example: For $y' = y^2 - x$, $y(0) = -1$, $h = \frac{1}{2}$, the Euler table gives $F(0, -1) = 1$ and the end of the Euler strut $(\frac{1}{2}, -\frac{1}{2})$, where $F = -\frac{1}{4}$. The first RK2 slope is the average $\frac{3}{8}$, which brings the polygon closer to the solution.

RK2 is second order: its error is at most $C_2 h^2$ for small h. There is also an RK4 method, polling the field four times per step, with error about $C_4 h^4$.

Each evaluation of the field costs time. Euler polls once per step, RK2 twice, RK4 four times, so a fair comparison pits Euler with step h against RK2 with step $2h$ and RK4 with step $4h$. Their errors are about

$$C_1 h, \qquad C_2 (2h)^2 = 4 C_2 h^2, \qquad C_4 (4h)^4 = 256 C_4 h^4,$$

and for small enough h the higher-order method wins whatever the constants. In practice the constants also decrease: $C_4 < C_2 < C_1$ is typical.

Example: For $y' = y$, $y(0) = 1$, approximate $y(1) = e$ with 1000 evaluations of the field:

method	step	error	error/(step)^{order}
RK1 (Euler)	0.001	1.36×10^{-3}	$C_1 \approx 1.36$
RK2 (Heun)	0.002	1.8×10^{-6}	$C_2 \approx 0.45$
RK4	0.004	5.8×10^{-12}	$C_4 \approx 0.023$

For good accuracy Euler is essentially useless. Still higher-order methods exist, but carry more overhead; RK4 is a good compromise.

For $y' = f(x)$, $y(a) = y_0$, the solution is $y(x) = y_0 + \int_a^x f(t)\,dt$, and the three methods become familiar quadrature rules: Euler's method is the left-endpoint Riemann sum, RK2 is the trapezoidal rule, and RK4 is Simpson's rule.

2.5 Catastrophic overshoot

Errors can be dramatic where the field funnels steeply. Euler's method with $h = 1$ on $y' = y^2 - x$, $y(0) = -1$ produces

$$y_k : \quad -1,\ 0,\ -1,\ -2,\ -1,\ -4,\ 7,\ 50,\ \ldots$$

at $x_k = 0, 1, \ldots, 7$. The field funnels solutions toward $-\sqrt{x}$, but each long strut overshoots across the funnel and lands where the field is steeper in the opposite direction, so the next strut overshoots further, until the polygon escapes upward. Practical ODE solvers guard against this, for instance by taking smaller steps where the field is steep.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 2, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 3: Linear equations, models

Linear equations are the most important general class of differential equations, and they occupy most of the course. This lecture models two systems evolving in time by first-order linear equations. The independent variable is time t, the dependent variable is $x = x(t)$, and x' (also written $\dot{x}$) is its time derivative.

3.1 First-order linear equations

Definition 3.1 [First-order linear equation]. A first-order linear ODE is one that can be put in the standard form

$$\boxed{r(t)\,x' + p(t)\,x = q(t)}.$$

A general solution method comes in Lecture 4. When r, p, q are constants, the equation is separable:

$$\frac{r\,dx}{q - px} = dt.$$

Example: A population with natural growth rate k that is harvested at rate h satisfies $x' = kx - h$. In a physical system both k and h may vary with time, and the model remains linear.

3.2 Bank accounts: rate and cumulative total

Let $x(t)$ be the balance of a bank account, with t in years. Treat a withdrawal as a negative deposit. If interest is paid at the end of each period of length Δt (for monthly statements, $\Delta t = \frac{1}{12}$) on the balance at its start, then

$$x(t + \Delta t) = x(t) + I\,x(t)\,\Delta t + \big[\text{deposits between } t \text{ and } t + \Delta t\big],$$

where the interest rate I has units $(\text{year})^{-1}$; a rate of 1% pays $\frac{1}{12}\%$ per month.

To pass to the limit $\Delta t \to 0$, record deposits as a *rate* $q(t)$ in dollars per year. The cumulative total $Q(t)$ of deposits up to time t then satisfies $Q'(t) = q(t)$. Depositing \$100 once a month makes Q a staircase; depositing continually at the constant rate $q = \$1200$ per year makes Q a straight line of slope 1200. The rate may vary, and is negative during withdrawals. In particular, deposits are being made exactly where Q is increasing, whatever the sign of Q itself.

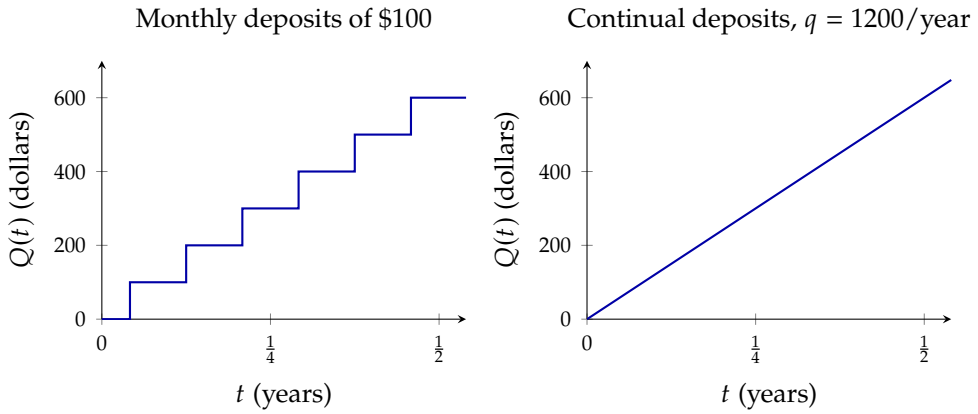


Figure 4: Cumulative deposits $Q(t)$ for \$100 deposited once a month (schematic deposit dates) and for continual deposits at \$1200 per year.

For continuous q, the deposits over a short period are about $q(t)\,\Delta t$, so

$$\frac{x(t+\Delta t)-x(t)}{\Delta t} \approx I\,x + q.$$

Letting the interest period Δt tend to zero gives

$$x' = I\,x + q, \qquad \text{or in standard form} \qquad \boxed{x' - I\,x = q}.$$

Both $I = I(t)$ and $q = q(t)$ may vary with time without affecting the derivation. If the interest rate also depended on the balance x, the equation would no longer be linear.

3.3 Systems and signals

In standard form, the left side represents the SYSTEM (the bank) and the right side an outside influence on it. A SIGNAL is simply a function of time; the right side is the INPUT SIGNAL (here the savings rate). The system responds to the input and yields the solution $x(t)$, the OUTPUT SIGNAL or SYSTEM RESPONSE. The initial condition is a further input.



Figure 5: Systems as signal processors: the bank, and the RC circuit below.

Recognizing linear equations is important. For instance, $\dot{x} = (t^2+1)(x-1)$ and $\dot{x} + x = t^2$ are linear, while $\dot{x} + x^2 = t$ is not.

3.4 RC circuits

An RC CIRCUIT is a series loop containing a variable voltage source, a resistor, and a capacitor. The current $I(t)$, in amperes, is the same everywhere in the loop but may vary in time; take clockwise as the positive direction, so counterclockwise flow gives a negative reading. Let $V(t)$ be the voltage *increase* across the source, and V_R, V_C the voltage *drops* across the resistor and the capacitor.

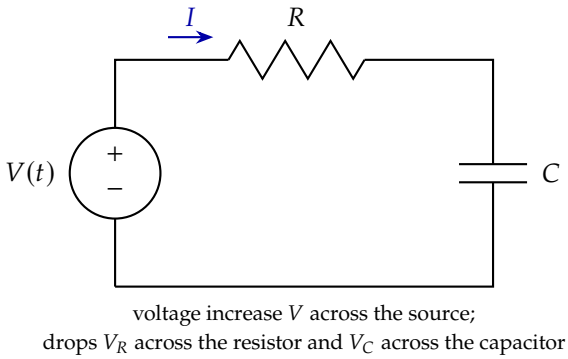


Figure 6: The series RC circuit.

Three physical laws govern the circuit:

$$\underbrace{V(t) = V_R(t) + V_C(t),}_{\text{Kirchhoff's voltage law}} \qquad \underbrace{V_R(t) = R\,I(t),}_{\text{resistor}} \qquad \underbrace{V_C'(t) = \tfrac{1}{C}\,I(t),}_{\text{capacitor}}$$

with constant RESISTANCE R and CAPACITANCE C. The drop across the resistor is proportional to the current; the drop across the capacitor is proportional to the integral of the current, since it comes from charge building up on its plates. A large capacitance leaves room for much charge, and a very large capacitor acts like no capacitor at all.

Differentiating Kirchhoff's law and substituting gives $V' = RI' + \frac{1}{C}I$, a first-order linear equation for the current:

$$\boxed{R\,I' + \frac{1}{C}\,I = V'(t)}.$$

The system is the circuit, the system response is the current $I(t)$, and the input signal is the source voltage $V(t)$. Note that its *derivative* is what appears on the right side.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 3, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 4: Solution of linear equations, integrating factors

4.1 Homogeneous linear equations

Recall the standard form $r(t)x' + p(t)x = q(t)$ of a first-order linear equation. The functions $r(t)$ and $p(t)$ are its coefficients; the left side represents the system, the right side comes from an input signal, and a solution is a system response (output signal). Dividing by $r(t)$ gives the reduced standard form

$$x' + p(t)\,x = q(t). \tag{4.1}$$

The equation is homogeneous if q is the null signal, $q(t) = 0$. This corresponds to letting the system evolve in isolation: no deposits or withdrawals for the bank, no voltage from the source in the RC circuit.

The homogeneous equation $x' + p(t)x = 0$ is separable. In general, and for the example $p(t) = 2t$:

	$x' + p(t)x = 0$	$x' + 2tx = 0$
separate	$dx/x = -p(t)\,dt$	$dx/x = -2t\,dt$
integrate	$\ln\lvert x\rvert = -\int p(t)\,dt + c$	$\ln\lvert x\rvert = -t^2 + c$
exponentiate	$\lvert x\rvert = e^c e^{-\int p(t)\,dt}$	$\lvert x\rvert = e^c e^{-t^2}$
drop $\lvert\cdot\rvert$, restore $x = 0$	$x = Ce^{-\int p(t)\,dt}$	$x = Ce^{-t^2}$

Here $\int p(t)\,dt$ denotes one fixed antiderivative (in the example, t^2); the constant of integration is absorbed into C. Thus the general solution of the homogeneous equation is

$$\boxed{x = C\,x_h}, \qquad x_h = e^{-\int p(t)\,dt},$$

and x_h may be replaced by *any* nonzero solution.

4.2 Newton's law of cooling

Let $x(t)$ be the temperature of a drink in a cooler, and $T_{\text{ext}}(t)$ the external temperature. The simplest (linear) model says that the temperature changes at a rate proportional to the temperature difference:

$$x' = k\big(T_{\text{ext}}(t) - x\big), \qquad \text{i.e.} \qquad x' + kx = k\,T_{\text{ext}}(t).$$

This is Newton's law of cooling. As a sanity check, with $k > 0$, if $T_{\text{ext}} > x$ then $x' > 0$. The system is the cooler, the input signal is T_{ext}, and the output signal is x. The right side is k times the input signal rather than the input signal itself: what counts as input and output is a matter of interpreting

the equation, not of the equation itself. The equation stays linear if k depends on t.

The constant k is a COUPLING CONSTANT. It is small when the insulation is good, large when it is bad, and zero for perfect insulation, in which case the inside temperature is decoupled from the outside.

Example: Take $k = \frac{1}{3}$ and an outside temperature rising at a constant rate, $T_{\text{ext}} = 60 + 6t$ (t in hours), with $x(0) = 32$:

$$x' + \tfrac{1}{3}x = 20 + 2t, \qquad x(0) = 32. \tag{4.2}$$

This equation is not separable.

4.3 Integrating factors

The method of INTEGRATING FACTORS (due to Euler) solves every first-order linear equation, using the product rule $(ux)' = ux' + u'x$.

Example: The equation $tx' + 2x = t$ is linear but not separable. Multiplying by t gives $t^2x' + 2tx = t^2$, whose left side is a derivative:

$$(t^2x)' = t^2, \qquad t^2x = \frac{t^3}{3} + c, \qquad x = \frac{t}{3} + \frac{c}{t^2}.$$

In reduced form the equation is $x' + \frac{2}{t}x = 1$, and the factor that works for it is t^2.

In general, multiply the reduced equation (4.1) by a function $u(t)$:

$$ux' + pux = uq.$$

The left side equals $(ux)' = ux' + u'x$ provided

$$u' = pu.$$

Any one nonzero solution u of this equation is an INTEGRATING FACTOR. With it, $(ux)' = uq$, which is solved by two integrations.

Example: For the cooler (4.2), $p = \frac{1}{3}$ is constant, so $u' = \frac{1}{3}u$ is the natural growth equation and $u = e^{t/3}$ works. Then

$$\left(e^{t/3}x\right)' = (20 + 2t)e^{t/3}, \qquad e^{t/3}x = 60e^{t/3} + \int 2te^{t/3}\,dt.$$

Integrating by parts with $U = 2t$, $dV = e^{t/3}dt$, so $dU = 2\,dt$, $V = 3e^{t/3}$:

$$e^{t/3}x = 60e^{t/3} + 6te^{t/3} - 18e^{t/3} + c = (42 + 6t)e^{t/3} + c,$$

so the general solution is

$$x = 42 + 6t + ce^{-t/3}.$$

The initial condition gives $32 = 42 + c$, so $c = -10$ and

$$\boxed{x = 42 + 6t - 10e^{-t/3}}.$$

4.4 Structure of the solution: particular solutions and transients

The general solution of the cooler equation has the form

$$x = x_p + c\,u^{-1},$$

where x_p is *any* one solution, called a PARTICULAR SOLUTION (here the simple $x_p = 42 + 6t$), and $u^{-1} = e^{-t/3}$ is the reciprocal of the integrating factor, which solves the homogeneous equation.

Very often, as here, $x_h \to 0$ as $t \to \infty$. It is then called a TRANSIENT: all solutions come to look alike as time goes on, a funnel. A drink that starts warmer than 60° first cools, crosses the nullcline $x = 60 + 6t$ (where $x' = 0$, momentarily in equilibrium with the outside), and then warms, becoming asymptotic to $42 + 6t$ like every other solution.

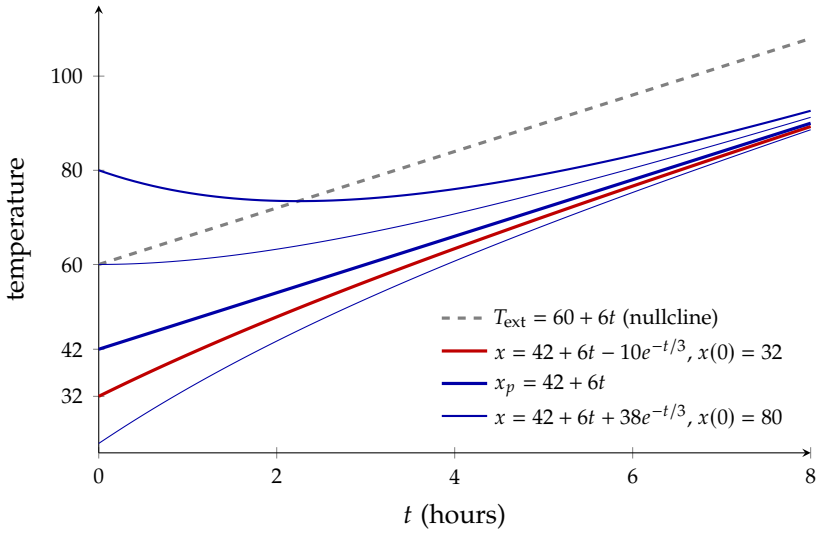


Figure 7: Solutions $x = 42 + 6t + ce^{-t/3}$ of the cooler equation for $c = -20, -10, 0, 18, 38$, with the external temperature. Every solution approaches $42 + 6t$; a solution meets T_{ext} only where its slope is zero.

4.5 The general formula

Theorem 4.1 [Integrating factor]. *For $x' + p(t)x = q(t)$ with continuous p, q, an integrating factor is*

$$u = e^{\int p(t)\,dt},$$

and any nonzero constant multiple also works. It is the reciprocal of a solution of the homogeneous equation, $u = x_h^{-1}$. The general solution is

$$\boxed{x = u^{-1} \int u\, q\, dt = x_p + c\, x_h}.$$

Proof. The equation $u' = pu$ is separable: $du/u = p\,dt$ gives $\ln|u| = \int p\,dt$, with the constant of integration inside the indefinite integral. So $|u| = e^{\int p\,dt}$; choose a sign, say $u > 0$. Comparing with the homogeneous solution $x_h = e^{-\int p\,dt}$ shows $u = x_h^{-1}$. Multiplying the equation by u gives $(ux)' = uq$, so $ux = \int uq\,dt$. Writing the constant of integration separately, $\int uq\,dt = F + c$ with F one antiderivative, gives $x = u^{-1}F + cu^{-1} = x_p + cx_h$ with $x_p = u^{-1}F$. □

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 4, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 5: Complex numbers, roots of unity

Anything with an amplitude and a phase is secretly a complex number, and it pays to make that identity explicit.

5.1 Complex algebra

The real numbers fill a line; the complex numbers fill a plane. The point one unit above 0 is i (written j by many engineers). Complex numbers add, and are multiplied by real numbers, as vectors in the plane. The new rule is

$$i^2 = -1,$$

and otherwise the usual rules of algebra apply; for example, $(1+i)(1+2i) = 1+2i+i-2 = -1+3i$. Every complex number can be written uniquely as $z = a + bi$ with a, b real, its real part $a = \operatorname{Re} z$ and imaginary part $b = \operatorname{Im} z$. Note that $\operatorname{Im} z$ is a real number. Complex numbers carry no units and have no ordering: there is no "$<$" on them.

Example: *Multiplication by i.* Since $i(a+bi) = -b + ai$, multiplication by i takes the point (a, b) to $(-b, a)$: it rotates the plane by 90° counterclockwise about the origin.

5.2 Complex conjugation

The identity

$$(a+bi)(a-bi) = a^2 - (bi)^2 = a^2 + b^2 \tag{5.1}$$

produces a real number, the square of the distance from $a+bi$ to the origin. It allows division by any nonzero complex number:

$$\frac{i}{1+2i} = \frac{i}{1+2i} \cdot \frac{1-2i}{1-2i} = \frac{i+2}{1+4} = \frac{2+i}{5}.$$

Definition 5.1 [Complex conjugate]. The complex conjugate of $z = a+bi$ is $\bar{z} = a - bi$, the reflection of z across the real axis.

Proposition 5.2 [Properties of conjugation]. *For complex w, z,*

$$\overline{w+z} = \bar{w} + \bar{z}, \qquad \overline{wz} = \bar{w}\,\bar{z}, \qquad \bar{\bar{z}} = z.$$

Proof. Write $w = a+bi$ and $z = c+di$. Then $\bar{w}+\bar{z} = (a-bi)+(c-di) = (a+c)-(b+d)i = \overline{w+z}$. For the product, $wz = (ac-bd)+(ad+bc)i$, while $\bar{w}\,\bar{z} = (a-bi)(c-di) = (ac-bd)-(ad+bc)i$,

which is the conjugate of wz. Reflecting twice returns z. □

For example, $\bar{z} = -z$ means $a - bi = -a - bi$, i.e. $a = 0$: z is purely imaginary.

5.3 Polar form and polar multiplication

As points of the plane, complex numbers have polar coordinates.

Definition 5.3 [Modulus and argument]. The modulus (or absolute value, or magnitude) $|z|$ is the distance from z to 0. The argument $\operatorname{Arg} z$ is the angle up from the positive real axis to z; it is defined only up to adding multiples of 2π (and not at all for $z = 0$).

For example $\operatorname{Arg} 1 = 0$, $\operatorname{Arg} i = \pi/2$, $\operatorname{Arg}(-1) = \pi$. For $z = a + bi$, by (5.1),

$$|z|^2 = z\bar{z} = a^2 + b^2, \qquad \tan(\operatorname{Arg} z) = \frac{b}{a} = \frac{\operatorname{Im} z}{\operatorname{Re} z}.$$

Theorem 5.4 [Polar multiplication]. *Magnitudes multiply and angles add:*

$$|wz| = |w|\,|z|, \qquad \operatorname{Arg}(wz) = \operatorname{Arg} w + \operatorname{Arg} z.$$

Proof. Absolute values involve square roots, but their squares are easy. By Proposition 5.2,

$$|wz|^2 = (wz)\overline{(wz)} = wz\bar{w}\bar{z} = (w\bar{w})(z\bar{z}) = |w|^2|z|^2,$$

and both $|wz|$ and $|w||z|$ are nonnegative, so they are equal. In particular, the product of two numbers on the unit circle lies on the unit circle.
For the angles, first take $w = \cos\alpha + i\sin\alpha$ and $z = \cos\beta + i\sin\beta$ on the unit circle. Then

$$wz = (\cos\alpha\cos\beta - \sin\alpha\sin\beta) + i(\cos\alpha\sin\beta + \sin\alpha\cos\beta) = \cos(\alpha+\beta) + i\sin(\alpha+\beta)$$

by the angle addition formulas. A general nonzero number is a positive real multiple of one on the unit circle, $w = |w|(\cos\alpha + i\sin\alpha)$, and positive real factors do not change angles, so the general case follows. □

Read backwards, complex multiplication contains the angle addition formulas for sin and cos, which therefore need not be memorized. The rotation by 90° under multiplication by i is the special case $|i| = 1$, $\operatorname{Arg} i = \pi/2$.

Example: *Powers of* $1 + i$. Since $|1 + i| = \sqrt{2}$ and $\operatorname{Arg}(1 + i) = \pi/4$, each further factor of $1 + i$

multiplies the modulus by $\sqrt{2}$ and adds $\pi/4$ to the argument:

	modulus	argument	$a + bi$
$(1+i)^0$	1	0	1
$(1+i)^1$	$\sqrt{2}$	$\pi/4$	$1+i$
$(1+i)^2$	2	$\pi/2$	$2i$
$(1+i)^3$	$2\sqrt{2}$	$3\pi/4$	$-2+2i$
$(1+i)^4$	4	π	-4

So $(1+i)^4 = -4$, and $1+i$ is a fourth root of -4. The powers lie on a spiral emanating from the origin.

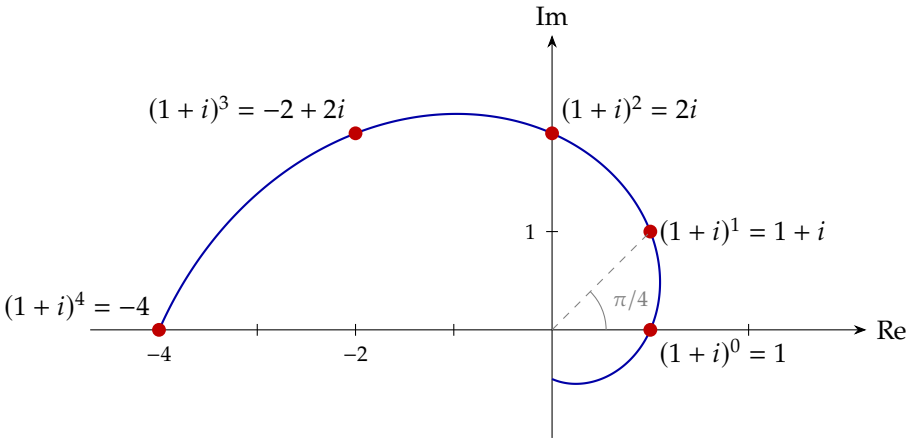


Figure 8: The powers $(1+i)^n$, $n = 0, \ldots, 4$, on the spiral $r = 2^{2\theta/\pi}$: each step multiplies the modulus by $\sqrt{2}$ and turns by $\pi/4$.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 5, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 6: Complex exponentials; sinusoidal functions

6.1 Powers and roots

Since magnitudes multiply and arguments add (Lecture 5),

$$|z^n| = |z|^n, \qquad \operatorname{Arg}(z^n) = n \operatorname{Arg} z.$$

For $z = 1 + i$, with $|z| = \sqrt{2}$ and $\operatorname{Arg} z = \pi/4$, the powers z^n march out along a spiral, for negative n as well as positive. The same rules find roots.

Example: *Cube roots of unity.* If $z^3 = 1$ then $|z|^3 = |z^3| = 1$, so $|z| = 1$ and z lies on the unit circle. Three times its argument must be $0, 2\pi, 4\pi, \ldots$, so the argument is 0, $2\pi/3$, or $4\pi/3$, giving

$$1, \qquad \frac{-1 + \sqrt{3}\,i}{2}, \qquad \frac{-1 - \sqrt{3}\,i}{2}.$$

In general, the nth roots of unity divide the unit circle into n equal arcs.

Example: *Cube roots of* -8. If $z^3 = -8$ then $|z|^3 = 8$; as $|z|$ is a positive real number, $|z| = 2$, so all cube roots lie on the circle of radius 2 about 0. The argument of -8 is π, 3π, or 5π (up to multiples of 6π), giving cube roots with arguments $\pi/3$, π, $5\pi/3$:

$$1 + \sqrt{3}\,i, \qquad -2, \qquad 1 - \sqrt{3}\,i.$$

Again they divide the circle evenly.

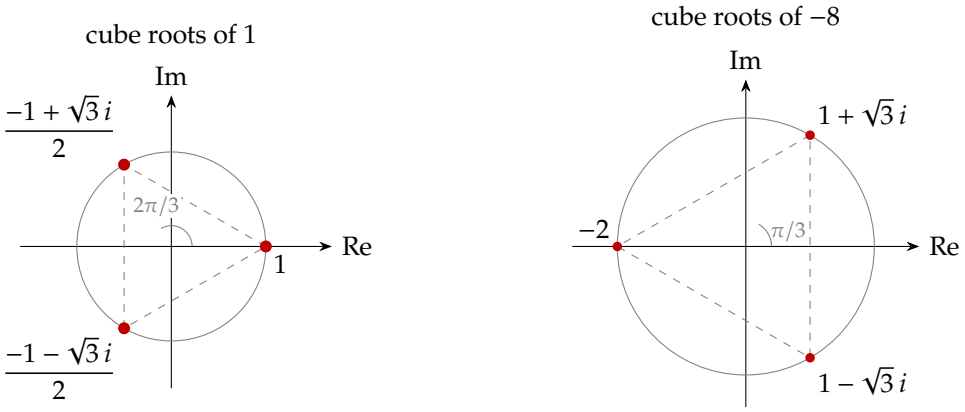


Figure 9: The cube roots of 1 on the unit circle and of -8 on the circle of radius 2; each set divides its circle into three equal arcs.

These are the roots of $z^n - a = 0$. More generally, a POLYNOMIAL is a function

$$f(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0, \qquad a_n \neq 0,$$

and its DEGREE is n, the index of the top nonzero coefficient.

Theorem 6.1 [FUNDAMENTAL THEOREM OF ALGEBRA]. *Every polynomial of degree $n > 0$ has a complex root. Consequently it factors as*

$$f(z) = a_n(z - r_1)(z - r_2)\cdots(z - r_n),$$

so it has n complex roots $r_1, \ldots, r_n$, some of which may coincide.

The factorization follows by repeatedly dividing off a factor $z - r$ for a root r, which lowers the degree by one.

6.2 The complex exponential

A complex-valued function $z(t) = a(t) + i\,b(t)$ of a real variable parametrizes a directed curve in the plane. Its derivative $z'(t) = a'(t) + i\,b'(t)$ is computed componentwise and is the velocity vector. For example, $z = t + i$ runs horizontally through i toward the right, with velocity $z' = 1$.

Consider the initial value problem

$$z' = iz, \qquad z(0) = 1. \tag{6.1}$$

For real k, e^{kt} is the solution of $x' = kx$, $x(0) = 1$, so the solution of (6.1) is written e^{it}. Multiplication by i rotates by $90°$, so along a solution the velocity is always perpendicular to the radius vector, and has the same length. The curve is therefore a circle about 0; with the initial condition, it is the unit circle traversed counterclockwise at unit speed: $z = \cos t + i \sin t$. Indeed $z' = -\sin t + i \cos t = iz$.

Definition 6.2 [EULER'S FORMULA]. For real t,

$$\boxed{e^{it} = \cos t + i \sin t}.$$

More generally, for a complex number $a + bi$, $e^{(a+bi)t}$ denotes the solution of $z' = (a + bi)z$, $z(0) = 1$, and

$$\boxed{e^{(a+bi)t} = e^{at}\big(\cos(bt) + i \sin(bt)\big)}.$$

To check the general formula, differentiate with the product rule:

$$\begin{aligned}\frac{d}{dt}\Big[e^{at}(\cos bt + i \sin bt)\Big] &= ae^{at}(\cos bt + i \sin bt) + be^{at}(-\sin bt + i \cos bt) \\ &= (a + bi)\, e^{at}(\cos bt + i \sin bt),\end{aligned}$$

since $b(-\sin bt + i\cos bt) = bi(\cos bt + i\sin bt)$; the value at $t = 0$ is 1. Thus $e^{(a+bi)t}$ has magnitude e^{at} and argument bt. For $a > 0$ and $b > 0$ it traces a spiral moving counterclockwise away from the origin. For example, $e^{(1+2\pi i)t}$ spirals outward counterclockwise, completing one turn per unit of time while its distance from 0 grows by the factor e.

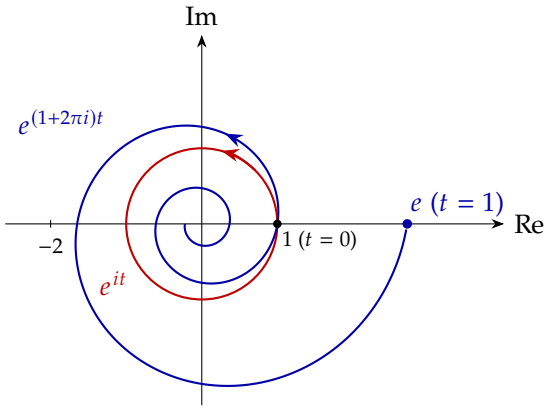


Figure 10: The unit circle e^{it} and the spiral $e^{(1+2\pi i)t}$ for $-1.5 \le t \le 1$; both pass through 1 at $t = 0$.

6.3 Consequences of Euler's formula

Proposition 6.3 [Exponential rule]. *For complex z, w and real t, $e^{(z+w)t} = e^{zt}e^{wt}$.*

Proof. This follows from the real exponential rule and the angle addition formulas, or from uniqueness: $f(t) = e^{zt}e^{wt}$ satisfies $f' = ze^{zt}e^{wt} + we^{zt}e^{wt} = (z + w)f$ and $f(0) = 1$, so it is the solution $e^{(z+w)t}$ of the same initial value problem. □

For any complex number z, adding and subtracting z and its reflection $\bar{z}$ gives

$$\operatorname{Re} z = \frac{z + \bar{z}}{2}, \qquad \operatorname{Im} z = \frac{z - \bar{z}}{2i}.$$

Conjugation preserves modulus and reverses angle, so $\overline{e^{it}} = e^{-it}$. Applied to $z = e^{it}$, these give the other forms of Euler's formulas:

$$\boxed{\cos t = \frac{e^{it} + e^{-it}}{2}, \qquad \sin t = \frac{e^{it} - e^{-it}}{2i}}.$$

Everything about sines and cosines can be derived from properties of the complex exponential. Some notable values are $e^{\pi i} = -1$, $e^{2k\pi i} = 1$ for integers k, and $e^{\pi i/2} = i$; the nth roots of unity are $e^{2\pi ik/n}$, $k = 0, 1, \ldots, n - 1$.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 6, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 7: Linear system response to exponential and sinusoidal input; gain, phase lag

Complex exponentials turn sinusoids into exponentials, and exponentials are easy to differentiate and integrate. This makes integrals of the form $e^{at}\cos(bt)$ and linear equations with sinusoidal input routine. Throughout, Euler's formula is used in the form

$$\operatorname{Re} e^{(a+bi)t} = e^{at}\cos(bt), \qquad \operatorname{Im} e^{(a+bi)t} = e^{at}\sin(bt).$$

7.1 Sinusoidal functions

A sinusoidal function is one whose graph is a cosine wave, possibly stretched and shifted. It is determined by three parameters:

- the amplitude A, the height of its maxima (and depth of its minima);
- the period P, the time until it repeats;
- the time lag t_0, the time of a maximum.

Write $f(t) = A\cos\theta$, where θ depends on t. Since $\theta = 0$ at $t = t_0$ and $\theta = 2\pi$ at $t = t_0 + P$, and θ must be a linear function of t (otherwise the wave would be distorted),

$$\theta = \frac{2\pi}{P}(t - t_0), \qquad f(t) = A\cos\Big(\frac{2\pi}{P}(t - t_0)\Big).$$

The frequency $\nu = 1/P$ counts cycles per unit time. More useful is the circular frequency $\omega = 2\pi/P$, in radians per unit time. With the phase lag $\phi = \omega t_0$,

$$\boxed{f(t) = A\cos(\omega t - \phi)}$$

For example, $\sin(\omega t) = \cos(\omega t - \pi/2)$: the sine lags a quarter cycle behind the cosine.

Example: The sinusoid in Figure 11 has $A = 2$, $P = 8$, and a maximum at $t_0 = -1$. So $\omega = 2\pi/8 = \pi/4$, $\phi = \omega t_0 = -\pi/4$, and

$$f(t) = 2\cos\Big(\frac{\pi}{4}(t + 1)\Big) = 2\cos\Big(\frac{\pi}{4}t + \frac{\pi}{4}\Big).$$

7.2 Sums of sines and cosines

A sum $a\cos(\omega t) + b\sin(\omega t)$ is again sinusoidal with the same circular frequency.

Proposition 7.1 [Trigonometric sum]. *Let (A, ϕ) be polar coordinates of the point (a, b), so that $A = \sqrt{a^2 + b^2}$, $a = A\cos\phi$, and $b = A\sin\phi$. Then*

$$a\cos(\omega t) + b\sin(\omega t) = A\cos(\omega t - \phi).$$

Proof. By the cosine difference identity, $A\cos(\omega t - \phi) = A\cos\phi\cos(\omega t) + A\sin\phi\sin(\omega t) = a\cos(\omega t) + b\sin(\omega t)$. □

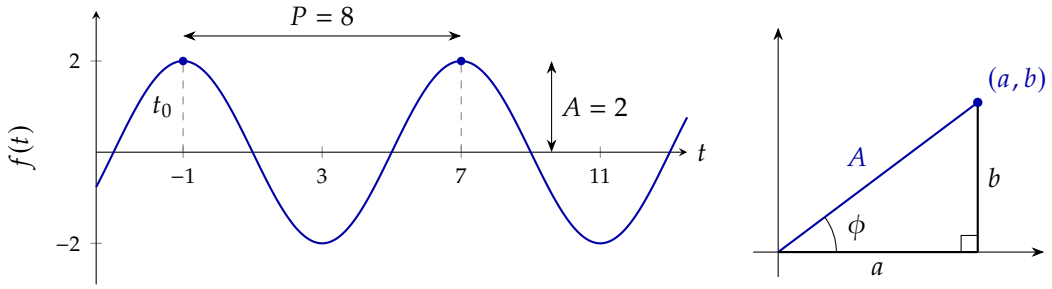


Figure 11: Left: the sinusoid $2\cos\left(\frac{\pi}{4}(t+1)\right)$ with amplitude $A = 2$, period $P = 8$, and time lag $t_0 = -1$. Right: the right triangle relating (a, b) to the amplitude A and phase lag ϕ in Proposition 7.1.

Example: For $\cos(\omega t) + \sqrt{3}\sin(\omega t)$, the point $(1, \sqrt{3})$ has $A = 2$ and $\phi = \pi/3$, so the sum is $2\cos(\omega t - \pi/3)$.

7.3 Integrating complex-valued functions

A complex-valued function of a real variable is differentiated and integrated separately in its real and imaginary parts. Since $\frac{d}{dt}e^{rt} = re^{rt}$ also holds for complex r, an antiderivative of e^{rt} is e^{rt}/r for $r \neq 0$. This replaces integration by parts twice:

$$\int e^{2t}\cos t\,dt = \operatorname{Re}\int e^{(2+i)t}\,dt = \operatorname{Re}\left(\frac{1}{2+i}e^{(2+i)t}\right) + c.$$

With $\frac{1}{2+i} = \frac{2-i}{5}$ and $e^{(2+i)t} = e^{2t}(\cos t + i\sin t)$, the real part of the product gives

$$\int e^{2t}\cos t\,dt = \frac{1}{5}e^{2t}(2\cos t + \sin t) + c.$$

7.4 Exponential input

Differentiation reproduces exponentials, so for a linear constant coefficient equation with exponential input one may hope for an exponential solution of the same form.

Example: For $x' + 2x = 4e^{3t}$, try $x_p = Ae^{3t}$. Then $x_p' + 2x_p = A(3+2)e^{3t}$, which equals $4e^{3t}$ when

$A = 4/5$. So $x_p = \frac{4}{5}e^{3t}$ is one solution, and adding the general solution of $x' + 2x = 0$ gives the general solution

$$x = \tfrac{4}{5}e^{3t} + ce^{-2t}.$$

The term ce^{-2t} is a transient.

7.5 Complex replacement

Example: Consider $x' + 2x = 2\cos t$. Since $2\cos t = \mathrm{Re}(2e^{it})$, replace it by the complex equation

$$z' + 2z = 2e^{it}.$$

If $z = x + iy$ solves this, then taking real parts (the coefficients are real) gives $x' + 2x = 2\cos t$. So a solution of the original equation is the real part of a solution of the complex one. Trying $z_p = Ae^{it}$ gives $A(i+2)e^{it} = 2e^{it}$, so

$$z_p = \frac{2}{2+i}e^{it} = \frac{2(2-i)}{5}(\cos t + i\sin t), \qquad x_p = \mathrm{Re}\, z_p = \frac{4}{5}\cos t + \frac{2}{5}\sin t.$$

This is the only sinusoidal solution; the general solution is $x = x_p + ce^{-2t}$.

The polar form of the complex factor gives amplitude and phase directly. Since $2 + i = \sqrt{5}\, e^{i\phi}$ with $\phi = \arctan(1/2)$,

$$z_p = \frac{2}{\sqrt{5}}e^{i(t-\phi)}, \qquad x_p = \frac{2}{\sqrt{5}}\cos(t-\phi).$$

The output is a sinusoid of the same frequency as the input $2\cos t$. Its amplitude is $1/\sqrt{5}$ times the input amplitude (the GAIN), and it lags behind the input by the phase lag $\phi = \arctan(1/2)$.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 7, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 8: Autonomous equations; the phase line, stability

A first order equation $y' = F(t, y)$ is autonomous when F does not depend on t:

$$y' = g(y).$$

The conditions the equation describes are constant in time, though they may depend on the current value of y. Autonomous equations are separable, but the aim here is a qualitative picture of the solutions.

8.1 The logistic equation

Natural growth $y' = k_0 y$ has a constant growth rate. A bank account with an interest rate depending on the balance but not on time, and a constant savings rate q, gives $y' = I(y)y + q$. Both are autonomous.

For a population, let the growth rate $k(y)$ depend on the current population. Suppose it is about k_0 for small y, decreases to zero as y approaches a maximal sustainable population p, and is negative for $y > p$. The simplest such rate is linear, $k(y) = k_0(1 - y/p)$, with $k(0) = k_0$ and $k(p) = 0$. This gives the logistic equation

$$y' = k_0\Big(1 - \frac{y}{p}\Big)y = g(y),$$

which is nonlinear but autonomous.

Since the slope depends only on y, the isoclines are horizontal lines; the nullclines are $y = 0$ and $y = p$. The graph of g is a downward parabola meeting the axis at $y = 0$ and $y = p$, so slopes are negative for $y < 0$, positive for $0 < y < p$, and negative for $y > p$. The bounded positive solutions are logistic curves (S-curves): they leave a neighborhood of 0 and approach p. A population above p falls back toward p.

Definition 8.1 [Critical points and stability]. A value c with $g(c) = 0$ is a critical point, or equilibrium, of $y' = g(y)$; the constant function $y = c$ is then a solution. The equilibrium is stable if nearby solutions move toward it on both sides, and unstable if nearby solutions move away from it.

For the logistic equation, $y = p$ is stable and $y = 0$ is unstable (Figure 12).

8.2 The phase line

Because the direction field is constant along horizontal lines, its content can be recorded on a single vertical line, the phase line: mark the equilibria, where $g(y) = 0$, and between them draw an

upward arrow where $g(y) > 0$ and a downward arrow where $g(y) < 0$.

Proposition 8.2 [Stability test]. *Let $g(c) = 0$. If $g'(c) < 0$, then c is stable; if $g'(c) > 0$, then c is unstable.*

Proof. If $g'(c) < 0$, then g changes sign from positive to negative at c, so the phase line arrows on both sides point toward c. If $g'(c) > 0$, the arrows on both sides point away from c. □

Example: For $g(y) = y^3 - y = y(y-1)(y+1)$, the equilibria are -1, 0, 1, and $g'(1) = 2 > 0$. So the rightmost critical point is unstable.

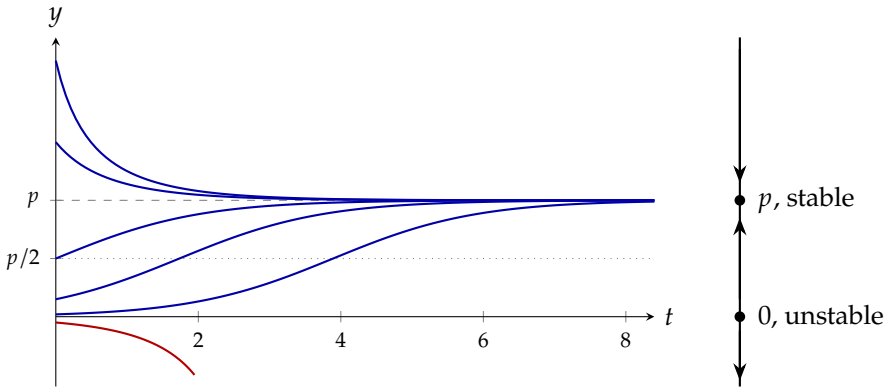


Figure 12: Solutions of the logistic equation with $k_0 = 1$ and $p = 1$, $y(t) = y_0/\big(y_0 + (1 - y_0)e^{-t}\big)$, and the phase line on the right. The S-curves have their inflection point at $y = p/2$. The solution starting slightly below 0 decreases without bound.

8.3 Harvesting and bifurcation

Without hunting, a population measured in thousands follows $y' = (a - y)y$, where the stable population a is proportional to the area of a preserve. A harvest of $1/4$ (thousand) per year gives

$$y' = g(y) = (a - y)y - \tfrac{1}{4} = -\Big(y^2 - ay + \tfrac{1}{4}\Big).$$

The graph of g is a downward parabola that rises as a increases. Its roots are

$$y = \frac{a}{2} \pm \frac{1}{2}\sqrt{a^2 - 1}.$$

- For $0 < a < 1$, $g < 0$ everywhere: the population crashes.
- At $a = 1$ there is a single equilibrium $y = 1/2$, which is semistable: solutions above it decrease to it, but any solution starting slightly below it crashes. So the smallest preserve allowing the harvest is not a safe choice.
- For $a > 1$ there are two equilibria; the upper one is stable and the lower one unstable. The farther apart they are, the more robust the population.

A BIFURCATION DIAGRAM assembles the phase lines for all values of the parameter a (Figure 13).

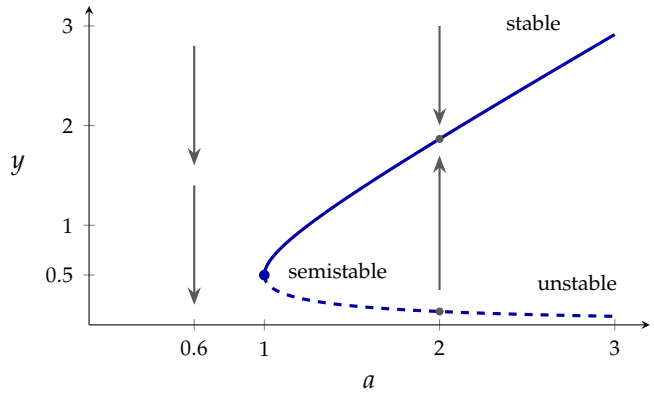


Figure 13: Bifurcation diagram of $y' = (a - y)y - \frac{1}{4}$. The equilibria $y = \frac{a}{2} \pm \frac{1}{2}\sqrt{a^2 - 1}$ exist for $a \geq 1$; the upper branch (solid) is stable and the lower branch (dashed) unstable. The arrows show the phase lines at $a = 0.6$ and $a = 2$.

8.4 Time translates, extrema, and inflection points

Since an autonomous equation is the same at every time, any TIME TRANSLATE $y(t - t_0)$ of a solution $y(t)$ is again a solution: its graph is shifted right by t_0.

Example: Every solution of $y' = k_0 y$ is a time translate of one of $e^{k_0 t}$, 0, or $-e^{k_0 t}$: it is $e^{k_0(t-t_0)}$, 0 (its only translate is itself), or $-e^{k_0(t-t_0)}$.

Proposition 8.3 [EXTREMA AND INFLECTION POINTS]**.** *Assume g is continuously differentiable, so solutions are uniquely determined by their initial values. Then nonconstant solutions of $y' = g(y)$ have no strict local maxima or minima, and their inflection points occur only at values y with $g'(y) = 0$.*

Proof. At an extremum, $y'(a) = 0$, so $g(y(a)) = 0$ and $y(a)$ is an equilibrium. The constant solution with this value agrees with y at $t = a$, so by uniqueness y is constant and has no strict extremum. By the chain rule, $y'' = g'(y)y' = g'(y)g(y)$. A nonconstant solution never takes an equilibrium value, so $g(y) \neq 0$ along it and $y'' = 0$ forces $g'(y) = 0$. □

For the logistic equation, $g'(y) = k_0(1 - 2y/p)$ vanishes at $y = p/2$, the inflection point of every S-curve.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 8, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 9: Linear vs. nonlinear

This lecture compares the two kinds of first order equation,

$$y' = F(t, y) \qquad \text{and} \qquad r(t)x' + p(t)x = q(t),$$

the general (nonlinear) equation and the linear one. Each viewpoint has its own strengths.

9.1 Linear equations: a solution procedure

A linear equation can always be reduced to integration: multiply by a factor that makes the left side the derivative of a product. Sometimes this is visible directly; for example, $t^2x' + 2tx = \frac{d}{dt}(t^2x)$. In reduced standard form ($r = 1$),

$$x' + p(t)x = q(t),$$

this is systematic. An integrating factor $u(t)$ satisfies $u(x' + px) = (ux)'$, that is, $u' = pu$. This is separable, with solution $u = e^{\int p(t)\,dt}$ (any constant of integration will do). Then $(ux)' = uq$, so

$$x = u^{-1} \int uq\,dt = x_p + cu^{-1}.$$

The constant of integration appears in a fixed place: the general solution is one particular solution plus a multiple of u^{-1}. The associated homogeneous equation $x' + px = 0$ is separable with solution $x_h = e^{-\int p\,dt} = u^{-1}$, the reciprocal of the integrating factor. In most applications $u^{-1} \to 0$ as $t \to \infty$, and cu^{-1} is a transient.

Proposition 9.1 [Solutions of linear equations do not blow up]. *If r, p, q are continuous on an interval I and $r \neq 0$ on I, then every solution of $r(t)x' + p(t)x = q(t)$ is defined on all of I.*

Proof. Divide by r to reach reduced form with continuous coefficients p/r and q/r. The integrating factor u and the integral $\int uq\,dt$ are then defined and finite on all of I, and $x = u^{-1}\int uq\,dt$ gives every solution. □

9.2 Nonlinear equations: qualitative information

Most nonlinear equations cannot be solved by formula, but much can be seen without integrating. The direction field of $y' = y^2 - x$ has a funnel along $y = -\sqrt{x}$: solutions near it are trapped and are asymptotic to $-\sqrt{x}$, and one can even argue that they ultimately lie slightly above it. The logistic

equation remains autonomous with a harvest rate that depends on the population,

$$y' = k_0\Big(1 - \frac{y}{p}\Big)y - a(y),$$

and its phase line shows the long-term behavior of the population directly (Lecture 8).

9.3 Linearization near equilibrium

Systems usually operate near equilibrium, and there the nonlinear equation behaves like a linear one. For the logistic equation, the equilibrium is $y = p$. Write $y = p + u$. Then $1 - y/p = -u/p$ and

$$u' = -\frac{k_0}{p}u(p + u) = -k_0 u - \frac{k_0}{p}u^2.$$

For small u the quadratic term is much smaller than the linear one. Dropping it LINEARIZES the equation near equilibrium:

$$u' = -k_0 u.$$

Near $y = p$, solutions behave like $p + u$ with $u = ce^{-k_0 t}$, so the population relaxes to equilibrium exponentially, at rate k_0.

9.4 The exponential response formula

For constant coefficient equations $x' + kx = q(t)$, the integrating factor is e^{kt}, but there is a faster method when the input is exponential. For $q(t) = Be^{rt}$, try $x_p = Ae^{rt}$. Then $x_p' + kx_p = A(r + k)e^{rt}$, so $A = B/(k + r)$.

Theorem 9.2 [EXPONENTIAL RESPONSE FORMULA, FIRST ORDER]. *If $k + r \neq 0$, then*

$$x_p = \frac{B}{k + r}e^{rt}$$

is a solution of $x' + kx = Be^{rt}$.

The formula holds for complex r as well, and Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ brings sinusoidal input within its reach.

Example: To solve $x' - x = 3\cos t$, replace it by $z' - z = 3e^{it}$, whose real part it is. The formula with $k = -1$, $r = i$, $B = 3$ gives

$$z_p = \frac{3}{-1 + i}e^{it} = \frac{3(-1 - i)}{2}(\cos t + i\sin t), \qquad x_p = \operatorname{Re} z_p = -\frac{3}{2}\cos t + \frac{3}{2}\sin t.$$

The general solution is $x = x_p + ce^t$. Here ce^t grows, so it is not a transient, but it still completes the general solution.

To see what the solution looks like, write $a\cos(\omega t) + b\sin(\omega t) = A\cos(\omega t - \phi)$ with (A, ϕ) the polar coordinates of (a, b). For $(a, b) = (-\frac{3}{2}, \frac{3}{2})$ this gives $A = \frac{3\sqrt{2}}{2}$ and $\phi = \frac{3\pi}{4}$:

$$x_p = \frac{3\sqrt{2}}{2}\cos\left(t - \frac{3\pi}{4}\right),$$

easy to draw and compare with the input $3\cos t$.

9.5 Blow-up of nonlinear solutions

Solutions of nonlinear equations can go to infinity in finite time. The equation $y' = y^2$ (with independent variable x) is separable: $y^{-2}\,dy = dx$ gives $-y^{-1} = x - c$, so

$$y = \frac{1}{c - x}.$$

The initial value problem $y(0) = 1$ has $c = 1$, so $y = 1/(1 - x)$, which tends to infinity as $x \to 1^-$. The formula $1/(1 - x)$ actually describes two solutions, one for $x < 1$ and one for $x > 1$. For a solution to be determined by its initial value, its graph must be connected: a solution is defined on an interval. The solution through $(0, 1)$ lives only on $(-\infty, 1)$.

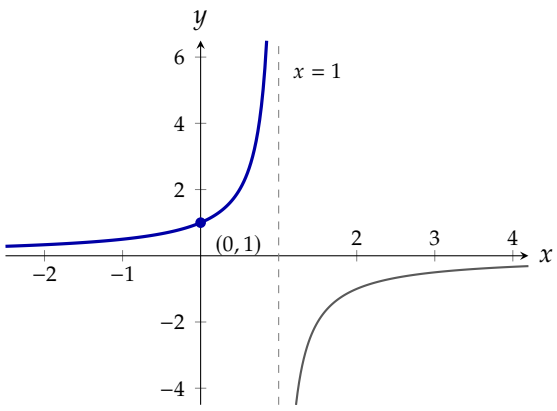


Figure 14: The function $1/(1 - x)$. The branch through the initial point $(0, 1)$ is the solution of $y' = y^2$, $y(0) = 1$, and blows up at $x = 1$; the branch for $x > 1$ is a different solution.

By Proposition 9.1, this cannot happen for linear equations: trouble in a solution can only come from trouble in the coefficients or the input. Physical laws can be nonlinear in this way; for Newtonian gravitation, Xia found a five-body system in which two of the bodies oscillate with increasing amplitude and frequency, roughly like $(1/t)\sin(1/t)$ as $t \to 0^-$.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 9, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 11: Modes and the characteristic polynomial

Second order linear equations model mechanical and electrical systems. This lecture sets up the spring, mass, and dashpot model and solves its homogeneous equation by exponentials when the characteristic polynomial has distinct real roots.

11.1 Springs and masses

A cart of mass m is attached to a wall by a spring. Choose the coordinate x so that the spring is relaxed, exerting no force, at $x = 0$. If an external force F_{ext} also acts on the cart, Newton's law gives

$$mx'' = F_{\text{spr}} + F_{\text{ext}}.$$

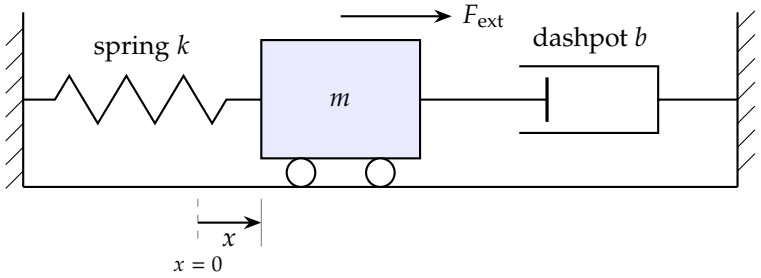


Figure 15: The spring, mass, and dashpot system. The coordinate x is the displacement of the cart from the position where the spring is relaxed.

The spring force depends only on position and pulls back toward equilibrium: $F_{\text{spr}}(x) < 0$ for $x > 0$, $F_{\text{spr}}(0) = 0$, and $F_{\text{spr}}(x) > 0$ for $x < 0$. The simplest model, valid for small x by the tangent line approximation, is Hooke's law

$$F_{\text{spr}}(x) = -kx, \qquad k > 0,$$

where k is the spring constant. This is a linear approximation like the linearization of Lecture 9. A rubber band is an example: its force is roughly linear for small extensions, nonlinear for larger ones, and the model fails completely when the band breaks.

11.2 Dashpots

Every real mechanical system has friction. Suppose it depends only on the velocity, as for a dashpot, a piston moving through an oil-filled cylinder (door dampers and shock absorbers often work this way). The damping force opposes the velocity: $F_{\text{dash}}(x') < 0$ for $x' > 0$, $F_{\text{dash}}(0) = 0$, and

$F_{\text{dash}}(x') > 0$ for $x' < 0$. The simplest model, valid for small x', is LINEAR DAMPING

$$F_{\text{dash}}(x') = -bx', \qquad b > 0,$$

with DAMPING CONSTANT b. Altogether,

$$\boxed{mx'' + bx' + kx = F_{\text{ext}}}$$

11.3 Second order linear equations

Definition 11.1 [LINEAR EQUATION]. A LINEAR DIFFERENTIAL EQUATION has the form

$$a_n(t)x^{(n)} + a_{n-1}(t)x^{(n-1)} + \cdots + a_1(t)x' + a_0(t)x = q(t),$$

with no term on the left that is free of x. The a_k are the COEFFICIENTS; they may depend on t but not on x. If $a_n \neq 0$, the equation has ORDER n.

The spring system is second order. The left side represents the SYSTEM and the coefficients its parameters (mass, damping constant, spring constant); the right side is the INPUT SIGNAL. Coefficients can vary in time (a rocket burning fuel loses mass, a spring softens with age), but they are usually taken *constant*, because they change on a much longer timescale than x. The input may vary, for example sinusoidally.

Many solutions of a second order equation pass through a given point (t_0, x_0). Physically, the initial position and the initial velocity together determine the motion. This is the existence and uniqueness theorem in this setting: for an equation of order n, the initial data are the n numbers $x(t_0), x'(t_0), \ldots, x^{(n-1)}(t_0)$.

11.4 The homogeneous case and superposition

With no external force, the system evolves on its own:

$$mx'' + bx' + kx = 0, \qquad m \neq 0. \tag{11.1}$$

This is a HOMOGENEOUS equation.

Example: *The harmonic oscillator.* With $b = 0$ and $k > 0$, set $\omega^2 = k/m$ to get $x'' + \omega^2 x = 0$. Since $\cos(\omega t)'' = -\omega^2 \cos(\omega t)$ and $\sin(\omega t)'' = -\omega^2 \sin(\omega t)$, both are solutions, and so is every sinusoid of circular frequency ω,

$$x = c_1 \cos(\omega t) + c_2 \sin(\omega t) = A \cos(\omega t - \phi).$$

These are all the solutions: $x(0) = c_1$ and $x'(0) = \omega c_2$, so c_1 and c_2 can be chosen uniquely to match any initial condition at $t = 0$, and uniqueness does the rest. For instance, every nonzero solution of

$x'' + 4x = 0$ has $\omega = 2$ and period $2\pi/\omega = \pi$.

Theorem 11.2 [Superposition I]. *If x_1 and x_2 are solutions of a homogeneous linear equation, then so is every linear combination $c_1x_1 + c_2x_2$.*

Proof. For (11.1), linearity of differentiation gives

$$\begin{aligned} &m(c_1x_1 + c_2x_2)'' + b(c_1x_1 + c_2x_2)' + k(c_1x_1 + c_2x_2) \\ &\quad = c_1(mx_1'' + bx_1' + kx_1) + c_2(mx_2'' + bx_2' + kx_2) = c_1 \cdot 0 + c_2 \cdot 0 = 0. \end{aligned}$$

The same computation works for any order. □

11.5 Exponential solutions: the characteristic polynomial

The first order equation $x' + kx = 0$ has the solution e^{-kt}, so try $x = e^{rt}$ in (11.1) with constant m, b, k and an undetermined constant r. Then $x' = re^{rt}$ and $x'' = r^2e^{rt}$, so

$$mx'' + bx' + kx = (mr^2 + br + k)e^{rt}.$$

An exponential is never zero, so ce^{rt} is a solution for every c exactly when r is a root of the characteristic polynomial

$$p(s) = ms^2 + bs + k.$$

Example: For $x'' + 5x' + 4x = 0$, the characteristic polynomial $s^2 + 5s + 4 = (s + 1)(s + 4)$ has roots -1 and -4. The exponential solutions are e^{-t} and e^{-4t}, and by superposition $x(t) = c_1e^{-t} + c_2e^{-4t}$ is a solution for all c_1, c_2. With $x(0) = 2$ and $x'(0) = -5$, using $x'(t) = -c_1e^{-t} - 4c_2e^{-4t}$, the conditions $c_1 + c_2 = 2$ and $-c_1 - 4c_2 = -5$ give $c_2 = 1$, $c_1 = 1$, and $x(t) = e^{-t} + e^{-4t}$.

This family is the general solution. For distinct real roots $r_1 \neq r_2$, the conditions $c_1 + c_2 = x_0$ and $r_1c_1 + r_2c_2 = x_0'$ have determinant $r_2 - r_1 \neq 0$, so every initial condition at $t = 0$ is met, and uniqueness shows there are no other solutions.

The same substitution works in any order, since $\frac{d^j}{dt^j}e^{rt} = r^je^{rt}$ turns the left side into $p(r)e^{rt}$:

Theorem 11.3 [Exponential solutions]. *For the constant coefficient equation $a_nx^{(n)} + \cdots + a_1x' + a_0x = 0$ with characteristic polynomial $p(s) = a_ns^n + \cdots + a_1s + a_0$, the function e^{rt} is a solution if and only if $p(r) = 0$. By superposition, every linear combination of such exponentials is also a solution.*

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 11, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 12: Good vibrations, damping conditions

This lecture completes the solution of the unforced spring, mass, and dashpot equation

$$mx'' + bx' + kx = 0, \tag{12.1}$$

a homogeneous constant coefficient linear second order equation. Its exponential solutions e^{rt} correspond to the roots r of the characteristic polynomial $p(s) = ms^2 + bs + k$; these are the modes of the system.

12.1 Summary of the three cases

Completing the square,

$$s^2 + \frac{b}{m}s + \frac{k}{m} = \left(s + \frac{b}{2m}\right)^2 - \left(\left(\frac{b}{2m}\right)^2 - \frac{k}{m}\right), \qquad r = -\frac{b}{2m} \pm \sqrt{\left(\frac{b}{2m}\right)^2 - \frac{k}{m}}.$$

The sign of the discriminant sorts the solutions into three cases. Write $\omega_d = \sqrt{k/m - (b/2m)^2}$ in the underdamped case.

Name	Condition	Characteristic roots	Basic real solutions
Overdamped	$b^2/4m > k$	distinct real r_1, r_2	$e^{r_1 t}$, $e^{r_2 t}$
Critically damped	$b^2/4m = k$	repeated $r = -b/2m$	e^{rt}, te^{rt}
Underdamped	$b^2/4m < k$	$-b/2m \pm i\omega_d$	$e^{-bt/2m}\cos(\omega_d t)$, $e^{-bt/2m}\sin(\omega_d t)$

Table 1: Unforced responses of $mx'' + bx' + kx = 0$. The names assume $b, k \geq 0$; the rest of the table holds for all real $m \neq 0$, b, k.

12.2 Overdamping: distinct real roots

For $x'' + 5x' + 4x = 0$ (Lecture 11), $p(s) = (s+1)(s+4)$ has roots -1 and -4. The modes e^{-t} and e^{-4t} are pure states, and the general solution $x = c_1 e^{-t} + c_2 e^{-4t}$ is a mixture of them. There is no vibration.

12.3 Underdamping and the real solution theorem

Make the spring a little stronger and the dashpot a little weaker: $x'' + 4x' + 5x = 0$. Completing the square, $s^2 + 4s + 5 = (s+2)^2 + 1$, so the roots are $s = -2 \pm i$ and the exponential solutions are

$e^{(-2+i)t}$ and $e^{(-2-i)t}$. These are complex valued; real solutions come from the following theorem.

Theorem 12.1 [Real solution theorem]. *If z is a complex-valued solution of $mz'' + bz' + kz = 0$, where m, b, k are real, then the real and imaginary parts of z are also solutions.*

Proof. Write $z = u + iv$ with u, v real. Then

$$0 = mz'' + bz' + kz = (mu'' + bu' + ku) + i(mv'' + bv' + kv).$$

Both expressions in parentheses are real, so both vanish. □

Here $e^{(-2+i)t}$ has real part $e^{-2t}\cos t$ and imaginary part $e^{-2t}\sin t$. The conjugate root gives the same pair up to sign, since $e^{(-2-i)t}$ is the complex conjugate of $e^{(-2+i)t}$. So the general real solution is

$$x = e^{-2t}(a\cos t + c\sin t) = Ae^{-2t}\cos(t - \phi),$$

a damped sinusoid with circular pseudo-frequency 1. In general, if the roots are $\alpha \pm i\omega$, go straight to the real solutions $e^{\alpha t}\cos(\omega t)$ and $e^{\alpha t}\sin(\omega t)$; the complex exponentials are only a convenient way to find them.

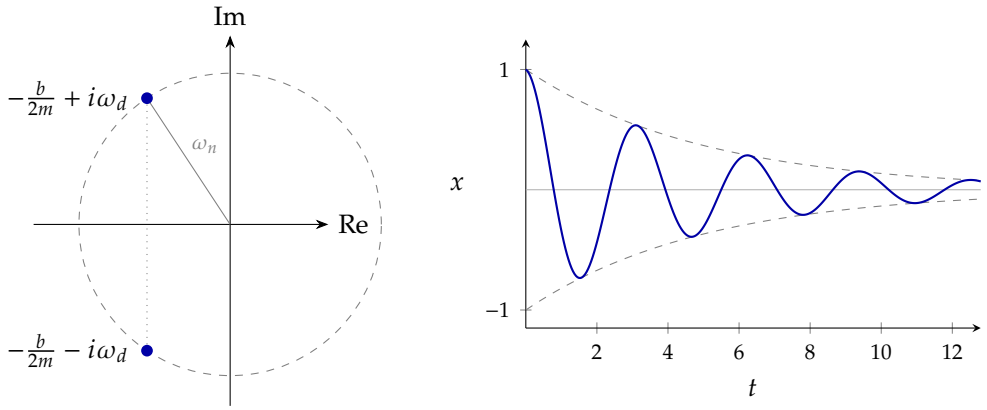


Figure 16: Left: underdamped characteristic roots $-b/2m \pm i\omega_d$, on the circle of radius ω_n. Right: the damped sinusoid $e^{-0.2t}\cos(2t)$ (roots $-0.2 \pm 2i$) between its envelope $\pm e^{-0.2t}$.

12.4 Natural and damped circular frequencies

When $(b/2m)^2 < k/m$, the root formula gives $r = -b/2m \pm i\omega_d$, using $\sqrt{-d} = i\sqrt{d}$.

Definition 12.2 [Natural and damped frequencies]. The natural circular frequency of the system is $\omega_n = \sqrt{k/m}$, whether or not $b = 0$. In the underdamped case, the damped circular frequency is

$$\omega_d = \sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2}.$$

With no damping, $mx'' + kx = 0$ is the HARMONIC OSCILLATOR: the roots are $\pm i\omega_n$, and the real and imaginary parts of $e^{i\omega_n t}$ recover the solutions $\cos(\omega_n t)$ and $\sin(\omega_n t)$.

As b increases from 0 with m and k fixed, the quantity under the square root decreases, so ω_d decreases and the period of oscillation $2\pi/\omega_d$ increases, from $2\pi/\omega_n$ at $b = 0$ to infinity as b approaches critical damping.

12.5 Critical damping

Between the two cases the roots coincide. For $x'' + 4x' + 4x = 0$, $p(s) = (s+2)^2$ has the repeated root -2, and the only exponential solutions are multiples of e^{-2t}. A second solution, not a multiple of the first, is te^{-2t}, so the general solution is

$$x = e^{-2t}(a + ct).$$

To check that te^{rt} solves the equation whenever r is a repeated root, differentiate the identity $L(e^{rt}) = p(r)e^{rt}$, where $L(x) = mx'' + bx' + kx$, with respect to r: $L(te^{rt}) = p'(r)e^{rt} + p(r)te^{rt}$, and both $p(r)$ and $p'(r)$ vanish at a repeated root. (Plugging in with the product rule works equally well.)

12.6 Transience

Theorem 12.3 [TRANSIENCE]. *If m, b, k are all positive, then every solution of (12.1) tends to 0 as $t \to \infty$.*

Proof. Use $r = -b/2m \pm \sqrt{(b/2m)^2 - k/m}$. In the underdamped case every solution has amplitude $Ae^{-bt/2m}$, which decays. In the overdamped case $k/m > 0$, so the square root is less than $b/2m$ and both roots are negative. In the critically damped case $r = -b/2m < 0$, and $te^{rt} \to 0$ as well, despite the factor t. □

12.7 The root diagram

Plot the characteristic roots in the complex plane. In the underdamped case they are the conjugate pair $-b/2m \pm i\omega_d$, and since $|r|^2 = (b/2m)^2 + \omega_d^2 = k/m$, they lie on the circle of radius ω_n.

- Roots farther left give faster decay. Roots on the imaginary axis give undamped oscillation, and roots in the right half-plane (negative damping) give oscillations growing exponentially in amplitude.
- Roots closer to the real axis give a smaller frequency and a longer period. Increasing b with m, k fixed moves the roots along the circle toward the real axis.

- When the roots meet on the real axis, damping is critical and the vibration stops. From then on, in the critically damped and overdamped regimes, no nonzero solution crosses the t-axis more than once.

For the last point: $e^{rt}(a + ct)$ vanishes only where $a + ct = 0$, and $c_1 e^{r_1 t} + c_2 e^{r_2 t} = 0$ means $e^{(r_1 - r_2)t} = -c_2/c_1$, which holds for at most one t since $r_1 \neq r_2$.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 12, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 13: Exponential response formula, spring drive

Now add an external force to the spring, mass, and dashpot system:

$$mx'' + bx' + kx = F_{\text{ext}}. \tag{13.1}$$

Placing the dashpot on the other side of the mass, in parallel with the spring, changes nothing: it still opposes the velocity, and the equation is the same. The associated homogeneous equation is

$$mx'' + bx' + kx = 0, \tag{13.2}$$

solved in Lecture 12. Dividing by m gives the reduced form $x'' + (b/m)x' + \omega_n^2 x = 0$, where $\omega_n = \sqrt{k/m}$ is the natural circular frequency. With $b = 0$, solutions oscillate at ω_n; with small $b > 0$ they are damped sinusoids of circular frequency $\omega_d < \omega_n$, and for large b they do not oscillate.

13.1 Superposition II

Recall Superposition I: linear combinations $c_1x_1 + c_2x_2$ of solutions of a homogeneous linear equation are solutions, and for a second order equation with neither x_1 nor x_2 a multiple of the other, they form the general solution.

Theorem 13.1 [Superposition II]. *If x_p is a solution of (13.1) and x_h is a solution of (13.2), then $x_p + x_h$ is a solution of (13.1). If x_h ranges over the general solution of (13.2), then $x_p + x_h$ is the general solution of (13.1).*

Proof. By linearity,

$$m(x_p + x_h)'' + b(x_p + x_h)' + k(x_p + x_h) = (mx_p'' + bx_p' + kx_p) + (mx_h'' + bx_h' + kx_h) = F_{\text{ext}} + 0.$$

Conversely, if x is any solution of (13.1), the same computation shows that $x - x_p$ solves (13.2), so $x = x_p + (x - x_p)$ has the stated form. □

The argument works for linear equations of any order; in first order it was already a consequence of integrating factors. The problem thus splits into finding *some* solution of (13.1), a particular solution, and the general solution of (13.2). The rest of the lecture finds particular solutions.

13.2 Spring drive: harmonic sinusoidal response

Take $b = 0$ and drive the mass through the spring: the far end of the spring is moved to position $y(t)$, and the coordinates are set so that the spring is relaxed when $x = y$. When $x > y$, the spring is stretched and pulls back, so

$$mx'' = k(y - x), \qquad \text{that is,} \qquad mx'' + kx = ky.$$

Here y is the input signal and x the system response, or output signal. For sinusoidal input $y = A\cos(\omega t)$,

$$x'' + \omega_n^2 x = \omega_n^2 A\cos(\omega t). \tag{13.3}$$

Two frequencies appear: the natural frequency ω_n of the system and the frequency ω of the input.

Try $x_p = B\cos(\omega t)$. Then $x_p'' + \omega_n^2 x_p = B(\omega_n^2 - \omega^2)\cos(\omega t)$, so for $\omega \neq \omega_n$,

$$B = \frac{\omega_n^2}{\omega_n^2 - \omega^2} A.$$

The output amplitude is a multiple of the input amplitude. The ratio is the GAIN

$$H(\omega) = \left|\frac{B}{A}\right| = \frac{\omega_n^2}{|\omega_n^2 - \omega^2|}.$$

With ω_n fixed and ω increasing from 0:

- at $\omega = 0$, $H = 1$: a slowly moved end is simply followed by the mass;
- H increases to a vertical asymptote at $\omega = \omega_n$. This is RESONANCE, where no sinusoidal solution of this form exists (a physical spring eventually behaves nonlinearly at large amplitude);
- for $\omega > \omega_n$, H falls back toward 0.

For $\omega < \omega_n$ the ratio B/A is positive and the output is in phase with the input. For $\omega > \omega_n$ it is negative, so the mass moves up when the driven end moves down: PHASE REVERSAL.

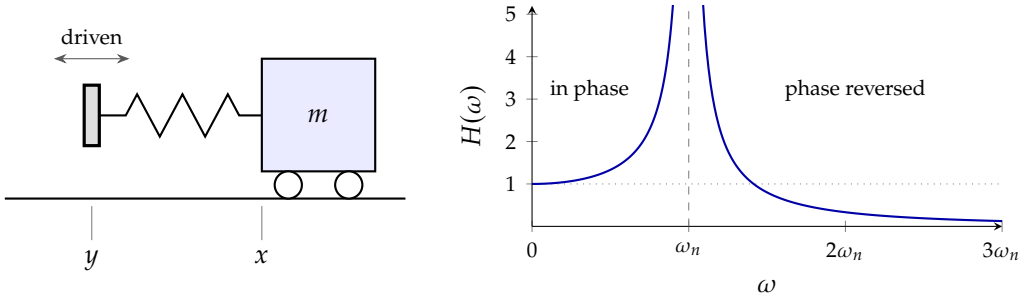


Figure 17: Left: the mass driven through the spring, whose far end is at position y. Right: the gain $H(\omega) = \omega_n^2/|\omega_n^2 - \omega^2|$ of (13.3), plotted with $\omega_n = 1$.

13.3 The exponential response formula

For exponential input $mx'' + bx' + kx = Ae^{rt}$, try $x_p = Be^{rt}$. Since $x_p' = Bre^{rt}$ and $x_p'' = Br^2e^{rt}$,

$$mx_p'' + bx_p' + kx_p = B(mr^2 + br + k)e^{rt} = Bp(r)e^{rt},$$

as in the first order case. The same computation works in any order.

Theorem 13.2 [Exponential response formula]. *Let $p(s) = a_n s^n + \cdots + a_1 s + a_0$ be the characteristic polynomial of the constant coefficient equation*

$$a_n x^{(n)} + \cdots + a_1 x' + a_0 x = Ae^{rt}.$$

If $p(r) \neq 0$, then a particular solution is

$$\boxed{x_p = \frac{A\, e^{rt}}{p(r)}}$$

If $p(r) = 0$, that is, r is a root of the characteristic polynomial, the system is in resonance with the input, and there is no solution of the form Be^{rt}.

Example: For $x'' + x' + 2x = 4e^{3t}$, $p(3) = 9 + 3 + 2 = 14$, so $x_p = \frac{4}{14}e^{3t} = \frac{2}{7}e^{3t}$.

13.4 Sinusoidal response by complex replacement

Example: Consider $x'' + x' + 2x = \cos t$. It is the real part of $z'' + z' + 2z = e^{it}$, and the real part of a solution of the complex equation solves the real one, because the coefficients are real. With $p(s) = s^2 + s + 2$, $p(i) = -1 + i + 2 = 1 + i$, so

$$z_p = \frac{1}{1+i}e^{it} = \frac{1-i}{2}(\cos t + i \sin t), \qquad x_p = \operatorname{Re} z_p = \frac{1}{2}(\cos t + \sin t).$$

The point $(\frac{1}{2}, \frac{1}{2})$ has polar coordinates $A = \sqrt{2}/2$ and $\phi = \pi/4$, so

$$x_p = \frac{\sqrt{2}}{2}\cos\left(t - \frac{\pi}{4}\right).$$

For input $\cos(\omega t)$, the denominator is $p(i\omega) = (k - m\omega^2) + ib\omega$. Without damping it is real, as in the spring drive. The damping term b gives it a nonzero imaginary part, which produces a phase lag in the response; it also increases $|p(i\omega)|$ and hence decreases the amplitude of the sinusoidal solution.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 13, Spring 2010.
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Lecture 14: Complex gain, dashpot drive

A stable linear system driven sinusoidally settles into a sinusoidal response at the input frequency. Two numbers describe it, the gain and the phase lag, and both can be read off from one complex number, the complex gain.

14.1 Transients and steady state

Consider the linear constant coefficient equation

$$a_n x^{(n)} + \cdots + a_1 x' + a_0 x = q(t), \qquad p(s) = a_n s^n + \cdots + a_1 s + a_0. \tag{14.1}$$

Solutions of the homogeneous equation ($q = 0$) are linear combinations of the exponentials e^{rt} for roots r of the characteristic polynomial p, multiplied by powers of t at repeated roots. Euler's formula gives $|e^{(a+bi)t}| = e^{at}$, so each such solution decays exactly when the real part of its root is negative.

Theorem 14.1 [Transience theorem]. *If every root of $p(s)$ has negative real part, then every solution of the homogeneous equation tends to 0 as $t \to \infty$.*

The homogeneous solutions are then called transients, and the equation is stable. By superposition, any two solutions of (14.1) differ by a transient, so all solutions converge together as t grows. After the transients die down the initial condition no longer matters: the system turns an input signal into a steady state output signal, and it suffices to find one particular solution x_p.

For sinusoidal input $A\cos(\omega t)$ (the clock is set so the input has no phase shift), the steady state output is again sinusoidal with the same circular frequency. By linearity its amplitude is proportional to A:

$$x_p = gA\cos(\omega t - \phi).$$

The gain g and the phase lag ϕ depend on the input circular frequency ω but not on A.

14.2 Gain and phase lag in polar form

Example: $x'' + x' + 2x = \cos t$. Here $p(s) = s^2 + s + 2$ and the right side is the input. The complex replacement $z'' + z' + 2z = e^{it}$ has, by the exponential response formula, the solution $z_p = e^{it}/p(i)$ with $p(i) = 1 + i$. In rectangular form $1/(1+i) = (1-i)/2$, so $x_p = \operatorname{Re} z_p = \frac{1}{2}\cos t + \frac{1}{2}\sin t$, which must then be converted to amplitude and phase. Passing to polar form first is more efficient:

$p(i) = \sqrt{2}\, e^{i\pi/4}$, so

$$z_p = \frac{1}{\sqrt{2}} e^{-i\pi/4} e^{it} = \frac{\sqrt{2}}{2} e^{i(t-\pi/4)}, \qquad x_p = \frac{\sqrt{2}}{2} \cos\Big(t - \frac{\pi}{4}\Big).$$

The gain is $\sqrt{2}/2 \approx 0.707$, and the response lags $\pi/4$ radians, or $1/8$ cycle, behind the input.

The polar form also shows how the parameters act. View the example as $mx'' + bx' + kx = \cos t$ with $m = 1$, $b = 1$, $k = 2$, so that $p(i) = (k - m) + bi = 1 + bi$. The phase lag is $\operatorname{Arg} p(i)$ and the gain is $1/|p(i)|$. If b increases with m and k fixed, then only the imaginary part of $p(i)$ grows. Since the real part $k - m = 1$ is positive, the argument of $p(i)$ increases, so the phase lag increases; and $|p(i)|$ increases, so the amplitude $1/|p(i)|$ of the response decreases. One divides by $p(i\omega)$, so a larger $|p(i\omega)|$ means a smaller response.

14.3 Driving through the dashpot

The spring system can also be driven through the dashpot. The spring is attached to a fixed wall, and the far end of the dashpot is moved, with position $y(t)$; as in Figure 18, x is the position of the mass. The dashpot exerts a force proportional to the relative velocity, $b(y' - x')$, and the spring exerts $-kx$, so

$$mx'' + bx' + kx = by'. \tag{14.2}$$

The input signal is y and the system response is x. The right side is not the input signal, nor even a multiple of it.

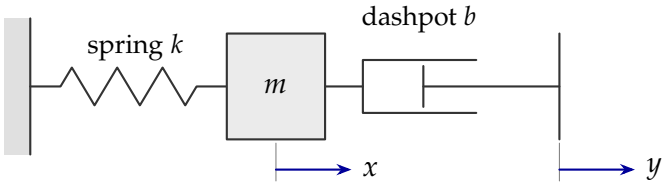


Figure 18: A spring, mass, and dashpot system driven through the dashpot. The input y is the position of the dashpot's far end; the response x is the position of the mass.

Drive sinusoidally, $y = A\cos(\omega t)$, and push the complex replacement back to the input signal itself: replace y by $y_{cx} = Ae^{i\omega t}$, so $\operatorname{Re} y_{cx} = y$. Solving (14.2) with y_{cx} in place of y,

$$mz'' + bz' + kz = by'_{cx} = bAi\omega e^{i\omega t},$$

the exponential response formula gives

$$z_p = \frac{bi\omega}{p(i\omega)} Ae^{i\omega t}, \qquad p(i\omega) = (k - m\omega^2) + bi\omega.$$

For $b > 0$ and $\omega > 0$, $p(i\omega) \neq 0$ because its imaginary part is nonzero.

14.4 Complex gain

Definition 14.2 [Complex gain]. For complex input $y_{cx} = Ae^{i\omega t}$, the complex gain $H(\omega)$ is the complex number by which the input is multiplied to give the exponential system response:

$$z_p = H(\omega)\, y_{cx}.$$

For the dashpot drive, $H(\omega) = bi\omega/p(i\omega)$. To return to the real equation, take real parts, $x_p = \mathrm{Re}\big(H(\omega)Ae^{i\omega t}\big)$, and compute in polar form. Write

$$H(\omega) = |H(\omega)|e^{-i\phi},$$

so that $-\phi = \operatorname{Arg} H(\omega)$. Then

$$z_p = A|H(\omega)|e^{-i\phi}e^{i\omega t} = A|H(\omega)|e^{i(\omega t-\phi)}, \qquad x_p = A|H(\omega)|\cos(\omega t - \phi).$$

Proposition 14.3 [Gain and phase lag from the complex gain]. *For sinusoidal input $A\cos(\omega t)$ the steady state response is $A|H(\omega)|\cos(\omega t - \phi)$. Thus the gain is $g(\omega) = |H(\omega)|$ and the phase lag is $\phi(\omega) = -\operatorname{Arg} H(\omega)$.*

Nothing in this calculation used the particular system: it holds for any linear time invariant system and any order. In the example of the previous subsection the input is the right side itself, so $H(\omega) = 1/p(i\omega)$ and $\phi = \operatorname{Arg} p(i\omega)$, consistent with what was found there.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 14, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 15: Operators, undetermined coefficients, resonance

Writing a linear constant coefficient equation as a polynomial in the differentiation operator handles every order at once. Differentiating the identity behind the exponential response formula with respect to the exponent gives the resonant case, and polynomial inputs have polynomial responses.

15.1 Differential operators

An operator takes functions to functions, as a function takes numbers to numbers. The differentiation operator D takes x to $Dx = x'$; for example $D\sin t = \cos t$, $Dt^n = nt^{n-1}$, and $D8 = 0$. It can be iterated, $D^2x = x''$, and together with the identity operator $Ix = x$ it generates linear combinations such as

$$(D^2 + 2D + 2I)x = x'' + 2x' + 2x.$$

The characteristic polynomial of this equation is $p(s) = s^2 + 2s + 2$, and the operator is $p(D)$. In general the equation

$$a_n x^{(n)} + \cdots + a_1 x' + a_0 x = q(t), \qquad p(s) = a_n s^n + \cdots + a_1 s + a_0,$$

is written $p(D)x = q(t)$, where $p(D) = a_n D^n + \cdots + a_1 D + a_0 I$ is a linear time invariant differential operator. It represents the system: an input signal determines $q(t)$, and the system response satisfies $p(D)x = q$. Much of the subject is about ways to invert such operators, $x = p(D)^{-1}q$.

The exponential response formula solves $p(D)x = Be^{rt}$ by $x_p = Be^{rt}/p(r)$, provided (1) r is not a root of p, and (2) B is constant. This lecture removes the first restriction.

15.2 The resonant response formula

Since $D^j e^{rt} = r^j e^{rt}$, multiplying by a_j and summing gives

$$p(D)e^{rt} = p(r)e^{rt}. \tag{15.1}$$

Dividing by $p(r)$ gives the exponential response formula, but (15.1) holds even when $p(r) = 0$.

Theorem 15.1 [Resonant response formula]. *If $p(r) = 0$ and $p'(r) \neq 0$, then*

$$\boxed{x_p = \frac{Bte^{rt}}{p'(r)}}$$

is a solution of $p(D)x = Be^{rt}$.

Proof. Regard (15.1) as an identity in two variables, t and r, and differentiate it with respect to r. Mixed partial derivatives of the smooth function e^{rt} commute, so $\partial/\partial r$ commutes with $D = \partial/\partial t$ and hence with $p(D)$:

$$\frac{\partial}{\partial r}\, p(D)e^{rt} = p(D)\frac{\partial}{\partial r}e^{rt} = p(D)\big(te^{rt}\big).$$

The right side of (15.1) differentiates to $p'(r)e^{rt} + p(r)te^{rt}$. Therefore

$$p(D)\big(te^{rt}\big) = p'(r)e^{rt} + p(r)te^{rt}.$$

When $p(r) = 0$ the second term vanishes, and dividing by $p'(r) \neq 0$ and multiplying by B gives the claim. □

Example: $x'' - 4x = e^{-2t}$. Here $p(s) = s^2 - 4$ vanishes at $r = -2$, and $p'(s) = 2s$ gives $p'(-2) = -4$. So

$$x_p = \frac{te^{-2t}}{-4} = -\frac{1}{4}te^{-2t}.$$

When the input exponent r is a root of $p(s)$, the system is said to be IN RESONANCE with the input signal.

Example: $x'' + 4x = \cos 2t$. Use the complex replacement $z'' + 4z = e^{2it}$. With $p(s) = s^2 + 4$, $p(2i) = 0$ and $p'(2i) = 4i$, so

$$z_p = \frac{te^{2it}}{4i}, \qquad x_p = \operatorname{Re} z_p = \frac{1}{4}t\sin 2t.$$

The oscillator is pushed in sync with its natural frequency, like a swing, and the amplitude grows linearly in t (Figure 19). In the first example the response does not grow: it still decays, but more slowly than the e^{-2t} that the exponential response formula would suggest.

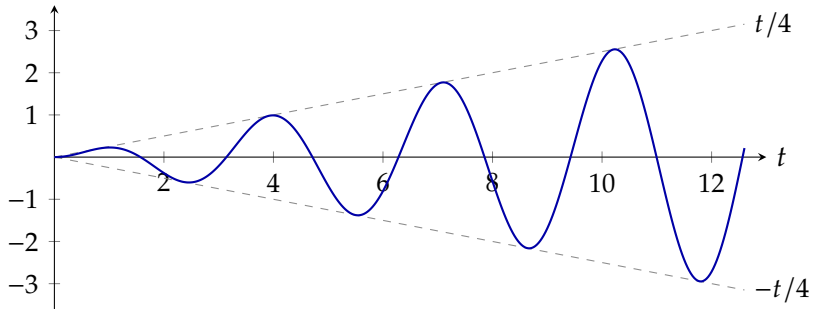


Figure 19: The resonant response $x_p = \frac{1}{4}t\sin 2t$ of $x'' + 4x = \cos 2t$, whose amplitude grows linearly between the lines $\pm t/4$.

15.3 Polynomial input: undetermined coefficients

Theorem 15.2 [Undetermined coefficients]. *Let $q(t) = b_k t^k + \cdots + b_1 t + b_0$. If $p(0) \neq 0$, then $p(D)x = q(t)$ has exactly one solution that is a polynomial of degree at most k.*

Proof. Note $p(0) = a_0$. The operator $p(D)$ maps the space of polynomials of degree at most k into itself, and

$$p(D)t^j = a_0 t^j + (\text{terms of degree less than } j).$$

In the basis $1, t, \ldots, t^k$ its matrix is therefore triangular with every diagonal entry equal to a_0. When $a_0 \neq 0$ this matrix is invertible, so exactly one polynomial x of degree at most k has $p(D)x = q$. In practice, plug in a polynomial with unknown coefficients and match coefficients from the top degree down. □

Example: $3x'' + 2x' + x = t^2 + 1$. Since $a_0 = 1 \neq 0$, there is exactly one solution $x = at^2 + bt + c$. Then $x' = 2at + b$ and $x'' = 2a$, so

$$3x'' + 2x' + x = at^2 + (b + 4a)t + (c + 2b + 6a) = t^2 + 1.$$

Matching coefficients gives $a = 1$, then $b = -4a = -4$, and $c = 1 - 2b - 6a = 3$. So

$$x_p = t^2 - 4t + 3.$$

If $a_0 = 0$ the theorem does not apply, but reduction of order does: substitute for the lowest derivative that appears.

Example: $x'' + x' = t$. Put $u = x'$, so $u' + u = t$. Trying $u = at + b$ gives $at + (a + b) = t$, so $a = 1$, $b = -1$, and $u_p = t - 1$. Integrating, $x_p = \frac{1}{2}t^2 - t$. The roots of $p(s) = s(s + 1)$ are 0 and -1, so $x_h = c_1 + c_2 e^{-t}$ and the general solution is $x = \frac{1}{2}t^2 - t + c_1 + c_2 e^{-t}$.

The same substitution handles higher order equations with $p(0) = 0$.

Example: $3x^{(4)} + 2x^{(3)} + x'' = t^2 + 1$. Here 1 and t solve the homogeneous equation, so polynomial solutions, if any, are not unique. With $u = x''$ the equation becomes $3u'' + 2u' + u = t^2 + 1$, solved above by $u = t^2 - 4t + 3$. Integrating twice gives the polynomial solution

$$x_p = \frac{1}{12}t^4 - \frac{2}{3}t^3 + \frac{3}{2}t^2,$$

and adding $c_1 + c_2 t$ gives others. So there are infinitely many polynomial solutions, of degree four, and not all solutions are polynomials (the roots $(-1 \pm i\sqrt{2})/3$ of $3s^2 + 2s + 1$ contribute non-polynomial homogeneous solutions).

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 15, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 16: Frequency response

Variation of parameters removes the second restriction on the exponential response formula. The rest of the lecture studies how the gain and phase lag of a system depend on the input frequency, through Bode and Nyquist plots.

16.1 Variation of parameters

To solve $p(D)x = u(t)e^{rt}$ with nonconstant u, try $x_p = ue^{rt}$ with u now an unknown function. This is variation of parameters.

Example: $3x'' + 8x' + 6x = (t^2 + 1)e^{-t}$. With $x_p = ue^{-t}$,

$$x_p' = (u' - u)e^{-t}, \qquad x_p'' = (u'' - 2u' + u)e^{-t},$$

and so

$$3x_p'' + 8x_p' + 6x_p = (3u'' + 2u' + u)e^{-t}.$$

Cancelling e^{-t} leaves $3u'' + 2u' + u = t^2 + 1$, already solved by undetermined coefficients in Lecture 15: $u_p = t^2 - 4t + 3$. Hence

$$x_p = (t^2 - 4t + 3)e^{-t}.$$

The method always works this way. Since $D(ue^{rt}) = e^{rt}(D + rI)u$, iterating gives $p(D)(ue^{rt}) = e^{rt}p(D + rI)u$. So the substitution replaces the equation by one for u whose right side is the old one without the factor e^{rt}.

16.2 Complex gain

The method of Lecture 14 does not depend on the order of the system. For input $y = A\cos(\omega t)$, replace it by $y_{cx} = Ae^{i\omega t}$ and let z_p be the exponential system response. Then:

complex gain	$z_p = H(\omega)\, y_{cx}$
polar form	$H(\omega) = \lvert H(\omega)\rvert e^{-i\phi}$, so $z_p = \lvert H(\omega)\rvert Ae^{i(\omega t - \phi)}$
real part	$x_p = \lvert H(\omega)\rvert A\cos(\omega t - \phi)$
gain	$g(\omega) = \lvert H(\omega)\rvert$
phase lag	$\phi(\omega) = -\operatorname{Arg} H(\omega)$

A Bode plot is the graph of the gain or of the phase lag against the input frequency ω. The Nyquist plot is the trajectory of the complex gain $H(\omega)$ in the complex plane as ω varies.

16.3 Second order: driving through the dashpot

Return to the spring system driven through the dashpot (Lecture 14),

$$mx'' + bx' + kx = by', \qquad y = A\cos(\omega t),$$

with $b > 0$. Its Bode plots (Figure 20) suggest:

- for small ω the gain is small and the phase lag is about $-\pi/2$ (the response leads);
- the gain reaches a maximum value 1, at a RESONANT FREQUENCY ω_r, where $\phi(\omega_r) = 0$, so the dashpot is "locked" and the mass moves with the input;
- for large ω the gain is small again and the phase lag is about $+\pi/2$;
- ω_r does not depend on b; as b decreases, the RESONANT PEAK gets sharper and the phase lag switches from $-\pi/2$ to $\pi/2$ more abruptly.

For $b \neq 0$ the peak has finite height, unlike the true resonance of Lecture 15. The same equation models an AM radio receiver: of all the incoming waves, only those with frequency near ω_r excite the receiver appreciably. This is tuning, and the sharper the peak, the better the reception.

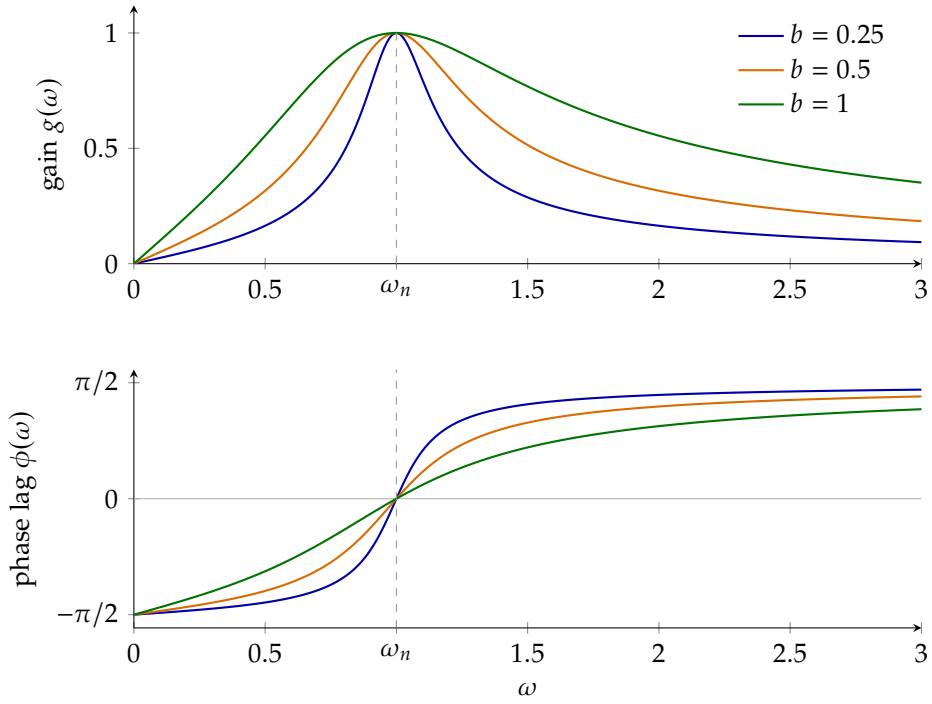


Figure 20: Bode plots of $mx'' + bx' + kx = by'$ for $m = k = 1$ (so $\omega_n = 1$) and $b = 0.25$, 0.5, 1, drawn from the exact formulas (16.1) and $\phi = -\operatorname{Arg} H(\omega)$. The peak is always at ω_n with height 1.

Proposition 16.1 [Dashpot-driven frequency response]. *The complex gain and gain are*

$$H(\omega) = \frac{bi\omega}{p(i\omega)}, \qquad g(\omega) = \frac{b\omega}{\sqrt{(k - m\omega^2)^2 + b^2\omega^2}}. \tag{16.1}$$

The maximal gain is 1*, attained exactly at* $\omega_r = \sqrt{k/m} = \omega_n$*, independent of* b*, and* $\phi(\omega_r) = 0$*. As* $\omega \to 0$*,* $\phi \to -\pi/2$*; as* $\omega \to \infty$*,* $\phi \to \pi/2$*.*

Proof. The formula for H comes from Lecture 14, with $p(i\omega) = (k - m\omega^2) + bi\omega$, and $g = |H|$. The denominator of g is at least the numerator $b\omega$, with equality exactly when $k - m\omega^2 = 0$, that is, $\omega = \sqrt{k/m}$. There $H(\omega_r) = bi\omega_r/(bi\omega_r) = 1$, so the gain is 1 and the phase lag is 0. For small ω, $H(\omega) \approx bi\omega/k$, which has argument $\pi/2$; for large ω, $H(\omega) \approx bi\omega/(-m\omega^2) = -ib/(m\omega)$, which has argument $-\pi/2$. □

Notably, the frequency of maximal response for this system is the natural frequency itself.

Proposition 16.2 [Nyquist circle]. *For every* $m, b, k > 0$*, the Nyquist plot of* $H(\omega) = bi\omega/p(i\omega)$ *lies on the circle of radius* $\frac{1}{2}$ *centered at* $\frac{1}{2}$*.*

Proof. Write $a = k - m\omega^2$ and $c = b\omega$, both real, so $H = ic/(a + ic)$. Then

$$\left|\frac{1}{2} - \frac{ic}{a + ic}\right| = \frac{1}{2}\left|\frac{a + ic - 2ic}{a + ic}\right| = \frac{1}{2}\,\frac{|a - ic|}{|a + ic|} = \frac{1}{2}.$$

□

As ω runs from 0 to ∞, H starts at 0, passes through the upper half plane while $\omega < \omega_n$ (there $a > 0$), reaches 1 at ω_n, and returns to 0 through the lower half plane (Figure 21).

16.4 First order: tides in a bay

A bay communicates with the ocean through a narrow channel. Let y be the ocean level and x the bay level, measured from the same mark. Water flows in when $x < y$ and out when $x > y$, so the linear model is

$$x' = k(y - x), \quad \text{or} \quad x' + kx = ky,$$

with coupling constant $k > 0$, just like Newtonian cooling. With sinusoidal input $y = A\cos(\omega t)$, the response crosses the input exactly at its own maxima and minima: there $x' = 0$, so $ky = kx$.

The complex replacement $z' + kz = kAe^{i\omega t}$ has $z_p = kAe^{i\omega t}/(i\omega + k)$, so

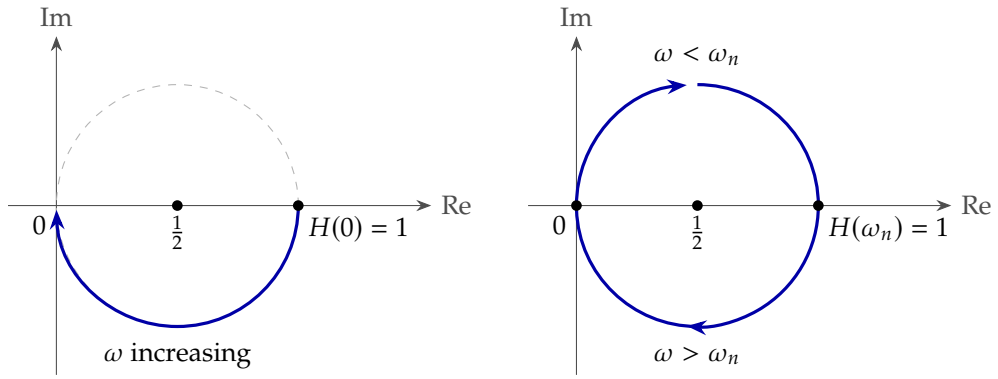
$$H(\omega) = \frac{k}{k + i\omega}, \qquad g(\omega) = \frac{k}{\sqrt{k^2 + \omega^2}}, \qquad \tan\phi(\omega) = \frac{\omega}{k}.$$

Since $\operatorname{Arg}(k + i\omega)$ lies between 0 and $\pi/2$, $\phi(\omega) = \arctan(\omega/k)$. These are the formulas behind the

Bode plots. The Nyquist plot is a semicircle independent of k:

$$\frac{1}{2} - H(\omega) = \frac{(i\omega + k) - 2k}{2(i\omega + k)} = \frac{1}{2}\,\frac{i\omega - k}{i\omega + k},$$

and k and $-k$ are equally far from $i\omega$, so $|\frac{1}{2} - H(\omega)| = \frac{1}{2}$. As ω runs from 0 to ∞, H moves from 1 to 0 through the lower half plane.



first order: $H = k/(k + i\omega)$ second order: $H = bi\omega/p(i\omega)$

Figure 21: Nyquist plots. Left: the first order system $x' + kx = ky$ traces the lower semicircle from $H(0) = 1$ toward 0. Right: the dashpot-driven second order system traces the whole circle, from 0 through $H(\omega_n) = 1$ and back to 0. Both circles have center $\frac{1}{2}$ and radius $\frac{1}{2}$ for all parameter values.

16.5 Driving through spring and dashpot

If the input y drives the mass through both the spring and the dashpot, both position and velocity of the input appear:

$$mx'' + bx' + kx = ky + by'.$$

This is a standard model of an automobile suspension. Its complex gain is $H(\omega) = (k + bi\omega)/p(i\omega)$. Now the gain can be well above 1, and the Nyquist plot depends heavily on the system parameters.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 16, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 17: LTI systems, superposition, RLC circuits

Second order constant coefficient equations model electrical circuits as well as springs. Two structural properties of the operator $p(D)$, linearity and time invariance, organize the methods for finding particular solutions.

17.1 RLC circuits

In a series circuit (Figure 22) the current I is the same everywhere. Let V be the voltage increase across the power source, the input signal; the current I is the system response. The components obey

$$V_R = RI, \qquad V_C' = \tfrac{1}{C} I, \qquad V_L = LI',$$

for the voltage drops across a resistor, a capacitor, and an inductor.

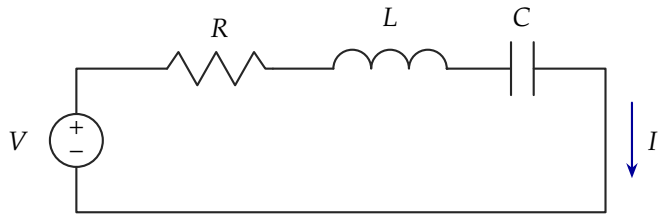


Figure 22: A series RLC circuit driven by a voltage source V. Without the inductor it is the RC circuit of Lecture 3.

For the RC circuit, Kirchhoff's voltage law says $V_R + V_C = V$; differentiating gives the first order equation

$$RI' + \tfrac{1}{C} I = V'.$$

Adding an inductor, the law becomes $V_L + V_R + V_C = V$, and differentiating gives

$$LI'' + RI' + \tfrac{1}{C} I = V'. \tag{17.1}$$

So the system is the operator $LD^2 + RD + \frac{1}{C}\,\mathrm{Id}$ (the identity is written Id to avoid confusion with the current), which takes the current, as a function of time, to the derivative of the impressed voltage. This is the spring equation with $L, R, 1/C$ in the roles of m, b, k.

17.2 Superposition

Theorem 17.1 [Superposition III]. *For constants c_1, c_2, $p(D)(c_1x_1 + c_2x_2) = c_1\,p(D)x_1 + c_2\,p(D)x_2$. Consequently, if $p(D)x_1 = q_1$ and $p(D)x_2 = q_2$, then $x = c_1x_1 + c_2x_2$ solves $p(D)x = c_1q_1 + c_2q_2$.*

Proof. Each derivative is linear, $(c_1x_1 + c_2x_2)^{(j)} = c_1x_1^{(j)} + c_2x_2^{(j)}$. Multiply the jth identity by a_j and add over $j = 0, \dots, n$. □

That is, $p(D)$ is a LINEAR OPERATOR. Taking $q_2 = 0$ gives Superposition II (particular plus homogeneous is a solution), and taking $q_1 = q_2 = 0$ gives Superposition I (combinations of homogeneous solutions are homogeneous solutions).

Example: $u'' - 4u = \cosh 2t$. Recall $\cosh x = \frac{1}{2}(e^x + e^{-x})$ and $\sinh x = \frac{1}{2}(e^x - e^{-x})$, so the right side is a combination of exponentials; solve for each separately. With $p(s) = s^2 - 4$, both $p(2) = 0$ and $p(-2) = 0$, so use the resonant response formula with $p'(2) = 4$ and $p'(-2) = -4$:

$$u_1 = \tfrac{1}{4}te^{2t} \text{ solves } u_1'' - 4u_1 = e^{2t}, \qquad u_2 = -\tfrac{1}{4}te^{-2t} \text{ solves } u_2'' - 4u_2 = e^{-2t}.$$

By superposition,

$$u = \frac{u_1 + u_2}{2} = \frac{t}{4} \cdot \frac{e^{2t} - e^{-2t}}{2} = \frac{t}{4} \sinh 2t.$$

17.3 Sine input and time invariance

Inputs of the form $A\cos(\omega t)$ are less special than they look. First, sine input is handled by taking imaginary parts.

Example: $x'' + 9x = \sin 2t$. This is the imaginary part of $z'' + 9z = e^{2it}$. With $p(s) = s^2 + 9$, $p(2i) = 9 - 4 = 5$, so $z_p = e^{2it}/5$ and $x_p = \operatorname{Im} z_p = \frac{1}{5}\sin 2t$.

In the same way, a harmonic oscillator driven away from resonance has no phase lag:

$$x'' + \omega_n^2 x = \cos(\omega t) \implies x_p = \frac{\cos(\omega t)}{\omega_n^2 - \omega^2}, \qquad x'' + \omega_n^2 x = \sin(\omega t) \implies x_p = \frac{\sin(\omega t)}{\omega_n^2 - \omega^2}.$$

The general reason is that a system with constant coefficients does not change in time: starting the input later only delays the response. Shifting the graph of $x(t)$ to the right by a gives the graph of $x(t - a)$.

Theorem 17.2 [TIME INVARIANCE]. *If $p(D)y(t) = q(t)$, then $p(D)\,y(t - a) = q(t - a)$. So to solve $p(D)x = q(t - a)$, solve $p(D)y = q(t)$ and set $x(t) = y(t - a)$.*

Proof. By the chain rule, the jth derivative of $y(t - a)$ is $y^{(j)}(t - a)$. Since the coefficients a_j are constant, $\sum_j a_j y^{(j)}(t - a)$ is the function $\sum_j a_j y^{(j)}$ evaluated at $t - a$, namely $q(t - a)$. With coefficients depending on t this fails. □

Example: $x'' + x' + 6x = \cos(2t - \pi/3)$. Write $\cos(2t - \pi/3) = \cos\big(2(t - \pi/6)\big)$, so the input is $f(t - a)$ with $f(t) = \cos 2t$ and $a = \pi/6$. Solve $y'' + y' + 6y = \cos 2t$ through $z'' + z' + 6z = e^{2it}$:

$p(s) = s^2 + s + 6$ and $p(2i) = 2 + 2i = 2\sqrt{2}\, e^{i\pi/4}$, so

$$z_p = \frac{1}{2\sqrt{2}} e^{i(2t-\pi/4)}, \qquad y_p = \frac{1}{2\sqrt{2}} \cos\Big(2t - \frac{\pi}{4}\Big).$$

Then

$$x_p(t) = y_p\Big(t - \frac{\pi}{6}\Big) = \frac{1}{2\sqrt{2}} \cos\Big(2\Big(t - \frac{\pi}{6}\Big) - \frac{\pi}{4}\Big) = \frac{1}{2\sqrt{2}} \cos\Big(2t - \frac{\pi}{3} - \frac{\pi}{4}\Big).$$

The shift $t - \pi/6$ replaces t, so it is multiplied by the circular frequency.

Corollary 17.3 [Gain and phase lag are intrinsic]. *The amplitude and phase lag of the steady state response to $A\cos(\omega t - \phi_0)$, measured relative to the input, depend only on ω and A, not on ϕ_0. Thus a system has a gain and a phase lag, functions of the input frequency, that are the same for every sinusoidal input of that frequency.*

Indeed, $A\cos(\omega t - \phi_0) = A\cos\big(\omega(t - a)\big)$ with $a = \phi_0/\omega$, so by time invariance the response is the response to $A\cos(\omega t)$ delayed by a.

17.4 Particular solutions by input type

For an LTI operator $p(D)$, homogeneous solutions are combinations of the modes e^{rt}, r a root of $p(s)$, together with $te^{rt}, \ldots$ at repeated roots; a non-real root $a + bi$ of a real polynomial comes with its conjugate, and the pair of modes combines into $e^{at}\cos(bt)$ and $e^{at}\sin(bt)$. If all roots have negative real part these are transients. The general solution of $p(D)x = q(t)$ is $x_p + x_h$, so all solutions converge together. A particular solution x_p is found according to the input:

Input $q(t)$	Method
e^{rt}	exponential (or, if $p(r) = 0$, resonant) response formula
$\cos(\omega t)$, or exponential times cosine	complex replacement
general sinusoid	time invariance, shifting to a cosine
polynomial	undetermined coefficients (reduce order if $p(0) = 0$)
(anything) $\times$ exponential	variation of parameters
linear combination	Superposition III

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 17, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 18: Engineering applications

The oscillator equation $mx'' + bx' + kx = F_{\text{ext}}$ is a working tool of vibration engineering: anything that makes noise vibrates. Engineers describe it through the natural frequency and a dimensionless damping ratio, which can be measured by counting cycles.

18.1 Natural frequency and damping ratio

Factor out m and complete the square in the characteristic polynomial:

$$p(s) = ms^2 + bs + k = m\left(\left(s + \frac{b}{2m}\right)^2 + \frac{k}{m} - \left(\frac{b}{2m}\right)^2\right).$$

If $k/m > (b/2m)^2$ the roots are not real, and the system is UNDERDAMPED; this is the only case in which the free system vibrates. The roots are then $-b/2m \pm i\omega_d$, with $\omega_d = \sqrt{k/m - (b/2m)^2}$, and the homogeneous solutions are $x_h = Ae^{-bt/2m}\cos(\omega_d t - \phi)$.

Definition 18.1 [NATURAL FREQUENCY AND DAMPING RATIO]. The UNDAMPED NATURAL FREQUENCY is $\omega_n = \sqrt{k/m}$. The DAMPING RATIO ζ is defined by

$$\frac{b}{2m} = \zeta\omega_n, \qquad \text{that is,} \qquad \zeta = \frac{b}{2m\omega_n} = \frac{b}{2\sqrt{mk}}.$$

Factoring ω_n out of ω_d gives

$$\omega_d = \omega_n\sqrt{1 - \zeta^2},$$

and dividing the equation by m puts it in the engineers' normal form

$$x'' + 2\zeta\omega_n x' + \omega_n^2 x = \frac{F_{\text{ext}}}{m}.$$

Critical damping is $\zeta = 1$ and underdamping is $\zeta < 1$. The free response is

$$x_h = Ae^{-\zeta\omega_n t}\cos(\omega_d t - \phi).$$

The exponent $\zeta\omega_n t$ is dimensionless and $\omega_n t$ already is (ω_n has units of 1/sec), so ζ is a pure number: it does not change if one switches from meters and seconds to feet and hours.

18.2 Measuring the damping ratio

A vibration engineer's rule of thumb is

$$\zeta \approx \frac{0.11}{n_{50\%}}, \tag{18.1}$$

where $n_{50\%}$ is the number of cycles it takes the amplitude of a free oscillation to decay by half.

Proposition 18.2 [Half-amplitude rule]. *Let $P = 2\pi/\omega_d$ be the period of the damped oscillation. Successive minima (or maxima) of x_h are P apart, and the amplitude is multiplied by $e^{-\zeta\omega_n P}$ from each to the next. Consequently*

$$\ln 2 = \zeta\omega_n P n_{50\%} = \frac{2\pi\zeta}{\sqrt{1-\zeta^2}}\, n_{50\%},$$

and for small ζ this gives $\zeta \approx \ln 2/(2\pi n_{50\%})$, with $\ln 2/(2\pi) = 0.1103\ldots$

Proof. The extrema do not occur where x_h touches the envelope $\pm Ae^{-\zeta\omega_n t}$, because the slope is not zero there. Instead differentiate: by the product rule, x_h' is $e^{-\zeta\omega_n t}$ times a linear combination of $\cos(\omega_d t - \phi)$ and $\sin(\omega_d t - \phi)$, which is again a sinusoid of circular frequency ω_d. So $x_h' = A' e^{-\zeta\omega_n t}\cos(\omega_d t - \phi')$, whose zeros are half a period apart; alternate zeros, the minima, are P apart. Over one period the factor $e^{-\zeta\omega_n t}$ is multiplied by $e^{-\zeta\omega_n P}$, and over n periods by $e^{-n\zeta\omega_n P}$. Setting this equal to $\frac{1}{2}$ for $n = n_{50\%}$ and taking logarithms gives $\ln 2 = \zeta\omega_n P n_{50\%}$. Finally $P = 2\pi/(\omega_n\sqrt{1-\zeta^2})$, and for small ζ, $\sqrt{1-\zeta^2} \approx 1$ ($\omega_d \approx \omega_n$), so $\ln 2 \approx 2\pi\zeta n_{50\%}$. □

Example: *A mass on a spring.* Ten cycles took 14.2 seconds, so the period is $P = 1.42$ seconds, the damped natural frequency is $1/P \approx 0.70$ cycles per second (Hz), and $\omega_d = 2\pi/P \approx 4.4$ radians per second. The amplitude halved in about 4 cycles, so $\zeta \approx 0.11/4 \approx 0.03$: roughly 2.5 to 3 percent of critical damping.

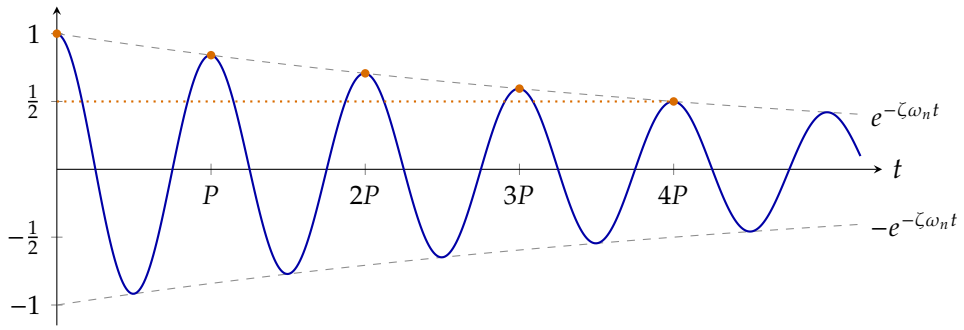


Figure 23: A free oscillation $e^{-\zeta\omega_n t}\cos(\omega_d t)$ with the values of the example, $P = 1.42$ seconds and ζ chosen by Proposition 18.2 so that $n_{50\%} = 4$ exactly ($\zeta \approx 0.0276$). The marked points, where $\cos(\omega_d t) = 1$, are one period apart and lie on the envelope; after four periods the amplitude has halved.

The same equation governs much more complicated systems. A vibrating string has infinitely many degrees of freedom and is described by a partial differential equation, but its fundamental mode, shaped like half a sine wave, obeys the oscillator equation, so its damping ratio can be measured

the same way. For a plucked string, $n_{50\%} \approx 5$ gave $\zeta \approx 0.11/5 \approx 0.02$.

18.3 Forced vibration and resonance by tuning

A cell phone vibrates by means of a small motor with an off-center mass on its shaft; the rotating mass exerts a sinusoidal force at the motor's frequency. Attach such a vibrator to the end of a cantilevered steel beam (a ruler). The beam's natural frequency depends on its length L as

$$\omega_n \propto \frac{1}{L^2},$$

with a constant determined by the beam's cross section and material (for the beam used, $\omega_n = 3.706/L^2$). Adjusting L until the beam oscillates with large amplitude brings ω_n to the driving frequency; reading off L then measures the driving frequency, so the beam is a frequency ruler.

This reverses the usual viewpoint of frequency response. There the system is fixed and the input frequency varies, tracing an amplitude response curve. Here the input frequency is fixed and a system parameter varies; still, near resonance appears when the two frequencies are close.

18.4 Vortex-induced vibration

Water flowing with speed U past a cylinder of diameter D sheds vortices into its wake at a frequency of approximately

$$f_S \approx 0.2\,\frac{U}{D}.$$

The shedding exerts a force on the cylinder transverse to the flow. For an oil well pipe with $D \approx 1$ meter in an ocean current of 1 meter per second, $f_S \approx 0.2$ Hz. A cylinder mounted on springs so that it can move transversely is a forced oscillator driven by this force. Replacing the dashpot by a generator, which extracts the energy the damping would otherwise dissipate, turns the device into a way of extracting power from a river.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 18 (with guest Kim Vandiver), Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 20: Fourier series

Periodic signals, such as a heartbeat, the sound of a violin, or many electronic signals, can be written as sums of sines and cosines. Parity and the orthogonality of the trigonometric functions determine the coefficients.

20.1 Periodic functions

Definition 20.1 [Periodic function]. A function $f(t)$ is PERIODIC if there is $L > 0$ such that $f(t + 2L) = f(t)$ for every t. The number $2L$ is a PERIOD of f.

Real signals agree with periodic functions only for a while; the methods accept this approximation, and the results approximate real behavior. A constant function has every period. All other periodic functions considered here have a minimal period, often called *the* period. Any WINDOW, an interval of length $2L$, determines the function; the usual choice is $[-L, L]$, which is why the period is written $2L$.

20.2 Sines, cosines, and Fourier series

Start with period 2π ($L = \pi$) and the window $[-\pi, \pi]$. The cosines $\cos(nt)$ for $n = 1, 2, 3, \ldots$ all have period 2π: $\cos(nt)$ has minimal period $2\pi/n$, so it also has period 2π. Likewise for $\sin(nt)$. These are the HARMONICS of the FUNDAMENTAL sinusoids with $n = 1$ (Figure 24, left). A linear combination $af + bg$ of functions of period $2L$ again has period $2L$, so one can form

$$f(t) = \frac{a_0}{2} + \sum_{n \geq 1} a_n \cos(nt) + \sum_{n \geq 1} b_n \sin(nt), \tag{20.1}$$

a FOURIER SERIES, consisting of a cosine series and a sine series. The a_n and b_n are the FOURIER COEFFICIENTS.

Theorem 20.2 [Fourier series]. *Any reasonable (for example, piecewise continuous) function of period 2π has exactly one expression as a Fourier series (20.1). Its Fourier coefficients are by definition the coefficients that make (20.1) true.*

Definition 20.3 [Square wave]. The standard SQUARE WAVE is the function of period 2π with

$$\mathrm{Sq}(t) = \begin{cases} 1, & 0 < t < \pi, \\ -1, & -\pi < t < 0, \\ 0, & t = 0, \pm\pi. \end{cases}$$

Cosines approximate Sq poorly: they are even and Sq is odd. Among sines, the even harmonics $\sin(2nt)$ break the symmetry of Sq about $t = \pi/2$ and do not help. Trial and error with odd harmonics shows that $b_1 = 1.00$, $b_3 = 0.33$, $b_5 = 0.20$ already approximate $\frac{\pi}{4}\,\mathrm{Sq}(t)$ well (Figure 24, right).

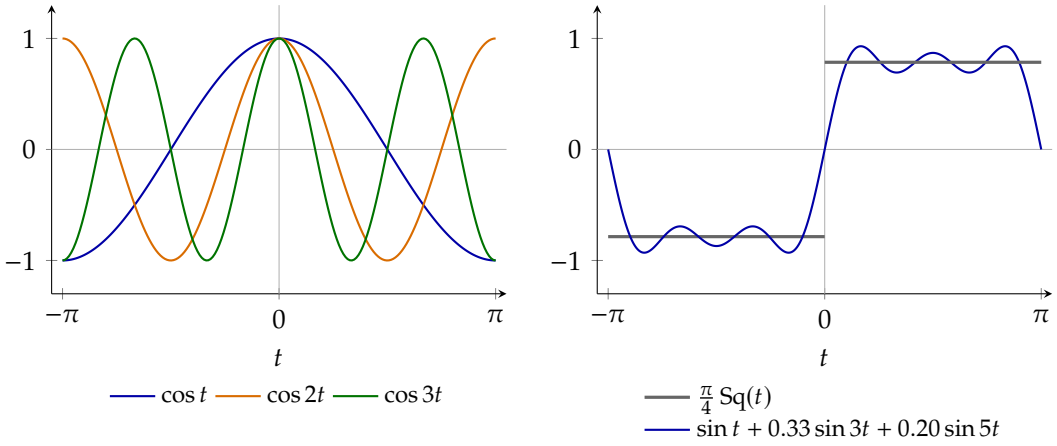


Figure 24: Left: the harmonics $\cos t$, $\cos 2t$, $\cos 3t$ on the window $[-\pi, \pi]$. Right: $\frac{\pi}{4}\,\mathrm{Sq}(t)$ and the sum $\sin t + 0.33\sin 3t + 0.20\sin 5t$.

20.3 Parity

A function is EVEN if $f(-t) = f(t)$ and ODD if $f(-t) = -f(t)$. Linear combinations of even functions are even, and of odd functions are odd; an even plus an odd function can be anything. Unlike products of odd numbers, the product of two odd functions is even: $f(-t)g(-t) = (-f(t))(-g(t)) = f(t)g(t)$. An even times an even function is even, and an even times an odd function is odd. The only function that is both even and odd is 0, since $f(t) = f(-t) = -f(t)$.

Each $\cos(nt)$ is even and each $\sin(nt)$ is odd.

Proposition 20.4 [PARITY OF FOURIER SERIES]. *The Fourier series of an even function is a cosine series ($b_n = 0$ for all n), and that of an odd function is a sine series ($a_n = 0$ for all n).*

Proof. Let f be even. Then f − (cosine series) is even, being a combination of even functions, but by (20.1) it equals the sine series, which is odd. So it is zero, and the sine series vanishes; by uniqueness, every $b_n = 0$. The odd case is the same. □

The same argument shows that an even polynomial is a sum of even powers of t, and an odd one a sum of odd powers.

20.4 Fourier coefficients by integration

The AVERAGE of a function of period 2π is

$$\operatorname{Ave}(f) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)\,dt.$$

It is linear, and $\operatorname{Ave}(\cos nt) = \operatorname{Ave}(\sin nt) = 0$ for $n > 0$. Applying it to (20.1) gives $\operatorname{Ave}(f) = a_0/2$, which explains the $a_0/2$:

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\,dt.$$

Lemma 20.5 [ORTHOGONALITY RELATIONS]. *For integers $m, n \geq 0$,*

$$\int_{-\pi}^{\pi} \cos(mt)\sin(nt)\,dt = 0, \qquad \int_{-\pi}^{\pi} \cos(mt)\cos(nt)\,dt = \begin{cases} 2\pi, & m = n = 0, \\ \pi, & m = n > 0, \\ 0, & m \neq n, \end{cases}$$

and, for $m, n \geq 1$, $\int_{-\pi}^{\pi} \sin(mt)\sin(nt)\,dt = \pi$ if $m = n$ and 0 if $m \neq n$.

Proof. The first integrand is odd and the interval is symmetric. For the others use the product formulas

$$\cos(mt)\cos(nt) = \tfrac{1}{2}\big[\cos((m-n)t) + \cos((m+n)t)\big],$$
$$\sin(mt)\sin(nt) = \tfrac{1}{2}\big[\cos((m-n)t) - \cos((m+n)t)\big],$$

and the fact that $\int_{-\pi}^{\pi} \cos(kt)\,dt$ is 2π for $k = 0$ and 0 for every other integer k. □

Theorem 20.6 [FOURIER COEFFICIENT FORMULAS]. *If f has period 2π and Fourier series (20.1), then for $n \geq 0$ and $n \geq 1$ respectively,*

$$\boxed{a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos(nt)\,dt}, \qquad \boxed{b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin(nt)\,dt}.$$

Proof. Let $n > 0$. Substitute (20.1) into $\int_{-\pi}^{\pi} f(t)\cos(nt)\,dt$ and integrate term by term. By Lemma 20.5, the constant term integrates to 0, each $a_m\cos(mt)$ contributes $a_m\pi$ if $m = n$ and 0 otherwise, and each $b_m\sin(mt)$ contributes 0. Only the term $m = n$ survives, so $\int_{-\pi}^{\pi} f(t)\cos(nt)\,dt = \pi a_n$. Since $\cos(0t) = 1$, the formula for $n = 0$ is the average computation above. Multiplying by $\sin(nt)$ instead gives b_n in the same way. □

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 20, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 21: Operations on Fourier series

21.1 The Fourier series theorem

A function f is periodic of period $2L$ if $f(t+2L) = f(t)$ for all t. Recall the basic statement for period 2π, with the coefficient formulas derived in Lecture 20.

Theorem 21.1 [Fourier series]. *A decent function f of period 2π (made precise in Definition 21.2) can be written in exactly one way as a Fourier series*

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \bigl(a_n \cos(nt) + b_n \sin(nt)\bigr). \tag{21.1}$$

The Fourier coefficients are

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos(nt)\,dt \quad (n \geq 0), \qquad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin(nt)\,dt \quad (n \geq 1).$$

Parity shortcuts. If f is odd, then $f(t)\cos(nt)$ is odd, so $a_n = 0$, while $f(t)\sin(nt)$ is even, so its integral over $[-\pi, \pi]$ is twice the integral over $[0, \pi]$. Similarly for even f:

$$f \text{ odd}: \quad b_n = \frac{2}{\pi}\int_0^{\pi} f(t)\sin(nt)\,dt, \qquad f \text{ even}: \quad a_n = \frac{2}{\pi}\int_0^{\pi} f(t)\cos(nt)\,dt.$$

21.2 The standard square wave

Recall the standard square wave $\mathrm{Sq}(t)$, of period 2π, with

$$\mathrm{Sq}(t) = \begin{cases} 1, & 0 < t < \pi, \\ -1, & -\pi < t < 0, \\ 0, & t = 0,\ t = \pi. \end{cases}$$

It is the basic building block for periodic on/off signals. Since it is odd, $a_n = 0$ for all n. On $(0, \pi)$ it equals 1, so

$$b_n = \frac{2}{\pi}\int_0^{\pi} \sin(nt)\,dt = \frac{2}{\pi}\Bigl[-\frac{\cos(nt)}{n}\Bigr]_0^{\pi} = \frac{2}{\pi n}\bigl(1 - \cos(n\pi)\bigr) = \begin{cases} \dfrac{4}{\pi n}, & n \text{ odd}, \\ 0, & n \text{ even}, \end{cases}$$

because $\cos(n\pi) = (-1)^n$. Therefore

$$\mathrm{Sq}(t) = \frac{4}{\pi}\Big(\sin t + \frac{1}{3}\sin(3t) + \frac{1}{5}\sin(5t) + \cdots\Big) = \frac{4}{\pi}\sum_{n\ \text{odd}} \frac{\sin(nt)}{n}. \tag{21.2}$$

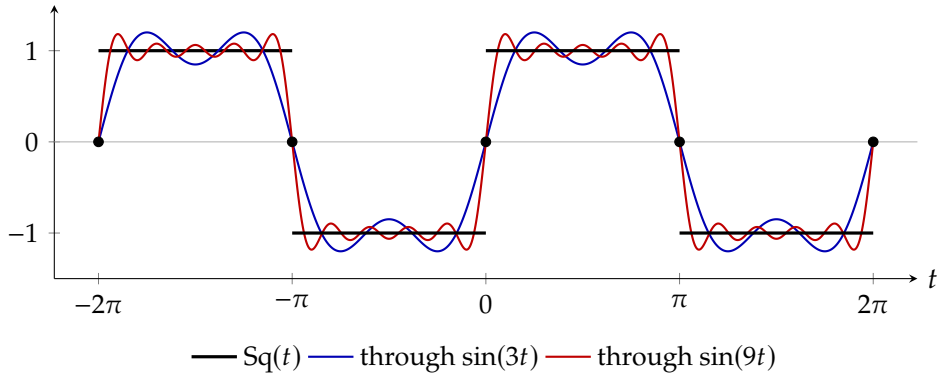


Figure 25: The square wave $\mathrm{Sq}(t)$ and the partial sums of (21.2) through $\sin(3t)$ and through $\sin(9t)$. At the jumps $t = 0, \pm\pi$ every partial sum equals 0, the average of the one-sided limits.

The whole function is recovered from a discrete sequence of numbers, the strengths of the harmonics above the fundamental tone. The sequence of Fourier coefficients is a *transform* of the periodic function; the Laplace transform will be another.

21.3 Piecewise continuity and convergence

Definition 21.2 [Piecewise continuous, decent]. A function f is piecewise continuous if it is made of continuous segments and at each point a the one-sided limits

$$f(a^-) = \lim_{t\to a^-} f(t), \qquad f(a^+) = \lim_{t\to a^+} f(t)$$

exist (at points of continuity they exist and are equal). It is decent if moreover $f(a) = \frac{1}{2}\big(f(a^+)+f(a^-)\big)$ at each discontinuity a.

For example, $1/t$ is not piecewise continuous, while $\mathrm{Sq}(t)$ is decent, as is every continuous function. The precise addendum to Theorem 21.1 is that at a jump the series cannot choose between the two sides: it converges to the average $\frac{1}{2}\big(f(a^+) + f(a^-)\big)$.

Example: *A series for $\pi/4$.* At $t = \pi/2$ we have $\sin(n\pi/2) = 1, -1, 1, -1, \ldots$ for $n = 1, 3, 5, 7, \ldots$, and $\mathrm{Sq}(\pi/2) = 1$. Evaluating (21.2) gives

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}.$$

21.4 Computing Fourier series without integrals

By uniqueness, any way of writing f in the form (21.1) gives its Fourier series. Often this avoids the integrals entirely.

Trigonometric identities. If (a, b) has polar coordinates (A, ϕ), that is $a = A\cos\phi$ and $b = A\sin\phi$, then $a\cos t + b\sin t = A\cos(t-\phi)$. With $A = 1$ and $\phi = \pi/4$,

$$\cos\Big(t - \frac{\pi}{4}\Big) = \frac{1}{\sqrt{2}}\cos t + \frac{1}{\sqrt{2}}\sin t,$$

so $a_1 = b_1 = 1/\sqrt{2}$ and every other coefficient is zero.

Linear combinations. Fourier series add and scale termwise:

$$1 + 2\,\mathrm{Sq}(t) = 1 + \frac{8}{\pi}\Big(\sin t + \frac{1}{3}\sin(3t) + \frac{1}{5}\sin(5t) + \cdots\Big).$$

Shifts. Substitute $t + \pi/2$ into (21.2). For odd n, $\sin\big(n(t+\pi/2)\big) = \sin(nt + n\pi/2)$ equals $\cos(nt)$ when $n \equiv 1 \pmod 4$ and $-\cos(nt)$ when $n \equiv 3 \pmod 4$, because $\sin(\theta + \pi/2) = \cos\theta$ and $\sin(\theta - \pi/2) = -\cos\theta$. Hence

$$\mathrm{Sq}\Big(t + \frac{\pi}{2}\Big) = \frac{4}{\pi}\Big(\cos t - \frac{1}{3}\cos(3t) + \frac{1}{5}\cos(5t) - \cdots\Big).$$

This shifted square wave is even, consistent with a pure cosine series.

Other periods. If $g(x)$ has period $2L$, the building blocks are $\cos(n\pi x/L)$ and $\sin(n\pi x/L)$, which have period $2L$, and

$$g(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\Big(a_n\cos\frac{n\pi x}{L} + b_n\sin\frac{n\pi x}{L}\Big),$$

with

$$a_n = \frac{1}{L}\int_{-L}^{L} g(x)\cos\frac{n\pi x}{L}\,dx, \qquad b_n = \frac{1}{L}\int_{-L}^{L} g(x)\sin\frac{n\pi x}{L}\,dx.$$

The substitution $t = \pi x/L$ reduces everything to period 2π. For example, $\mathrm{Sq}(\pi x/2)$ has period 4 ($L = 2$), and substituting $t = \pi x/2$ into (21.2) gives

$$\mathrm{Sq}\Big(\frac{\pi}{2}x\Big) = \frac{4}{\pi}\Big(\sin\frac{\pi x}{2} + \frac{1}{3}\sin\frac{3\pi x}{2} + \frac{1}{5}\sin\frac{5\pi x}{2} + \cdots\Big).$$

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 21, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 22: Periodic solutions; resonance

22.1 Differentiating and integrating Fourier series

Fourier series can be differentiated and integrated term by term. One caution concerns integration: an antiderivative of a periodic function need not be periodic (integrate the constant 1). Term by term, $\int \sin(nt)\,dt$ is always periodic and $\int \cos(nt)\,dt$ is periodic unless $n = 0$. So the integral $\int_0^t f(u)\,du$ of a periodic f is periodic exactly when the average value $a_0/2$ of f is zero.

Example: *The triangle wave.* Let f have period 2π with $f(t) = |t|$ for $-\pi \le t \le \pi$. Its average value is $a_0/2 = \pi/2$. One could compute the coefficients by integration by parts. Instead, note that $f'(t) = \mathrm{Sq}(t)$ wherever f' exists; equivalently $f(t) = \int_0^t \mathrm{Sq}(u)\,du$. The square wave has average zero, and its series from Lecture 21 integrates term by term:

$$\begin{aligned} f(t) &= \int_0^t \frac{4}{\pi} \sum_{n \text{ odd}} \frac{\sin(nu)}{n}\,du = \frac{4}{\pi} \sum_{n \text{ odd}} \Big[-\frac{\cos(nu)}{n^2}\Big]_0^t \\ &= \frac{4}{\pi} \sum_{n \text{ odd}} \frac{1}{n^2} - \frac{4}{\pi} \sum_{n \text{ odd}} \frac{\cos(nt)}{n^2}. \end{aligned}$$

This is the Fourier series of f. Its constant term must be the average value $\pi/2$, so

$$\sum_{n \text{ odd}} \frac{1}{n^2} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}, \tag{22.1}$$

and

$$\boxed{|t| = \frac{\pi}{2} - \frac{4}{\pi}\Big(\cos t + \frac{\cos(3t)}{9} + \frac{\cos(5t)}{25} + \cdots\Big) \quad \text{for } |t| \le \pi.}$$

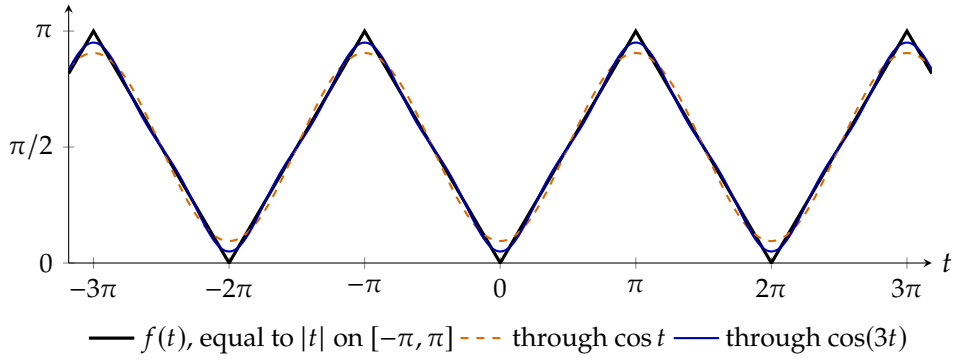


Figure 26: The 2π-periodic extension of $|t|$ and two partial sums of its Fourier series: $\frac{\pi}{2} - \frac{4}{\pi}\cos t$ and $\frac{\pi}{2} - \frac{4}{\pi}\big(\cos t + \frac{1}{9}\cos(3t)\big)$.

Proposition 22.1 [Sum of reciprocal squares]**.**

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Proof. Every positive integer is uniquely $2^k m$ with $k \geq 0$ and m odd. The terms with a fixed k sum to $\sum_{m \text{ odd}} 1/(4^k m^2) = (1/4)^k \, \pi^2/8$ by (22.1). Summing the geometric series over k,

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \left(1 + \frac{1}{4} + \frac{1}{4^2} + \cdots\right) \frac{\pi^2}{8} = \frac{4}{3} \cdot \frac{\pi^2}{8} = \frac{\pi^2}{6}.$$

Rearranging is legitimate because all terms are positive. □

22.2 Harmonic oscillator with periodic input

Drive an undamped harmonic oscillator with a periodic signal $f(t)$ of period 2π, such as a square wave:

$$x'' + \omega_n^2 x = f(t).$$

Think of a radio tuner: f is the incoming wave and x the receiver's output. For a single sinusoid with $\omega \neq \omega_n$,

$$x'' + \omega_n^2 x = \cos(\omega t) \text{ has } x_p = \frac{\cos(\omega t)}{\omega_n^2 - \omega^2}, \qquad x'' + \omega_n^2 x = \sin(\omega t) \text{ has } x_p = \frac{\sin(\omega t)}{\omega_n^2 - \omega^2}.$$

When $\omega = \omega_n$ the denominator vanishes: this is resonance, and there is no periodic solution.

By superposition, each harmonic of the input is handled separately. For $f = \mathrm{Sq}$,

$$x_p = \frac{4}{\pi} \sum_{n \text{ odd}} \frac{\sin(nt)}{n(\omega_n^2 - n^2)} = \frac{4}{\pi}\left(\frac{\sin t}{\omega_n^2 - 1} + \frac{\sin(3t)}{3(\omega_n^2 - 9)} + \frac{\sin(5t)}{5(\omega_n^2 - 25)} + \cdots\right),$$

a periodic solution whenever ω_n is not an odd integer.

- *Resonance* occurs for $\omega_n = 1, 3, 5, \ldots$, but *not* for $\omega_n = 2, 4, 6, \ldots$: the square wave has no even harmonics, so nothing in the input matches those natural frequencies.
- When ω_n is slightly below an odd integer k, the term $\sin(kt)/\big(k(\omega_n^2 - k^2)\big)$ is a large *negative* multiple of $\sin(kt)$ and dominates the response. As ω_n passes k, the dominant term flips sign and becomes a large positive multiple of $\sin(kt)$.

A spring driven through the spring obeys $x'' + \omega_n^2 x = \omega_n^2 f(t)$, which multiplies every term above by ω_n^2; for $f(t) = \sin t$ the response $\omega_n^2 \sin t/(\omega_n^2 - 1)$ has amplitude $\omega_n^2/|\omega_n^2 - 1|$.

22.3 What you hear

Let f have period $2L$ and write $\omega = \pi/L$ for its fundamental circular frequency, so that

$$f(t) = \frac{a_0}{2} + \sum_{n\geq 1} \big(a_n \cos(n\omega t) + b_n \sin(n\omega t)\big).$$

The POLAR FORM comes from writing each pair (a_n, b_n) in polar coordinates, $a_n = A_n \cos\phi_n$ and $b_n = A_n \sin\phi_n$:

$$f(t) = A_0 + \sum_{n\geq 1} A_n \cos(n\omega t - \phi_n), \qquad A_0 = \frac{a_0}{2}, \quad A_n = \sqrt{a_n^2 + b_n^2}.$$

If f is the pressure at the eardrum, A_0 is atmospheric pressure and the rest is the sound. Changing the phases ϕ_n alters the waveform dramatically, but the ear hears only the amplitudes A_n of the harmonics, not their relative phases.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 22, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 23: Step functions and delta functions

Two additions to the modeling toolkit describe events that happen on a time scale much shorter than the one of interest: the step function (Heaviside) and the delta function (Dirac).

23.1 The unit step function

A light turning on is dark, then lit. On a fine time scale the switch is not instantaneous: the brightness may rise smoothly, overshoot, or come up in stages. On the longer time scale of interest these details do not matter, and the model ignores them.

Definition 23.1 [Unit step function]. The Heaviside unit step function is

$$u(t) = \begin{cases} 0, & t < 0, \\ 1, & t > 0, \end{cases}$$

with $u(0)$ left undefined.

The shift $u(t-a)$ turns on at $t = a$. For $a < b$, the difference $u(t-a) - u(t-b)$ turns on at a and off at b; it is a window. Multiplying by a step turns another function on: $u(t-a)f(t)$ is 0 for $t < a$ and equals $f(t)$ for $t > a$. Likewise

$$\big(u(t-a) - u(t-b)\big)f(t)$$

agrees with $f(t)$ for $a < t < b$ and is zero outside this window.

23.2 Rates and the delta function

A bank account with balance $x(t)$ (in K\$), interest rate I (per year), and savings rate $q(t)$ (K\$/yr) satisfies $x' + Ix = q(t)$. The rate is clarified by the cumulative total deposit $Q(t) = \int_0^t q(\tau)\,d\tau$, so that $Q'(t) = q(t)$.

Save at the rate 1 starting at $t = 0$, and at $t = 1$ deposit a windfall of 2. Then $Q(t) = t\,u(t) + 2u(t-1)$. To give the corresponding rate we need a derivative of $u(t)$ whose integral recovers $u(t)$; no ordinary function has this property. Approximate u by functions u_ε that rise from 0 to 1 in a short interval around 0. Their derivatives are rates with two properties that persist as $\varepsilon \to 0$, regardless of the details:

- the integral of u_ε' is always 1;
- u_ε' is concentrated very near 0.

These functions approximate an object $\delta(t)$, the DELTA FUNCTION, with

$$\delta(t) = u'(t).$$

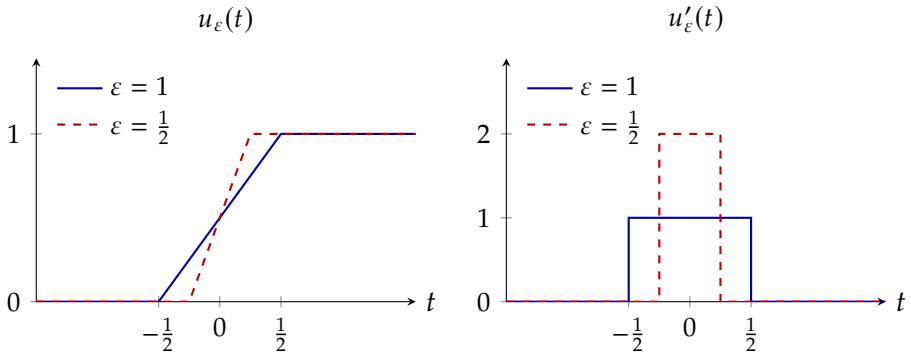


Figure 27: Piecewise linear approximations u_ε of the step, rising from 0 to 1 on $[-\varepsilon/2, \varepsilon/2]$, for $\varepsilon = 1$ and $\varepsilon = \frac{1}{2}$ (left), and their derivatives u'_ε, boxes of height $1/\varepsilon$ and area 1 (right).

With this, the savings rate is $q(t) = u(t) + 2\delta(t-1)$. If at $t = 2$ a camera costing 3 is paid from the account, then

$$q(t) = u(t) + 2\delta(t-1) - 3\delta(t-2), \qquad Q(t) = t\,u(t) + 2u(t-1) - 3u(t-2).$$

A multiple $c\,\delta(t-a)$ is drawn as a HARPOON at $t = a$ labeled c, the area under it. A negative multiple is drawn either pointing up with label -3 or pointing down with label 3.

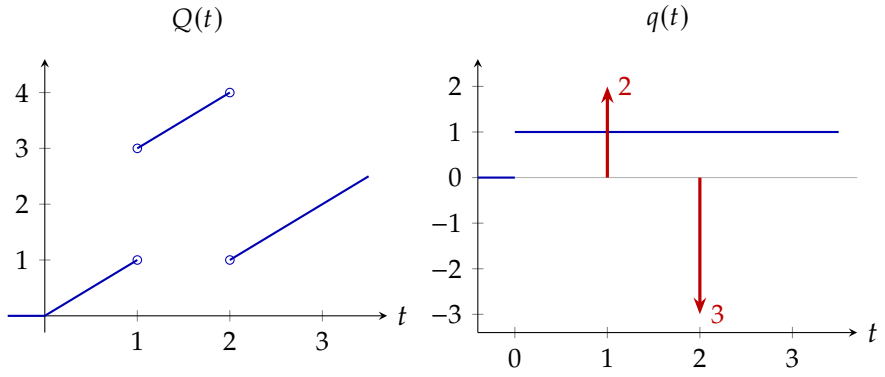


Figure 28: The cumulative deposit $Q(t) = t\,u(t) + 2u(t-1) - 3u(t-2)$ (left) and its rate $q(t) = u(t) + 2\delta(t-1) - 3\delta(t-2)$ (right). The harpoons have areas 2 and -3; the downward harpoon labeled 3 represents $-3\delta(t-2)$.

23.3 Regular, singular, and generalized functions

Definition 23.2 [GENERALIZED FUNCTION]. A function is PIECEWISE SMOOTH, or REGULAR, if it is made of segments on each of which it has derivatives of all orders, and at each endpoint all derivatives

have left and right limits. A SINGULARITY FUNCTION is a linear combination of shifted delta functions $\delta(t-a)$. A GENERALIZED FUNCTION is a sum

$$f(t) = f_r(t) + f_s(t)$$

of a regular part f_r and a singular part f_s.

For the savings rate above, $q_r(t) = u(t)$ and $q_s(t) = 2\delta(t-1) - 3\delta(t-2)$.

Differentiating a regular function whose graph has breaks produces a generalized function. At each a with $Q(a^+) \neq Q(a^-)$, the derivative acquires the delta contribution

$$\big(Q(a^+) - Q(a^-)\big)\,\delta(t-a).$$

The GENERALIZED DERIVATIVE is the ordinary derivative (away from the breaks) plus these singular terms. Keeping them lets integration reconstruct the function up to a constant. *From now on, the derivative of a discontinuous function always means its generalized derivative.*

23.4 Impulses

Firing a gun exerts a very large force on the bullet over a very short time. Integrating $F = mx''$ over that time shows that the force changes the momentum mx' by $\int F\,dt$, the IMPULSE, essentially instantaneously.

Example: *A bullet fired straight up.* The elevation is zero for $t < 0$, then follows an inverted parabola, and stays zero again after the bullet lands. Its velocity $v(t)$ is zero for $t < 0$, jumps to v_0 at $t = 0$, decreases with constant slope $-g$, and returns abruptly to zero when the bullet lands at $T = 2v_0/g$, where $v(T^-) = -v_0$. Both jumps are $+v_0$, so the generalized derivative is

$$v'(t) = v_0\,\delta(t) - g\big(u(t) - u(t-T)\big) + v_0\,\delta(t-T).$$

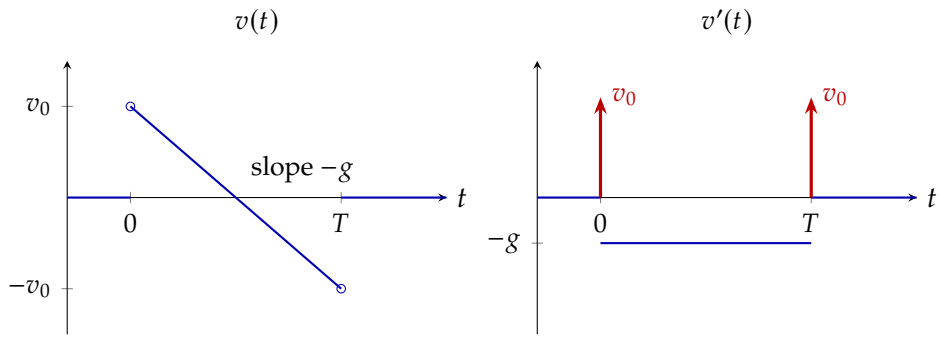


Figure 29: Velocity $v(t)$ of the bullet (left) and its generalized derivative (right): two upward harpoons of area v_0 at $t = 0$ and $t = T$, and the constant acceleration $-g$ in between.

23.5 What a delta function is

One answer is that $\delta(t)$ is a symbol, standing for an approximation to reality and obeying certain rules. Another comes from measurement. An instrument never reads a value $f(t)$ at an instant; it integrates over a short time against a smooth SENSITIVITY PROFILE $m(t)$ that rises to a peak and falls again, and reports

$$M(f;m) = \int f(t)\, m(t)\, dt.$$

The most one can know about f is the collection of all these measurements. So f determines a new kind of function, sending each m to the number $M(f;m)$; this is a DISTRIBUTION. Changing f at a single point does not change any $M(f;m)$. There are other ways to assign a number to each measuring device, and the delta function is the one that sends m to $m(0)$:

$$\int \delta(t)\, m(t)\, dt = m(0).$$

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 23, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 24: Step response, impulse response

24.1 Generalized derivatives and integrals

The step $u(t)$ and the delta $\delta(t)$ model events on a time scale much shorter than the one of interest: $u(t)$ is 0 for $t < 0$ and 1 for $t > 0$, $\delta(t)$ is zero away from $t = 0$ with total area 1, and $u'(t) = \delta(t)$. Recall from Lecture 23 that a generalized function is a sum $g = g_r + g_s$ of a piecewise smooth regular part g_r and a singular part g_s, a linear combination of shifted delta functions.

Every regular function f has a generalized derivative $f' = f_r' + f_s'$. The regular part f_r' is the ordinary derivative, undefined at the break points. The singular part has one delta for each break in the graph:

$$f_s'(t) = \sum_a \big(f(a^+) - f(a^-)\big)\,\delta(t-a).$$

No separate notation is used; we simply write $f'(t)$ or $\dot{f}(t)$. For example,

$$f(t) = t + 2u(t), \qquad f'(t) = 1 + 2\delta(t), \qquad f_r'(t) = 1, \qquad f_s'(t) = 2\delta(t).$$

Conversely, a generalized function g can be integrated: in $f(t) = \int_a^t g(\tau)\,d\tau$, integrate the regular part as usual and add the coefficient of each delta located between a and t. With these conventions the fundamental theorem of calculus still holds,

$$f(b) - f(a) = \int_a^b f'(t)\,dt,$$

for a, b that are not break points of f.

24.2 Rest initial conditions

For equations with discontinuous input signals, the standing assumptions, called rest initial conditions, are:

1. the input signal is zero for $t < 0$;
2. the system response is zero for $t < 0$;
3. the solution has as many derivatives as possible.

Definition 24.1 [Unit step and unit impulse responses]. For a linear time-invariant system, the unit step response $v(t)$ is the response to the input $u(t)$, and the unit impulse response $w(t)$ is the response to $\delta(t)$, both with rest initial conditions.

24.3 First order

Example: *Step input:* $x' + kx = r\,u(t)$. Uranium is added to a reactor at the constant rate r starting at $t = 0$, and decays at rate k. The input has no delta, so look for a continuous solution: $x(0) = 0$, and for $t > 0$ solve $x' + kx = r$. Its general solution is $x = r/k + ce^{-kt}$, and $x(0) = 0$ forces $c = -r/k$:

$$x(t) = \frac{r}{k}\bigl(1 - e^{-kt}\bigr), \quad t > 0.$$

With $r = 1$ this is the unit step response,

$$\boxed{v(t) = u(t)\,\frac{1}{k}\bigl(1 - e^{-kt}\bigr).}$$

Its slope jumps at 0: $v(0) = 0$, so $v'(0^+) = u(0^+) - kv(0^+) = 1$, while $v'(0^-) = 0$.

Example: *Impulse input:* $x' + kx = \delta(t)$. Now one kilogram of uranium is inserted at $t = 0$ and then left alone. The solution cannot be continuous: $x(0^-) = 0$ but $x(0^+) = 1$. For $t > 0$ the equation is the homogeneous $x' + kx = 0$, with solution e^{-kt}. The unit impulse response is

$$\boxed{w(t) = u(t)\,e^{-kt}.}$$

In a sense it is the simplest nontrivial solution: give the system a unit kick at $t = 0$, stand back, and watch. Indeed $w' = \delta(t) - ku(t)e^{-kt}$, so $w' + kw = \delta(t)$. Since $w(0^+) = 1$ and $\delta(0^+) = 0$, the equation gives $w'(0^+) = -k$.

24.4 Second order

Let $\omega_n = \sqrt{k/m}$.

Example: *Step input:* $mx'' + kx = u(t)$. At $t = 0$ a steady force starts to act on the mass. The mass changes neither its position nor its velocity abruptly, so expect x and x' to be continuous, with only the acceleration jumping. A constant solution is $x = 1/k$, so for $t > 0$

$$x(t) = \frac{1}{k} + a\cos(\omega_n t) + b\sin(\omega_n t).$$

The conditions $0 = x(0) = 1/k + a$ and $0 = x'(0) = b\,\omega_n$ give $a = -1/k$ and $b = 0$:

$$\boxed{v(t) = u(t)\,\frac{1}{k}\bigl(1 - \cos(\omega_n t)\bigr).}$$

Example: *Impulse input:* $mx'' + kx = \delta(t)$. A sudden, very brief, very intense force, like a hammer blow, changes the momentum instantaneously without changing the position. Since $x(0) = 0$ and x is continuous, x is small for small t, so near $t = 0$ the equation is approximately $mx'' = \delta(t)$. Integrating across $t = 0$ shows that the velocity jumps by $1/m$ (the momentum by 1, a UNIT IMPULSE).

So for $t > 0$ the problem is equivalent to

$$mx'' + kx = 0, \qquad x(0^+) = 0, \qquad x'(0^+) = \frac{1}{m}.$$

The condition $x(0^+) = 0$ leaves $x = c\sin(\omega_n t)$, and $x'(0^+) = c\,\omega_n = 1/m$ gives $c = 1/(m\omega_n)$:

$$\boxed{w(t) = u(t)\,\frac{1}{m\omega_n}\sin(\omega_n t),} \qquad w'(0^+) = \frac{1}{m}.$$

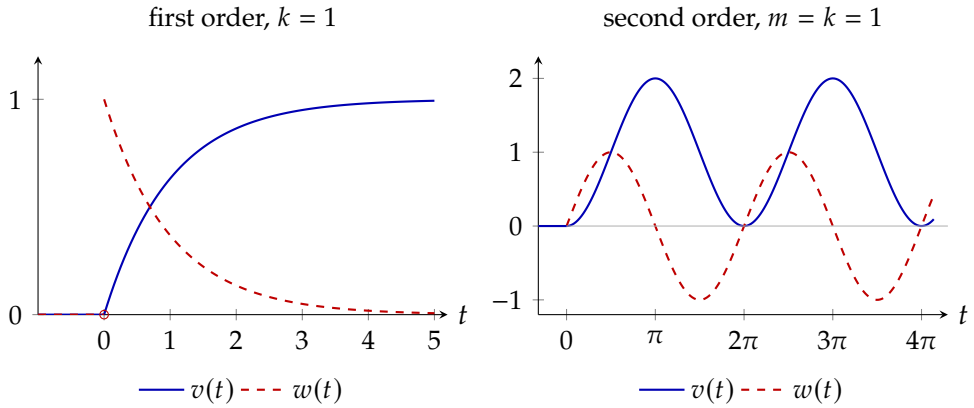


Figure 30: Unit step and impulse responses. Left: first order $x' + kx$ with $k = 1$, where $v = u(t)(1 - e^{-t})$ and $w = u(t)e^{-t}$. Right: second order $mx'' + kx$ with $m = k = 1$ ($\omega_n = 1$), where $v = u(t)(1 - \cos t)$ and $w = u(t)\sin t$.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 24, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 25: Convolution

Throughout, $p(D) = a_n D^n + \cdots + a_1 D + a_0$ with constant coefficients and $a_n \neq 0$, and all equations carry rest initial conditions.

25.1 The unit impulse response of $p(D)$

The unit impulse response $w(t)$ of $p(D)$ solves $p(D)w = \delta(t)$ with rest initial conditions.

Proposition 25.1 [Equivalent initial conditions]. *For $t > 0$, the equation $p(D)x = \delta(t)$ with rest initial conditions is equivalent to the homogeneous equation $p(D)x = 0$ with*

$$x(0) = x'(0) = \cdots = x^{(n-2)}(0) = 0, \qquad x^{(n-1)}(0^+) = \frac{1}{a_n}.$$

Proof. Keep x as smooth as possible. The delta on the right must come from the highest derivative alone: if $x^{(n-1)}$ jumps by c at 0, then $x^{(n)}$ contains $c\,\delta(t)$ while $x, \ldots, x^{(n-1)}$ contain no deltas. Matching coefficients of $\delta(t)$ gives $a_n c = 1$. The derivatives $x, \ldots, x^{(n-2)}$ do not jump, so they stay 0 at $t = 0$, and $x^{(n-1)}(0^+) = x^{(n-1)}(0^-) + c = 1/a_n$. Away from $t = 0$ the input is zero. □

For example, $mx'' + bx' + kx = \delta(t)$ with rest initial conditions is equivalent, for $t > 0$, to $mx'' + bx' + kx = 0$ with $x(0) = 0$, $x'(0) = 1/m$.

25.2 Two consequences of time invariance

Differentiation. If $p(D)x = q(t)$, then $p(D)x' = q'(t)$. Indeed, because the coefficients are constant, $Dp(D) = p(D)D$, so $p(D)x' = p(D)Dx = Dp(D)x = Dq = q'$. Since $u'(t) = \delta(t)$, the derivative of the unit step response is the unit impulse response:

$$\boxed{v'(t) = w(t).}$$

For $D+k$, the step response $v = u(t)\frac{1}{k}(1-e^{-kt})$ is continuous, so its generalized derivative has no delta term, and $v' = u(t)e^{-kt} = w$. For $mD^2 + k$, likewise $\frac{d}{dt}\,\frac{1}{k}(1 - \cos\omega_n t) = \frac{\omega_n}{k}\sin(\omega_n t) = \frac{1}{m\omega_n}\sin(\omega_n t)$ for $t > 0$, using $k = m\omega_n^2$.

Shifts. It does not matter when the clock starts. Since the unit impulse response of $D + k$ is $u(t)e^{-kt}$, the solution of $x' + kx = \delta(t - a)$ with $x(t) = 0$ for $t < a$ is $x(t) = u(t - a)\, e^{-k(t-a)}$.

25.3 Superposition of impulse responses

The unit impulse response contains complete information about $p(D)$, and hence about the system it models: by linearity and time invariance, the response to any input is a kind of product of w with the input.

Definition 25.2 [Convolution]. Let $f(t) = g(t) = 0$ for $t < 0$. Their convolution is

$$(f * g)(t) = \int_0^t f(t-\tau)\, g(\tau)\, d\tau.$$

Example: *Phosphates in a lake.* Phosphates run off farm fields into a lake at a rate that varies with the seasons, say $q(t) = u(t)(1 + \cos bt)$. In the lake the load decays, carried out by the stream at a rate proportional to the amount present:

$$x' + ax = q(t), \qquad x(0) = 0.$$

The unit impulse response $w(t) = u(t)e^{-at}$ is the amount left after time t of one pound deposited at time 0. By time invariance, if c pounds go in at time τ, then $w(t-\tau)\,c$ pounds remain at time $t > \tau$.

Fix t and divide $[0, t]$ into intervals of width h. Between τ and $\tau + h$ about $q(\tau)h$ pounds go in, and $w(t-\tau)q(\tau)h$ of them remain at time t. Adding the contributions from $\tau = 0, h, 2h, \ldots, t-h$,

$$x(t) \approx w(t)q(0)h + w(t-h)q(h)h + w(t-2h)q(2h)h + \cdots + w(h)q(t-h)h.$$

This is a Riemann sum, and letting $h \to 0$ gives

$$x(t) = \int_0^t w(t-\tau)\, q(\tau)\, d\tau = \int_0^t e^{-a(t-\tau)}\big(1 + \cos b\tau\big)\, d\tau.$$

Evaluating with $\int_0^t e^{a\tau}\cos b\tau\, d\tau = \left[e^{a\tau}(a\cos b\tau + b\sin b\tau)\right]_0^t/(a^2+b^2)$ gives, for $t > 0$,

$$x(t) = \frac{1 - e^{-at}}{a} + \frac{a\cos bt + b\sin bt - ae^{-at}}{a^2 + b^2}.$$

The unit impulse response acts as a weight: it records the long-term impact of an action, and is also called the weight function.

Theorem 25.3 [Superposition integral]. *Let w be the unit impulse response of $p(D)$, and let $q(t) = 0$ for $t < 0$. The solution of $p(D)x = q(t)$ with rest initial conditions is*

$$\boxed{x(t) = (w * q)(t) = \int_0^t w(t-\tau)\, q(\tau)\, d\tau.}$$

The argument above works for any order: by time invariance the input $q(\tau)h$ near τ produces

$w(t-\tau)q(\tau)h$ at time t, and by linearity these add. In the second order case, think of the input as many little blows, each producing a decaying ringing into the future; the convolution integral, also called the SUPERPOSITION INTEGRAL, adds them up.

25.4 The convolution product

The system is thus represented by its unit impulse response: the system w sends an input q to $w * q$. The input δ gives $w * \delta$, which by definition is also w, so $w * \delta = w$.

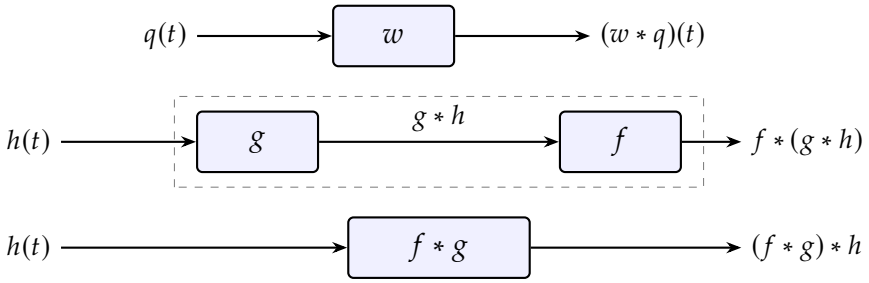


Figure 31: Top: the system with unit impulse response w sends q to $w * q$. Middle: the cascade of the systems g and f, fed h. Bottom: the same cascade viewed as one system, whose unit impulse response is $f * g$.

The convolution $f * g$ is not the pointwise product $f(t)g(t)$, but it deserves to be called a product, with δ as its unit:

$$f * g = g * f, \qquad (f * g) * h = f * (g * h),$$
$$f * (c_1 g_1 + c_2 g_2) = c_1 f * g_1 + c_2 f * g_2, \qquad f * \delta = f.$$

Distributivity is the linearity of the integral, and commutativity follows from the substitution $\sigma = t - \tau$, which turns $\int_0^t f(t-\tau)g(\tau)\,d\tau$ into $\int_0^t g(t-\sigma)f(\sigma)\,d\sigma$.

Associativity via systems. Feed the output of the system g into the system f. This composite system is again linear and time invariant. Its input δ produces g from the first stage and then $f * g$, so its unit impulse response is $f * g$, and by Theorem 25.3 the input h produces $(f * g) * h$. Following h through the two stages instead gives $g * h$ and then $f * (g * h)$. The outputs agree, so $(f * g) * h = f * (g * h)$. □

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 25, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 26: Laplace transform: basic properties

26.1 Definition

We continue to consider functions $f(t)$ that are zero for $t < 0$ (a factor $u(t)$ is often left implicit). The Laplace transform turns a function of time $f(t)$ into a function $F(s)$ of a complex variable s. It makes the long-term behavior of f explicit, it answers how the unit impulse response $w(t)$ determines the characteristic polynomial $p(s)$, and it converts differential equations into algebraic equations; these virtues appear over the next lectures.

Definition 26.1 [Laplace transform]. The Laplace transform of $f(t)$ is the improper integral

$$F(s) = \int_0^\infty e^{-st} f(t)\, dt,$$

for those s where it converges. We write $f(t) \rightsquigarrow F(s)$ or $\mathcal{L}[f(t)] = F(s)$.

For each s, $F(s)$ is a weighted integral of all of f. At $s = 0$ it is just $\int_0^\infty f(t)\, dt$; for real $s > 0$ the values of f at large t receive less weight.

Example: *The constant* 1. For real $s \neq 0$,

$$\int_0^\infty e^{-st}\, dt = \lim_{T\to\infty} \left[\frac{e^{-st}}{-s}\right]_0^T = -\frac{1}{s}\Big(\lim_{T\to\infty} e^{-sT} - 1\Big).$$

If $s > 0$, then $e^{-sT} \to 0$ and the integral is $1/s$; if $s < 0$, then $e^{-sT} \to \infty$ and the integral diverges. For complex $s = x + iy$, $|e^{-st}| = |e^{-xt}e^{-iyt}| = e^{-xt}$, so the same argument gives

$$\boxed{\mathcal{L}[1] = \frac{1}{s}, \qquad \operatorname{Re} s > 0.}$$

26.2 Exponential order and the region of convergence

The integral converges for a given s when $|f(t)e^{-st}| = |f(t)|e^{-(\operatorname{Re} s)t}$ is integrable, which holds if $|f(t)|$ is eventually dominated by $e^{(\operatorname{Re} s - \varepsilon)t}$ for some $\varepsilon > 0$.

Definition 26.2 [Exponential order]. A function f is of exponential order if there are constants M and a such that $|f(t)| \leq Me^{at}$ for all large t.

In that case $F(s)$ converges for $\operatorname{Re} s > a$. Not every function qualifies: e^{t^2} grows too fast and has no Laplace transform. So the definition is refined to hold for $\operatorname{Re} s$ large, in a half-plane called the region

of convergence. It carries important information and should be declared with each transform: 1 has Laplace transform $1/s$ with region of convergence $\operatorname{Re} s > 0$.

The expression obtained by integration usually makes sense on all of $\mathbb{C}$ except a few points, like $s = 0$ for $1/s$. This formula is used to define $F(s)$ also where $\operatorname{Re} s$ is too small for the integral to converge, a process called analytic continuation.

26.3 Powers

For $n \geq 1$ and $\operatorname{Re} s > 0$, integrate by parts with $u = t^n$ and $dv = e^{-st}\,dt$:

$$\mathcal{L}[t^n] = \int_0^\infty t^n e^{-st}\,dt = \Big[-\frac{t^n e^{-st}}{s}\Big]_0^\infty + \frac{n}{s}\int_0^\infty t^{n-1}e^{-st}\,dt = \frac{n}{s}\,\mathcal{L}[t^{n-1}],$$

since the boundary term vanishes at both ends. Starting from $\mathcal{L}[1] = 1/s$, this gives $\mathcal{L}[t] = 1/s^2$, $\mathcal{L}[t^2] = 2/s^3$, $\mathcal{L}[t^3] = 3!/s^4$, and by induction

$$\boxed{\mathcal{L}[t^n] = \frac{n!}{s^{n+1}}, \qquad \operatorname{Re} s > 0.}$$

26.4 Linearity and the s-shift rule

Rules like the following, rather than the defining integral, are the normal way to compute Laplace transforms.

Proposition 26.3 [Linearity]. *$\mathcal{L}[af(t) + bg(t)] = aF(s) + bG(s)$, wherever both transforms converge.*

This is the linearity of the integral. For example,

$$\mathcal{L}\big[(1+t)^2\big] = \mathcal{L}[1 + 2t + t^2] = \frac{1}{s} + \frac{2}{s^2} + \frac{2}{s^3},$$

which is not $\big(\mathcal{L}[1+t]\big)^2 = 1/s^2 + 2/s^3 + 1/s^4$.

Proposition 26.4 [s-shift]. *For any complex number z,*

$$\mathcal{L}\big[e^{zt}f(t)\big] = F(s - z).$$

If F converges for $\operatorname{Re} s > a$, this converges for $\operatorname{Re} s > a + \operatorname{Re} z$.

Proof. Combine the exponentials: $\int_0^\infty e^{zt}f(t)e^{-st}\,dt = \int_0^\infty f(t)e^{-(s-z)t}\,dt = F(s-z)$, and the last integral converges when $\operatorname{Re}(s-z) > a$. □

With $f(t) = 1$,

$$\boxed{\mathcal{L}\left[e^{zt}\right] = \frac{1}{s-z}, \qquad \operatorname{Re} s > \operatorname{Re} z.}$$

26.5 Sines and cosines

For real ω, Euler's formula gives $\cos(\omega t) = \frac{1}{2}\left(e^{i\omega t} + e^{-i\omega t}\right)$ and $\sin(\omega t) = \frac{1}{2i}\left(e^{i\omega t} - e^{-i\omega t}\right)$. By linearity and the transform of e^{zt} with $z = \pm i\omega$,

$$\mathcal{L}[\cos(\omega t)] = \frac{1}{2}\left(\frac{1}{s-i\omega} + \frac{1}{s+i\omega}\right) = \frac{s}{s^2+\omega^2}, \qquad \mathcal{L}[\sin(\omega t)] = \frac{1}{2i}\left(\frac{1}{s-i\omega} - \frac{1}{s+i\omega}\right) = \frac{\omega}{s^2+\omega^2}.$$

Both converge for $\operatorname{Re} s > 0$, since $\pm i\omega$ lie on the imaginary axis.

26.6 The t-domain and the s-domain

The Laplace transform and its inverse pass between two worlds, matching each kind of object and operation on one side with one on the other.

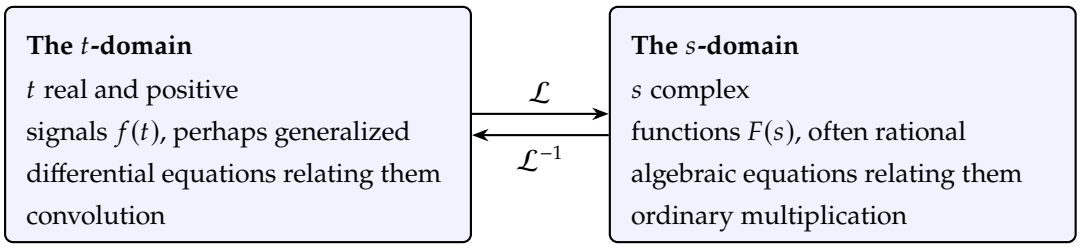


Figure 32: The t-domain and the s-domain, related by $\mathcal{L}$ and $\mathcal{L}^{-1}$. Corresponding rows match.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 26, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 27: Application to ODEs

The Laplace transform turns an initial value problem into an algebraic equation. This lecture adds the transform of the delta function and of a derivative, explains what inverting the transform means, and solves first order equations, using partial fractions to recognize the answer.

27.1 Transforms so far

All functions vanish for $t < 0$. The Laplace transform of $f(t)$ is

$$F(s) = \mathcal{L}[f(t)] = \int_0^\infty f(t)e^{-st}\,dt,$$

written also $f(t) \longmapsto F(s)$. The improper integral may converge only for $\mathrm{Re}(s) > a$ for some a depending on f; the smallest such a gives the region of convergence. The computations and rules so far are

$$1 \longmapsto \frac{1}{s}, \qquad t^n \longmapsto \frac{n!}{s^{n+1}}, \qquad e^{rt} \longmapsto \frac{1}{s-r}, \qquad \cos(\omega t) \longmapsto \frac{s}{s^2+\omega^2}, \qquad \sin(\omega t) \longmapsto \frac{\omega}{s^2+\omega^2},$$

together with

- *linearity*: $af(t) + bg(t) \longmapsto aF(s) + bG(s)$;
- the *s-shift rule*: $e^{rt}f(t) \longmapsto F(s-r)$.

Example: To transform $e^{-t}\cos(3t)$, one could write it as $\frac{1}{2}\left(e^{(-1+3i)t} + e^{(-1-3i)t}\right)$, but the s-shift is quicker: $\cos(3t) \longmapsto s/(s^2+9)$, so

$$e^{-t}\cos(3t) \longmapsto \frac{s+1}{(s+1)^2+9} = \frac{s+1}{s^2+2s+10}.$$

The last step "uncompletes the square."

27.2 The delta function

If f is continuous at $t = b$, then $\delta(t-b)f(t) = f(b)\delta(t-b)$, so $\int \delta(t-b)f(t)\,dt = f(b)u(t-b) + c$. Equivalently, for $a < b < c$,

$$\int_a^c \delta(t-b)f(t)\,dt = f(b),$$

which some accounts take as the definition of δ. For $b > 0$ this gives

$$\mathcal{L}[\delta(t-b)] = \int_0^\infty \delta(t-b)e^{-st}\,dt = e^{-bs}.$$

The region of convergence is the whole complex plane: the integral up to T is constant once $T > b$.

At $b = 0$ there is a problem: $\delta(t)e^{-st} = \delta(t)$, but $\int_0^\infty \delta(t)\,dt = u(\infty) - u(0)$, and $u(0)$ is indeterminate. We want the formula for $b > 0$ to persist, $\mathcal{L}[\delta(t)] = 1$, so the lower limit is also taken as a limit from below.

Definition 27.1 [Laplace transform, refined].

$$\mathcal{L}[f(t)] = \lim_{\substack{c\to\infty \\ a\uparrow 0}} \int_a^c f(t)e^{-st}\,dt = \int_{0^-}^\infty f(t)e^{-st}\,dt.$$

With this refinement,

$$\boxed{\mathcal{L}[\delta(t-b)] = e^{-bs} \quad \text{for } b \geq 0},$$

with region of convergence the whole plane. In particular $\delta(t) \longmapsto 1$.

27.3 The t-derivative rule

Integrate by parts with $u = e^{-st}$ and $dv = f'(t)\,dt$:

$$\mathcal{L}[f'(t)] = \int_{0^-}^\infty f'(t)e^{-st}\,dt = e^{-st}f(t)\Big|_{0^-}^\infty + s\int_{0^-}^\infty f(t)e^{-st}\,dt.$$

The boundary term vanishes: $f(0^-) = 0$, and $e^{-st}f(t) \to 0$ in the region of convergence. Hence

$$\boxed{\mathcal{L}[f'(t)] = sF(s)}.$$

Here f' must be the generalized derivative; only then does the integral of the derivative behave correctly. Even if f has no breaks for $t > 0$, it typically jumps at $t = 0$, since $f(0^-) = 0$ but $f(0^+)$ need not vanish. So f' should be expected to contain a delta function at $t = 0$.

Example: Transforming a function and its generalized derivative:

$$u(t) \longmapsto \frac{1}{s}, \qquad \delta(t) \longmapsto s\cdot\frac{1}{s} = 1;$$

$$t^n \longmapsto \frac{n!}{s^{n+1}}, \qquad nt^{n-1} \longmapsto s\cdot\frac{n!}{s^{n+1}} = \frac{n\,(n-1)!}{s^n};$$

$$\cos t \longmapsto \frac{s}{s^2+1}, \quad \delta(t) - \sin t \longmapsto \frac{s^2}{s^2+1} = 1 - \frac{1}{s^2+1}.$$

27.4 The inverse transform

Solving an initial value problem by Laplace transform follows the scheme of Figure 33. For it to work, $f(t)$ must be recovered from $F(s)$. There is no formula for $\mathcal{L}^{-1}$ in practice; one looks for pieces of $F(s)$ in the table of computations, an art like integration.

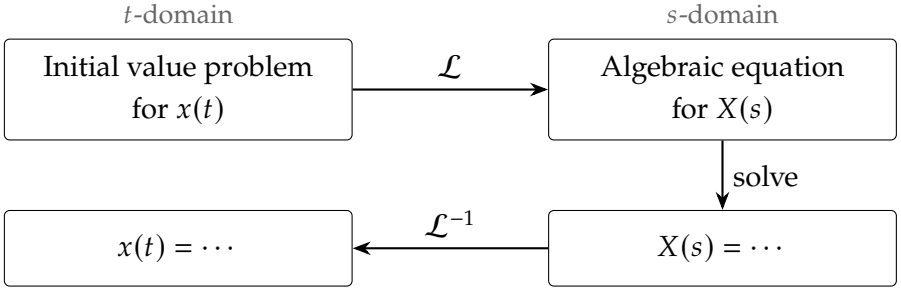


Figure 33: Solving an initial value problem with the Laplace transform.

Nor can f be recovered exactly when it is not required to be continuous: $F(s)$ is an integral, unchanged if any single value of f is altered. What does hold is the following.

Theorem 27.2 [Uniqueness]. *If generalized functions $f(t)$ and $g(t)$ have the same Laplace transform, then for every $a \geq 0$, $f(a^+) = g(a^+)$ and $f(a^-) = g(a^-)$, and any delta functions occur identically in f and g. In particular, if f and g are continuous at a, then $f(a) = g(a)$.*

27.5 Unit impulse response

Example: Find the unit impulse response of $D + 3I$, that is, solve $w' + 3w = \delta(t)$ with rest initial conditions. Apply $\mathcal{L}$: $sW + 3W = 1$. Solve: $W = 1/(s+3)$. Invert: $w(t) = u(t)e^{-3t}$.

The Laplace transform is a good way to find unit impulse responses. The unit impulse response is also called the weight function, and its Laplace transform is the transfer function.

27.6 Partial fractions and the cover-up method

Example: Solve $x' + 3x = e^{-t}$ with $x(0^+) = 0$. By linearity, the s-shift rule, and the t-derivative rule, $sX + 3X = 1/(s+1)$, so

$$X = \frac{1}{(s+1)(s+3)}.$$

To recognize X, write it as a combination of known transforms by partial fractions:

$$\frac{1}{(s+1)(s+3)} = \frac{a}{s+1} + \frac{b}{s+3}.$$

Instead of cross-multiplying and matching coefficients, use the cover-up method. Multiply through

by $s+1$:

$$\frac{1}{s+3} = a + (s+1)\frac{b}{s+3}.$$

Now set $s=-1$: the second term vanishes and $a = 1/(-1+3) = 1/2$. This "covers up" the factor $s+1$ together with every unwanted coefficient. In the same way, covering up $s+3$ and setting $s=-3$ gives $b = 1/(-3+1) = -1/2$. So

$$X = \frac{1/2}{s+1} - \frac{1/2}{s+3}, \qquad x(t) = u(t)\left(\tfrac{1}{2}e^{-t} - \tfrac{1}{2}e^{-3t}\right).$$

This agrees with earlier methods: the exponential response formula gives $\frac{1}{2}e^{-t}$, the homogeneous solutions are ce^{-3t}, and $x(0)=0$ forces $c=-1/2$.

27.7 Nonzero initial values and $\mathcal{L}[f_r']$

Consider $x' + 3x = e^{-t}$ with $x(0) = 5$. Since $x(t) = 0$ for $t < 0$, the solution must jump at $t = 0$, so the intended condition is $x(0^+) = 5$. But then the equation, with x' the generalized derivative, has no solution: $x(0^-) = 0$, so x' contains the singular part $5\delta(t)$, and there is no $5\delta(t)$ on the right side. What is really intended is

$$x_r' + 3x = e^{-t}, \qquad x(0^+) = 5,$$

where x_r' is the REGULAR PART of the derivative.

Suppose $f(t)$ is continuous for $t > 0$. Then the only singular part of the generalized derivative is at $t = 0$, namely $f_s'(t) = f(0^+)\delta(t)$, and $f' = f_r' + f_s'$. By linearity and $\mathcal{L}[\delta(t)] = 1$, $\mathcal{L}[f'(t)] = f(0^+) + \mathcal{L}[f_r'(t)]$, so

$$\boxed{\mathcal{L}[f_r'(t)] = sF(s) - f(0^+)}$$

when f has a continuous derivative for $t > 0$. This is the form found in most books and most often used in practice.

Example: For $x_r' + 3x = e^{-t}$ with $x(0^+) = 5$, the rule gives $(sX - 5) + 3X = 1/(s+1)$, so

$$X = \frac{1}{(s+1)(s+3)} + \frac{5}{s+3}.$$

The first term is the previous solution; the second adds $\mathcal{L}^{-1}[5/(s+3)] = 5e^{-3t}$, which corrects the transient:

$$x(t) = \tfrac{1}{2}e^{-t} - \tfrac{1}{2}e^{-3t} + 5e^{-3t} = \tfrac{1}{2}e^{-t} + \tfrac{9}{2}e^{-3t} \qquad (t > 0).$$

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 27, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 28: Second order equations; completing the squares

This lecture revisits the generalized derivative, solves a first order problem whose partial fractions involve a repeated factor, adds the s-derivative rule, and extends the method to second order equations.

28.1 Generalized derivatives and the fundamental theorem

Example: Let $f(t) = u(t)\cos(\omega t)$. Its graph is continuous except for a jump from 0 to 1 at $t = 0$. The regular part of the derivative is $f_r'(t) = -\omega u(t)\sin(\omega t)$ and the singular part is $f_s'(t) = \delta(t)$. The product rule gives the same answer:

$$f'(t) = -\omega u(t)\sin(\omega t) + \delta(t)\cos(\omega t) = -\omega u(t)\sin(\omega t) + \delta(t),$$

since $\delta(t)g(t) = g(0)\delta(t)$ and $\cos(0) = 1$.

The fundamental theorem of calculus,

$$\int_a^c f'(t)\,dt = f(c) - f(a),$$

holds only if f' is the generalized derivative. For example, with $a = -1$, $c = 1$, and $f = u$: taking $u'(t) = 0$ would give $\int_{-1}^{1} u'(t)\,dt = 0$, not 1, while $\int_{-1}^{1} \delta(t)\,dt = 1$.

28.2 Initial values and repeated factors

Let f vanish for $t < 0$ and have its only jump at $t = 0$. Then

$$f' = f_r' + \big(f(0^+) - f(0^-)\big)\delta(t) = f_r' + f(0^+)\delta(t).$$

Transforming, $sF(s) = \mathcal{L}[f_r'(t)] + f(0^+)$, that is, $\mathcal{L}[f_r'(t)] = sF(s) - f(0^+)$.

Example: Solve $x' + 2x = 4t$ with $x(0) = 1$. Since $x(t) = 0$ for $t < 0$, the solution jumps at $t = 0$, which would produce a delta function in the generalized derivative; there is none in the equation, so it has no solution in that sense. What is intended is the ordinary derivative:

$$x_r' + 2x = 4t, \qquad x(0^+) = 1.$$

Transforming, $sX - 1 + 2X = 4/s^2$, so

$$X = \frac{4}{s^2(s+2)} + \frac{1}{s+2}.$$

The second term is $\mathcal{L}[e^{-2t}]$. The first needs partial fractions with a repeated factor:

$$\frac{4}{s^2(s+2)} = \frac{as+b}{s^2} + \frac{c}{s+2} = \frac{a}{s} + \frac{b}{s^2} + \frac{c}{s+2}.$$

- Cover up s^2: $4/(s+2) = b + s(\cdots)$; at $s = 0$, $b = 2$.
- Cover up $s+2$: $4/s^2 = (s+2)(\cdots) + c$; at $s = -2$, $c = 1$.
- Cover-up cannot isolate a; set s to any other value, say $s = 1$: $4/3 = a + 2 + 1/3$, so $a = -1$.

Hence $X = -1/s + 2/s^2 + 2/(s+2)$ and, for $t > 0$,

$$x(t) = -1 + 2t + 2e^{-2t}.$$

Undetermined coefficients would have found this more easily. Figure 34 shows the jump at $t = 0$.

28.3 The s-derivative rule

Differentiating under the integral sign,

$$F'(s) = \frac{d}{ds}\int_0^\infty e^{-st} f(t)\,dt = \int_0^\infty \left(-te^{-st}\right) f(t)\,dt,$$

which is the Laplace transform of $-tf(t)$. Thus

$$\boxed{\mathcal{L}[tf(t)] = -F'(s)},$$

a counterpart of the t-derivative rule $\mathcal{L}[f'(t)] = sF(s)$.

Example: For $f(t) = \sin(\omega t)$, $F(s) = \omega/(s^2+\omega^2)$, so

$$\mathcal{L}[t\sin(\omega t)] = -F'(s) = \frac{2\omega s}{(s^2+\omega^2)^2}.$$

28.4 Second order equations

Let $g = f'$, so $G(s) = sF(s)$. Then

$$\mathcal{L}[f''(t)] = \mathcal{L}[g'(t)] = sG(s) = s^2F(s).$$

This is used only when $f(0^+) = 0$, so that f' contains no delta function; derivatives of the delta function are not used in this course.

Example: Find the unit impulse response of $D^2 + 2D + 2I$: solve $w'' + 2w' + 2w = \delta(t)$ with rest

initial conditions. Transforming, $s^2W + 2sW + 2W = 1$, so

$$W = \frac{1}{s^2 + 2s + 2} = \frac{1}{(s+1)^2 + 1}$$

after COMPLETING THE SQUARE. Since $\sin t \longmapsto 1/(s^2+1)$, the s-shift rule gives $e^{-t}\sin t \longmapsto W(s)$, so

$$w(t) = u(t)e^{-t}\sin t.$$

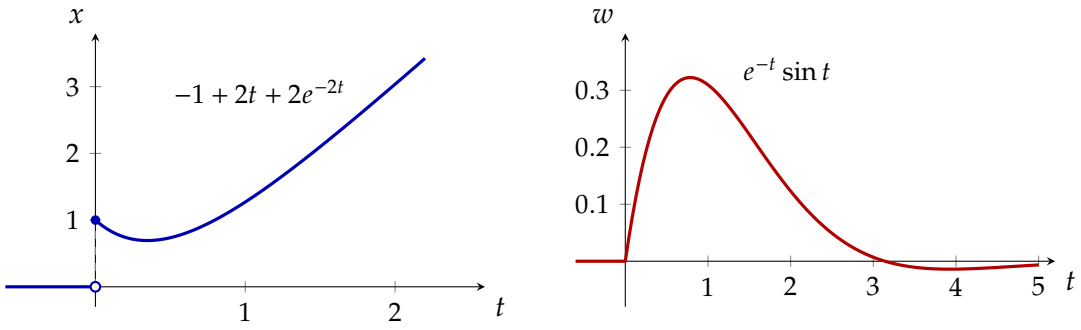


Figure 34: Left: the solution $x(t) = -1 + 2t + 2e^{-2t}$ of $x' + 2x = 4t$, $x(0^+) = 1$, which jumps from 0 to 1 at $t = 0$. Right: the unit impulse response $w(t) = u(t)e^{-t}\sin t$ of $D^2 + 2D + 2I$, continuous at $t = 0$ with $w'(0^+) = 1$.

Recall the terminology: the unit impulse response is the weight function, and its Laplace transform is the transfer function.

Theorem 28.1 [TRANSFER FUNCTION OF $p(D)$]. *The transfer function of $p(D) = a_nD^n + \cdots + a_1D + a_0I$ is*

$$W(s) = \frac{1}{p(s)}, \qquad \textit{that is,} \qquad w(t) = \mathcal{L}^{-1}\left[\frac{1}{p(s)}\right].$$

Proof. The weight function solves $a_nw^{(n)} + \cdots + a_0w = \delta(t)$ with rest initial conditions. Applying $\mathcal{L}$ with $\mathcal{L}[w^{(k)}] = s^kW$ gives $a_ns^nW + \cdots + a_0W = 1$, that is, $p(s)W(s) = 1$. □

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 28, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 29: The pole diagram

In the s-domain the response of a system is a product with the transfer function. Delaying a signal multiplies its transform by an exponential. The poles of $F(s)$, the points where it becomes infinite, are easy to see and describe the long-term behavior of $f(t)$.

29.1 System response in the s-domain

If $p(D)w = \delta(t)$, that is, $a_n w^{(n)} + \cdots + a_1 w' + a_0 w = \delta(t)$ with rest initial conditions, then $p(s)W(s) = 1$, so

$$W(s) = \frac{1}{p(s)}, \qquad w(t) = \mathcal{L}^{-1}\Big[\frac{1}{p(s)}\Big],$$

where w is the weight function and W the transfer function. The same computation applies to any input signal.

Theorem 29.1 [Response in the s-domain]. *The equation $p(D)x = f(t)$ with rest initial conditions has transformed equation $p(s)X(s) = F(s)$, with solution*

$$\boxed{X(s) = W(s)F(s)}.$$

In the s-domain, the system response is obtained by multiplying by the transfer function.

Example: Solve $x'' + 4x = \cos(2t)$ with rest initial conditions. The signal contains no delta functions, so $x(0^+) = x'(0^+) = 0$, but this comes out automatically. Here $W(s) = 1/(s^2+4)$ and $\cos(2t) \longmapsto s/(s^2+4)$, so

$$X(s) = \frac{s}{(s^2+4)^2}.$$

By the s-derivative rule, $t\sin(2t) \longmapsto 4s/(s^2+4)^2$, hence

$$x(t) = \tfrac{1}{4}u(t)\, t\sin(2t),$$

an oscillation with linearly growing amplitude: resonance.

29.2 The t-shift rule

For $a > 0$ define the delayed function

$$f_a(t) = \begin{cases} 0, & t < a, \\ f(t-a), & t > a, \end{cases}$$

that is, $f_a(t) = u(t-a)f(t-a)$. If the singular part of f contains $c\,\delta(t)$, then f_a is to contain $c\,\delta(t-a)$; for example $\delta_a(t) = \delta(t-a)$.

Proposition 29.2 [t-SHIFT RULE].

$$\boxed{f_a(t) \longmapsto e^{-as}F(s)}.$$

Proof. Substitute $v = t - a$, $dv = dt$:

$$\int_{0^-}^{\infty} f_a(t)e^{-st}\,dt = \int_{a^-}^{\infty} f(t-a)e^{-st}\,dt = \int_{0^-}^{\infty} f(v)e^{-s(v+a)}\,dv = e^{-as}F(s). \qquad \square$$

For example, $u_a(t) = u(t-a) \longmapsto e^{-as}/s$, and $\delta_a(t) \longmapsto e^{-as}$, as found before.

29.3 Poles

$F(s)$ essentially determines $f(t)$, but the path from F to f can be difficult. Certain features of f, those concerning its long-term behavior, can be read off from F easily. Since F assigns a complex number to each complex number s, look at its absolute value $|F(s)|$: a real function on the plane, whose graph is a surface in space.

For $F(s) = 1/s$, $|F(s)| = 1/|s|$ is the reciprocal of the distance to the origin. Its graph is radially symmetric, the surface of revolution of one branch of the hyperbola $z = 1/x$: it rises to infinity as $s \to 0$ and falls to zero as $|s|$ grows.

Definition 29.3 [POLE]. A POLE of $F(s)$ is a point at which $F(s)$ becomes infinite. The set of poles, marked in the complex plane, is the POLE DIAGRAM of F.

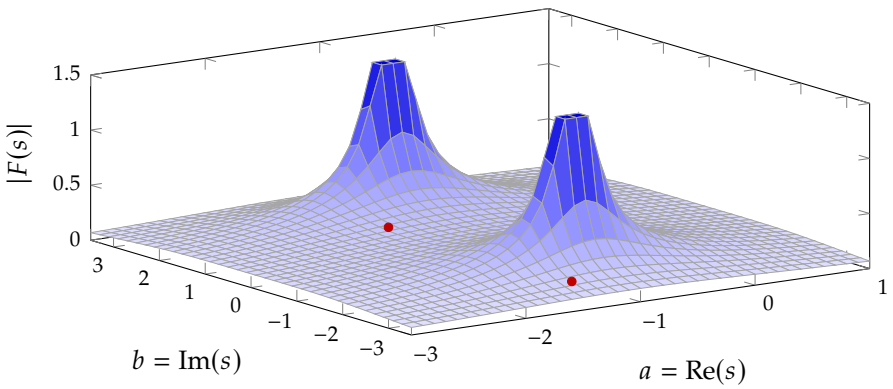


Figure 35: The graph of $|F(s)|$ for $F(s) = 1/(s^2 + 2s + 5)$ over the s-plane, $s = a + bi$, cut off at height 1.5. It rises to infinity at the poles $-1 \pm 2i$ (dots).

The pole $s = 0$ of $1/s$ is a simple gross feature of F; seen from a distance, it is the only one. The poles of $1/(s^2 + 2s + 5)$ are the roots of $s^2 + 2s + 5 = (s+1)^2 + 4$, namely $s = -1 \pm 2i$ (Figure 35), and

the sum $1/s + 1/(s^2 + 2s + 5)$ has all three. These transforms come from

$$\mathcal{L}[1] = \frac{1}{s}, \qquad \mathcal{L}\left[\tfrac{1}{2}e^{-t}\sin(2t)\right] = \frac{1}{(s+1)^2+4}.$$

Region of convergence. The poles explain the region of convergence of $F(s) = \int_{0^-}^{\infty} e^{-st} f(t)\,dt$. Since $|e^{-st}| = e^{-\mathrm{Re}(s)t}$ depends only on $\mathrm{Re}(s)$, the region of convergence is always a half-plane $\mathrm{Re}(s) > a$, or empty, or the whole plane. At a pole F is infinite, reflecting divergence of the integral. So the region of convergence is the half-plane to the right of the rightmost pole.

29.4 What the poles of $F(s)$ say about $f(t)$

The pole diagram is only a small part of $F(s)$ and cannot determine $f(t)$:

- If $f(t) = 0$ for $t > A$, then $F(s) = \int_{0^-}^{A} e^{-st} f(t)\,dt$ converges for all s: there are no poles at all.
- e^{-as} is never zero and has no poles, so $e^{-as}F(s)$ has the same poles as $F(s)$. Time translation of f does not change the pole diagram; it cannot see phase.

But the pole diagram does say a lot about the long-term behavior of f. Figure 36 shows the pole diagrams of three examples.

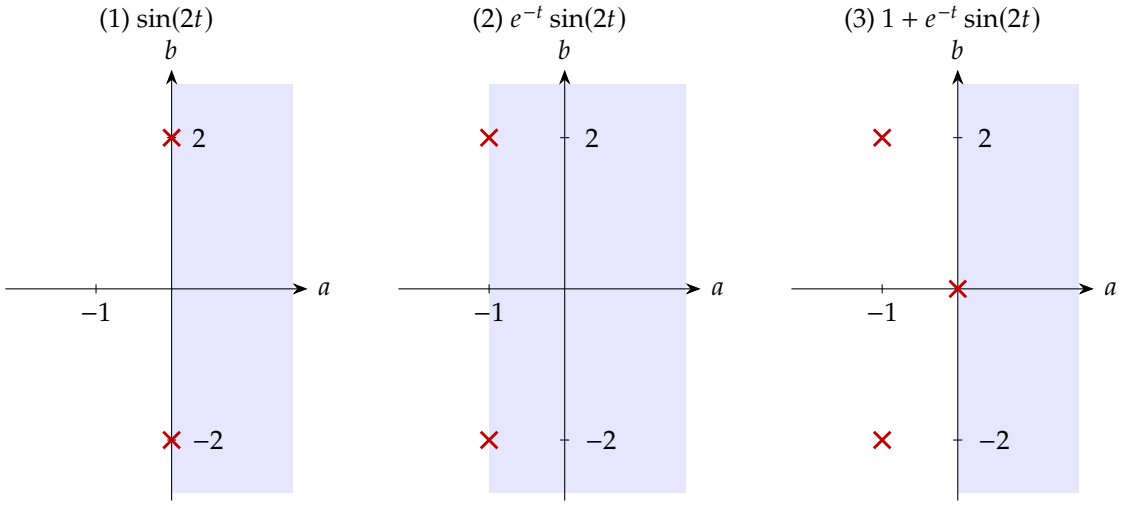


Figure 36: Pole diagrams of the three examples below. The shaded half-plane right of the rightmost pole is the region of convergence.

Example:

(1) $f(t) = \sin(2t)$, $F(s) = 2/(s^2 + 4)$, poles $\pm 2i$. For large t, any f whose transform has this pole diagram oscillates with circular frequency 2 and neither grows nor decays exponentially. For

instance $t\sin(2t)$, with transform $4s/(s^2+4)^2$, has the same poles; it grows, but less than exponentially.

(2) $f(t) = e^{-t}\sin(2t)$, $F(s) = 2/((s+1)^2+4)$, poles $-1 \pm 2i$. Any f with this pole diagram oscillates with circular frequency 2 and decays exponentially, on the order of e^{-t}.

(3) $f(t) = 1 + e^{-t}\sin(2t)$, $F(s) = 1/s + 2/((s+1)^2+4)$, poles 0 and $-1 \pm 2i$. Any f with this pole diagram shows sub-exponential growth or decay, meaning $e^{-at} < |f(t)| < e^{at}$ for large t and every $a > 0$, plus an oscillation with circular frequency 2 whose amplitude decays like e^{-t}.

The rightmost poles carry the dominant information. In particular, if all poles of $F(s)$ have negative real part, then $f(t)$ decays to zero exponentially. The position of the poles gives the long-term rates of growth and oscillation of f, which can be hard to see from the graph of f, since all exponentials look rather alike. So when solving an equation by Laplace transform, it may be enough to stop at $X(s)$: its poles already describe the long-run behavior of the solution.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 29, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 30: The transfer function and frequency response

The pole diagram of a transfer function governs stability, and its restriction to the imaginary axis is the amplitude response. Together these tie the Laplace transform to the earlier study of exponential and sinusoidal input.

30.1 Pole diagrams and long-term behavior

A Laplace transform $F(s)$ can be understood, in broad terms, through its pole diagram, the set of points at which it becomes infinite. Each of the following, with $a \neq 0$, has pole diagram $\{2i, -2i\}$:

$F(s)$	$f(t)$
$a/(s^2+4)$	$(a/2)\sin(2t)$
$as/(s^2+4)$	$a\cos(2t)$
$e^{-bs}/(s^2+4)$	$\frac{1}{2}u(t-b)\sin\big(2(t-b)\big)$
$1 + a/(s^2+4)$	$\delta(t) + (a/2)\sin(2t)$
$4s/(s^2+4)^2$	$t\sin(2t)$

All functions whose transforms have this pole diagram share two features for large t: they oscillate with circular frequency 2, and they may grow or shrink, but less than exponentially.

Adding a pole at 1, with $c \neq 0$, gives pole diagram $\{2i, -2i, 1\}$; for instance

$$\frac{c}{s-1} + \cdots \longmapsto ce^t + \cdots, \qquad \frac{c}{(s-1)^2} + \cdots \longmapsto cte^t + \cdots,$$

where the dots stand for any of the transforms above. All such f grow "like e^t": faster than e^{kt} for every $k < 1$ and slower than e^{lt} for every $l > 1$. They still oscillate, but the oscillations become insignificant relative to the size of $f(t)$.

The rightmost poles of $F(s)$ determine the dominant features of $f(t)$ for large t. In particular, if all poles of $F(s)$ have negative real part, so lie left of the imaginary axis, then $f(t) \to 0$ as $t \to \infty$.

30.2 Stability

The unit impulse response $w(t)$ of $p(D)$ has transform $W(s) = 1/p(s)$, so the poles of W are the roots of $p(s)$. For $t > 0$, $w(t)$ solves $p(D)x = 0$. If all roots of p have negative real part, then all solutions of $p(D)x = 0$ die off exponentially. So when all poles of W lie in the left half-plane, not just the unit impulse response but every solution of the homogeneous equation dies off: all are transients.

30.3 Exponential and sinusoidal signals

The reciprocal of the characteristic polynomial appeared long ago, in the exponential response formula: $p(D)x = e^{rt}$ has the exponential solution

$$x_p = W(r)e^{rt} = \frac{e^{rt}}{p(r)} \qquad (p(r) \neq 0).$$

So the transfer function, the Laplace transform of the weight function, is also the factor that multiplies an exponential input.

The same holds for sinusoidal response. For input $\cos(\omega t)$, solve the complex replacement $p(D)z = e^{i\omega t}$:

$$z_p = W(i\omega)e^{i\omega t}, \qquad x_p = \mathrm{Re}\big[W(i\omega)e^{i\omega t}\big] = \mathrm{gain}(\omega)\cos(\omega t - \phi),$$

where

$$\mathrm{gain}(\omega) = |W(i\omega)|, \qquad -\phi = \mathrm{Arg}\big(W(i\omega)\big).$$

$W(i\omega)$ is the COMPLEX GAIN, with gain and phase lag measured relative to the input signal $\cos(\omega t)$. Thus the gain is the restriction of $|W(s)|$ to the imaginary axis.

30.4 The unifying picture

Consider a system, such as a black box of springs, masses, and dashpots, whose response to various input signals we want to predict. The systems analyzed here are LINEAR TIME INVARIANT: superposition holds, and delaying the input delays the response without otherwise changing it.

Of special interest is the periodic response to periodic signals. By Fourier series, periodic signals decompose into sinusoids, so it is enough to study sinusoidal response, which involves a gain and a phase lag; focus on the gain. The procedure is:

(1) Hit the system with $\delta(t)$; what comes out is $w(t)$.

(2) Apply $\mathcal{L}$ to get $W(s)$, and graph $|W(s)|$, a surface over the complex plane, probably with some poles.

(3) Restrict to $s = i\omega$. Since $p(D)x = e^{i\omega t}$ has solution $W(i\omega)e^{i\omega t}$, the height $|W(i\omega)|$ is the gain.

The intersection of the graph of $|W(s)|$ with the vertical plane over the imaginary axis is the amplitude response curve, extended to an even function by allowing negative ω.

Near resonance occurs because $i\omega$ comes close to a pole of $W(s)$. Increasing the damping moves the poles deeper into the left half-plane, and eventually the two humps of the amplitude response curve merge into one, as in Figure 37. A higher order system has more poles and a more complicated

amplitude response curve.

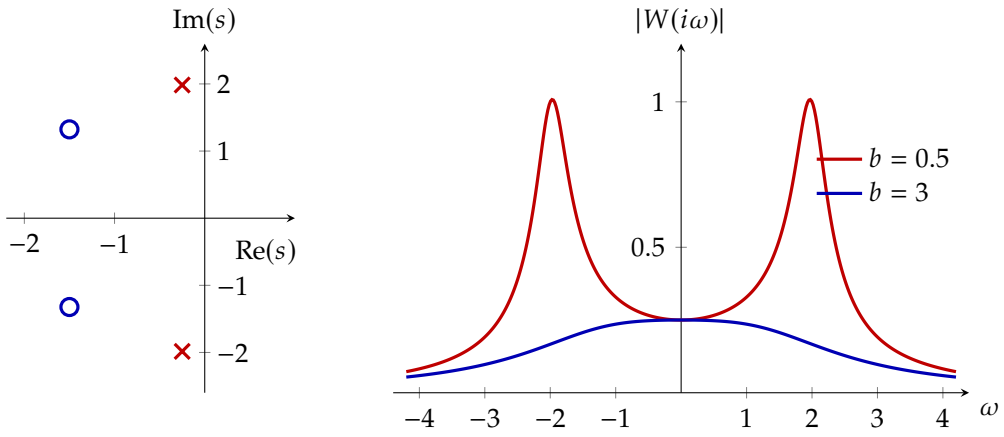


Figure 37: Illustration with $W(s) = 1/(s^2 + bs + 4)$. Left: the poles for $b = 0.5$ (crosses) and $b = 3$ (circles). Right: the amplitude response $|W(i\omega)|$, extended to negative ω. For $b = 0.5$ the poles $-0.25 \pm 1.98i$ lie near the imaginary axis and the curve has two sharp humps near $\omega = \pm 2$; for $b = 3$ the poles $-1.5 \pm 1.32i$ are farther left and the humps have merged.

30.5 Laplace and Fourier

A periodic function neither grows nor decays as $t \to \infty$, so its rightmost poles lie on the imaginary axis. Precisely, a periodic function of period $2L$ has a Fourier series; with $\omega = 2\pi/2L = \pi/L$,

$$f(t) = \frac{a_0}{2} + a_1 \cos(\omega t) + a_2 \cos(2\omega t) + \cdots + b_1 \sin(\omega t) + b_2 \sin(2\omega t) + \cdots .$$

The poles of $\mathcal{L}[\cos(n\omega t)]$ and $\mathcal{L}[\sin(n\omega t)]$ are $\pm n\omega i$. So the poles of $u(t)f(t)$ lie on the imaginary axis, at multiples of ωi.

This gives some feeling for the meaning of the s-domain. If $F(s)$ has poles right of the imaginary axis, then $f(t)$ grows like e^{at}, where a is the largest real part of a pole. The imaginary part of a pole gives the circular frequency of exponentially damped or growing oscillations.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 30, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 32: Linear systems and matrices

Two interacting quantities lead to a first order linear system. Elimination reduces it to a second order equation; matrix notation packages it as $\mathbf{u}' = A\mathbf{u}$; and conversely every second order equation becomes a first order system through its companion matrix.

32.1 Elimination: two fields of rabbits

Farmer Jones's field contains $x(t)$ rabbits and Farmer McGregor's field contains $y(t)$. In each field the net growth rate is 0.5 rabbits per rabbit per month. Rabbits also hop over the hedge: Jones's at the rate 0.2 per month, McGregor's at the rate 0.1 per month (Figure 38).

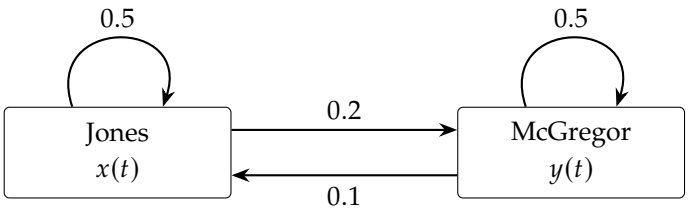


Figure 38: Flow diagram for the two rabbit populations. Loops show the net growth rate in each field; arrows show the hopping rates.

The equations are

$$x' = 0.5x - 0.2x + 0.1y = 0.3x + 0.1y, \tag{32.1}$$

$$y' = 0.5y - 0.1y + 0.2x = 0.2x + 0.4y. \tag{32.2}$$

This is a homogeneous linear system. Solving it seems circular, since x requires y and y requires x, but it can be solved by elimination. By (32.1), $y = 10x' - 3x$. Substituting into (32.2),

$$10x'' - 3x' = 0.2x + 4x' - 1.2x, \qquad \text{so} \qquad x'' - 0.7x' + 0.1x = 0.$$

The characteristic polynomial $s^2 - 0.7s + 0.1$ has roots $r_1 = 0.5$ and $r_2 = 0.2$ (recall $(s - r_1)(s - r_2) = s^2 - (r_1 + r_2)s + r_1 r_2$). Each basic solution x gives a matching $y = 10x' - 3x$:

$$x_1 = e^{0.5t},\ y_1 = 2e^{0.5t}; \qquad x_2 = e^{0.2t},\ y_2 = -e^{0.2t}.$$

Some of these solutions are not biologically meaningful. Packaging x and y into the vector $\mathbf{u}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$, a solution traces a curve in the plane as time increases. For example, $\mathbf{u}_1 = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$ traces the ray through $(1, 2)$, passing $(1, 2)$ at $t = 0$ and moving off toward infinity like $e^{0.5t}$.

32.2 Matrix form

The general homogeneous linear system is

$$x' = ax + by, \qquad y' = cx + dy. \tag{32.3}$$

A 2×2 MATRIX is an array of numbers $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, written in Matlab notation as $[a\ b;\ c\ d]$. A COLUMN VECTOR $\begin{bmatrix} x \\ y \end{bmatrix} = [x;\ y]$ is a matrix with one column. Matrix multiplication runs along a row of the first factor and down a column of the second, multiplying and adding:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix} = x \begin{bmatrix} a \\ c \end{bmatrix} + y \begin{bmatrix} b \\ d \end{bmatrix}.$$

The second form says that $A\mathbf{u}$ is the combination of the columns of A weighted by the entries of $\mathbf{u}$. With $\mathbf{u} = [x;\ y]$ and $\mathbf{u}' = [x';\ y']$, the system (32.3) is exactly

$$\boxed{\mathbf{u}' = A\mathbf{u}}.$$

In the rabbit example, $A = \begin{bmatrix} 0.3 & 0.1 \\ 0.2 & 0.4 \end{bmatrix}$, and there are two independent basic solutions,

$$\mathbf{u}_1 = e^{0.5t} \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad \mathbf{u}_2 = e^{0.2t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

The general solution is a linear combination $c_1\mathbf{u}_1 + c_2\mathbf{u}_2$ (Figure 39, left).

32.3 Companion matrices

Conversely, every second order linear equation arises this way, and the first order system gives an important way to visualize it. Given

$$x'' + bx' + kx = 0, \tag{32.4}$$

set $y = x'$. Then $y' = x'' = -kx - by$, so $\mathbf{u} = [x;\ y]$ satisfies $\mathbf{u}' = A\mathbf{u}$ with the COMPANION MATRIX

$$A = \begin{bmatrix} 0 & 1 \\ -k & -b \end{bmatrix}.$$

Solutions have the form $\mathbf{u} = [x;\ x']$: they record both a solution of (32.4) and its derivative.

Example: The equation $x'' - \frac{1}{2}x' + \frac{17}{16}x = 0$ has companion matrix $\begin{bmatrix} 0 & 1 \\ -17/16 & 1/2 \end{bmatrix}$. Its characteristic polynomial is $p(s) = s^2 - \frac{1}{2}s + \frac{17}{16} = (s - \frac{1}{4})^2 + 1$, with roots $\frac{1}{4} \pm i$. So the general solution is

$$x = Ce^{t/4}\cos(t - \phi).$$

These oscillate under an exponentially growing envelope. The derivative does the same, out of phase, so the trajectory of $(x, y) = (x, x')$ is an expanding spiral.

32.4 Romeo and Juliet

The same system models a love affair. Let R be Romeo's love for Juliet and J Juliet's love for Romeo, with

$$R' = J, \qquad J' = -\tfrac{17}{16}R + \tfrac{1}{2}J.$$

Romeo's feelings change only in response to how Juliet feels about him. Juliet's own love feeds itself (the term $\frac{1}{2}J$), but Romeo's love frightens her (the term $-\frac{17}{16}R$).

Start at $(R, J) = (1, 0)$: Romeo is fond and Juliet neutral. His fondness makes her hostile, so his affection wanes; his distance then makes her warm to him, and his decline reverses. The trajectory returns to $J = 0$ with R larger than before: the relationship is cyclic, but each cycle is more intense. With $R(0) = 1$ and $R'(0) = J(0) = 0$, the solution is

$$R = e^{t/4}\left(\cos t - \tfrac{1}{4}\sin t\right), \qquad J = -\tfrac{17}{16}e^{t/4}\sin t,$$

a clockwise spiral that expands by the factor $e^{\pi/2}$ in each turn (Figure 39, right).

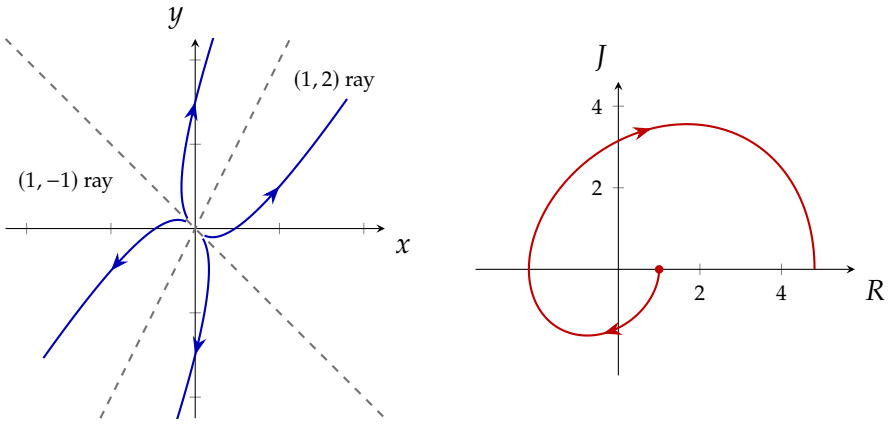


Figure 39: Left: rabbit trajectories $c_1e^{0.5t}(1, 2) + c_2e^{0.2t}(1, -1)$ for $c_1, c_2 \in \{\pm 1\}$ and $t \le 1.6$, with the two ray solutions (dashed); ticks at ± 2, ± 4. Right: the Romeo and Juliet trajectory from $(1, 0)$ for $0 \le t \le 2\pi$.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 32, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 33: Eigenvalues, eigenvectors

Some solutions of $\mathbf{u}' = A\mathbf{u}$ move along rays from the origin. Looking for them leads to eigenvalues and eigenvectors, which in turn give exponential solutions and, by superposition, the general solution.

33.1 When does $A\mathbf{u} = \mathbf{0}$ have a nonzero solution?

Recall that a matrix times a column vector is the combination of the columns of the matrix weighted by the entries of the vector:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = x \begin{bmatrix} a \\ c \end{bmatrix} + y \begin{bmatrix} b \\ d \end{bmatrix}.$$

With $A = [a\ b;\ c\ d]$ and $\mathbf{u} = [x;\ y]$, the equation $A\mathbf{u} = \mathbf{0}$ always has the TRIVIAL SOLUTION $\mathbf{u} = \mathbf{0}$. A nonzero solution exists when the two columns lie along a single line, so that some combination of them cancels. In terms of slopes, this is $c/a = d/b$, or $ad - bc = 0$. This combination of entries is the DETERMINANT

$$\det A = ad - bc.$$

Theorem 33.1 [NONTRIVIAL SOLUTIONS]. *$A\mathbf{u} = \mathbf{0}$ has a nontrivial solution exactly when* $\det A = 0$.

Proof. If $ax + by = 0$ and $cx + dy = 0$, then $d(ax + by) - b(cx + dy) = (ad - bc)x = 0$ and $a(cx + dy) - c(ax + by) = (ad - bc)y = 0$, so a nonzero (x, y) forces $ad - bc = 0$. Conversely, if $ad - bc = 0$, then $A[d;\ -c] = \mathbf{0}$ and $A[b;\ -a] = \mathbf{0}$; one of these vectors is nonzero unless $A = 0$, in which case every $\mathbf{u}$ works. □

For larger square matrices the same theorem holds, with the appropriate definition of $\det A$.

33.2 Ray solutions

Consider $\mathbf{u}' = A\mathbf{u}$ with

$$A = \begin{bmatrix} -2 & 1 \\ -4 & 3 \end{bmatrix}.$$

Phase portraits show that some trajectories run along rays from the origin, as in the rabbit example of Lecture 32 (but not in the Romeo and Juliet example). So look for solutions

$$\mathbf{u}(t) = r(t)\mathbf{v}, \qquad \mathbf{v} \neq \mathbf{0}.$$

Then $\mathbf{u}' = r'\mathbf{v}$ points along $\mathbf{v}$ as well, while the equation gives $\mathbf{u}' = A(r\mathbf{v}) = rA\mathbf{v}$. So $r'\mathbf{v} = rA\mathbf{v}$, and $A\mathbf{v}$ must point in the same or the opposite direction as $\mathbf{v}$:

$$A\mathbf{v} = \lambda\mathbf{v}$$

for some number λ.

Definition 33.2 [Eigenvalue, eigenvector]. A number λ is an eigenvalue of a square matrix A if $A\mathbf{v} = \lambda\mathbf{v}$ for some nonzero vector $\mathbf{v}$. A vector $\mathbf{v}$ with $A\mathbf{v} = \lambda\mathbf{v}$ is an eigenvector for λ. Thus $\mathbf{0}$ is an eigenvector for every λ, and λ is an eigenvalue exactly when it has a nonzero eigenvector.

33.3 Eigenvalues

Finding eigenvalues is a pure linear algebra problem. Write $\lambda\mathbf{v} = (\lambda I)\mathbf{v}$, where $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is the identity matrix and λI has λ down the diagonal. Then $A\mathbf{v} = \lambda\mathbf{v}$ becomes

$$(A - \lambda I)\mathbf{v} = \mathbf{0},$$

which by Theorem 33.1 has a nonzero solution exactly when

$$\boxed{\det(A - \lambda I) = 0}.$$

This is an equation for λ alone. Find λ first, then solve for $\mathbf{v}$, knowing in advance that a nonzero solution exists.

Definition 33.3 [Characteristic polynomial]. The characteristic polynomial of A is $p_A(\lambda) = \det(A - \lambda I)$. Its roots are the eigenvalues of A.

Example: Subtract λ from both diagonal entries and take the determinant:

$$p_A(\lambda) = \det \begin{bmatrix} -2 - \lambda & 1 \\ -4 & 3 - \lambda \end{bmatrix} = (-2 - \lambda)(3 - \lambda) + 4 = \lambda^2 - \lambda - 2 = (\lambda + 1)(\lambda - 2).$$

The eigenvalues are $\lambda_1 = -1$ and $\lambda_2 = 2$ (in either order).

33.4 Eigenvectors and normal modes

Each eigenvalue gives one special line. For $\lambda_1 = -1$,

$$A - \lambda_1 I = \begin{bmatrix} -1 & 1 \\ -4 & 4 \end{bmatrix},$$

whose columns are negatives of each other, so $(A + I)[1; 1] = \mathbf{0}$. That is, $A[1; 1] = -[1; 1]$, as can be checked directly. Along $\mathbf{v} = [1; 1]$, the equation $r'\mathbf{v} = rA\mathbf{v} = r\lambda\mathbf{v}$ gives, since $\mathbf{v} \neq \mathbf{0}$,

$$r' = \lambda r, \qquad r = ce^{\lambda t},$$

straight back to the first equation of the course. Here $r = ce^{-t}$, so all solutions along this line are

$$\mathbf{u} = ce^{-t}\begin{bmatrix}1\\1\end{bmatrix}.$$

Each of these is called a NORMAL MODE.

Proposition 33.4 [RAY SOLUTIONS]. *If $A\mathbf{v} = \lambda\mathbf{v}$, then $\mathbf{u} = e^{\lambda t}\mathbf{v}$ solves $\mathbf{u}' = A\mathbf{u}$. The ray solutions are exactly these exponential solutions, with λ an eigenvalue and $\mathbf{v}$ a nonzero eigenvector for λ.*

Proof. $\mathbf{u}' = \lambda e^{\lambda t}\mathbf{v} = e^{\lambda t}A\mathbf{v} = A\mathbf{u}$. Conversely, the computation above shows that a ray solution $r(t)\mathbf{v}$ has $A\mathbf{v} = \lambda\mathbf{v}$ and $r = ce^{\lambda t}$. □

So the eigenvalue plays a larger role than it first seemed: it is the growth rate of the exponential solution. For $\lambda_2 = 2$,

$$A - \lambda_2 I = \begin{bmatrix}-4 & 1\\ -4 & 1\end{bmatrix},$$

and $(A - 2I)\mathbf{v} = \mathbf{0}$ has the nonzero solution $\mathbf{v} = [1; 4]$. This gives the second ray solution $e^{2t}[1; 4]$.

33.5 Initial values

As long as there are two distinct eigenvalues, the general solution of $\mathbf{u}' = A\mathbf{u}$ is a linear combination of the two normal modes. In the example,

$$\mathbf{u} = c_1 e^{-t}\begin{bmatrix}1\\1\end{bmatrix} + c_2 e^{2t}\begin{bmatrix}1\\4\end{bmatrix}.$$

Example: For $\mathbf{u}(0) = [1; 0]$ we need $c_1[1; 1] + c_2[1; 4] = [c_1 + c_2; c_1 + 4c_2] = [1; 0]$. Subtracting, $3c_2 = -1$, so $c_2 = -1/3$ and $c_1 = -4c_2 = 4/3$:

$$\mathbf{u}(t) = \frac{4}{3}e^{-t}\begin{bmatrix}1\\1\end{bmatrix} - \frac{1}{3}e^{2t}\begin{bmatrix}1\\4\end{bmatrix}.$$

For very negative t the first term is huge and the second tiny, so the solution lies far out near the line through $[1; 1]$. Near $t = 0$ the terms are comparable and the solution curves around. For large t the second term dominates, and the solution becomes asymptotic to the line through $[1; 4]$, moving off

in the direction of $-[1;\ 4]$ (Figure 40).

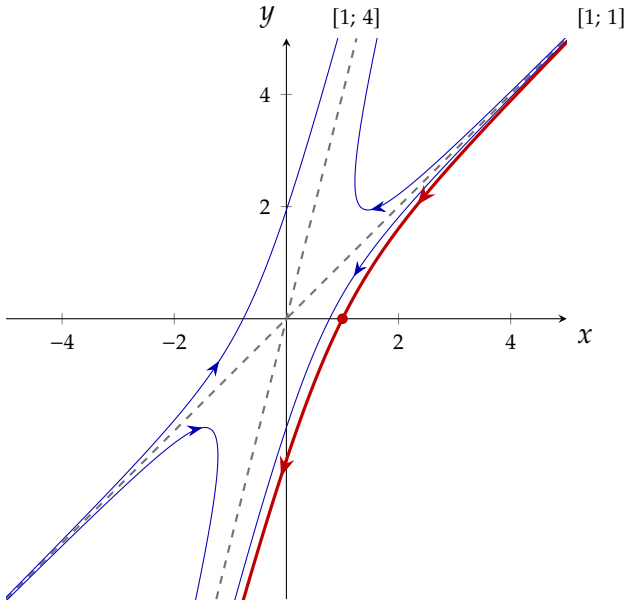


Figure 40: Solutions of $\mathbf{u}' = A\mathbf{u}$ for $A = [-2\ 1;\ -4\ 3]$. Dashed: the normal-mode lines through $[1;\ 1]$ (eigenvalue -1, solutions approach the origin) and $[1;\ 4]$ (eigenvalue 2, solutions move away). Thick: the solution with $\mathbf{u}(0) = [1;\ 0]$ (dot). Thin: the solutions $\pm 3e^{-t}[1;\ 1] \pm \frac{3}{100}e^{2t}[1;\ 4]$.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 33, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 34: Complex or repeated eigenvalues

The eigenvalue method for $\mathbf{u}' = A\mathbf{u}$ produces normal modes $e^{\lambda t}\mathbf{v}$. It needs two adjustments: when the eigenvalues are not real, and when an eigenvalue is repeated.

34.1 Eigenvalues and the characteristic polynomial

For a 2×2 matrix A (the method works the same way for $n \times n$):

1. find the eigenvalues λ_1, λ_2, the roots of the CHARACTERISTIC POLYNOMIAL $p_A(\lambda) = \det(A - \lambda I)$;
2. for each eigenvalue find a nonzero eigenvector, $(A - \lambda_1 I)\mathbf{v}_1 = \mathbf{0}$ and $(A - \lambda_2 I)\mathbf{v}_2 = \mathbf{0}$; this gives the NORMAL MODES $\mathbf{u}_1 = e^{\lambda_1 t}\mathbf{v}_1$ and $\mathbf{u}_2 = e^{\lambda_2 t}\mathbf{v}_2$;
3. the general solution is $\mathbf{u} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2$.

Any nonzero multiple of an eigenvector is again an eigenvector for the same eigenvalue; together with $\mathbf{0}$ they form a line, the EIGENLINE. The zero vector satisfies $A\mathbf{0} = \lambda\mathbf{0}$ for every λ, so λ is an eigenvalue exactly when there is a *nonzero* eigenvector for it.

For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, the TRACE $\operatorname{tr} A = a + d$ is the sum of the diagonal entries, and

$$p_A(\lambda) = (a - \lambda)(d - \lambda) - bc = \lambda^2 - (\operatorname{tr} A)\lambda + \det A.$$

For example, $A = \left(\begin{smallmatrix} -2 & 1 \\ -4 & 3 \end{smallmatrix}\right)$ has $\operatorname{tr} A = 1$, $\det A = -2$, and $p_A(\lambda) = \lambda^2 - \lambda - 2$.

34.2 Complex eigenvalues

Let $A = \left(\begin{smallmatrix} 1 & 2 \\ -2 & 1 \end{smallmatrix}\right)$. Then $\operatorname{tr} A = 2$, $\det A = 5$, and

$$p_A(\lambda) = \lambda^2 - 2\lambda + 5 = (\lambda - 1)^2 + 4, \qquad \lambda_{1,2} = 1 \pm 2i.$$

There are no real eigenvalues, hence no real ray solutions. Proceed anyway with complex numbers. For $\lambda_1 = 1 + 2i$,

$$A - (1 + 2i)I = \begin{pmatrix} -2i & 2 \\ -2 & -2i \end{pmatrix}.$$

A vector killed by the top row is obtained by reversing its entries and changing one sign, giving $\left(\begin{smallmatrix} 2 \\ 2i \end{smallmatrix}\right)$, or more simply $\mathbf{v}_1 = \left(\begin{smallmatrix} 1 \\ i \end{smallmatrix}\right)$. The bottom row must then give zero as well, $-2 \cdot 1 - 2i \cdot i = 0$, which checks the eigenvalue computation. Since A is real and $\lambda_2 = \bar{\lambda}_1$, conjugating $A\mathbf{v}_1 = \lambda_1\mathbf{v}_1$ shows that

$\mathbf{v}_2 = \bar{\mathbf{v}}_1 = \left(\begin{smallmatrix} 1 \\ -i \end{smallmatrix}\right)$ is an eigenvector for λ_2. The complex normal modes are

$$\mathbf{u}(t) = e^{(1+2i)t}\begin{pmatrix} 1 \\ i \end{pmatrix}, \qquad \bar{\mathbf{u}}(t) = e^{(1-2i)t}\begin{pmatrix} 1 \\ -i \end{pmatrix}.$$

As for second-order equations, the real and imaginary parts of a complex solution of a real system are solutions: $\operatorname{Re}\mathbf{u} = (\mathbf{u} + \bar{\mathbf{u}})/2$ and $\operatorname{Im}\mathbf{u} = (\mathbf{u} - \bar{\mathbf{u}})/(2i)$. So only one complex normal mode is needed. Expanding,

$$\mathbf{u} = e^t(\cos 2t + i \sin 2t)\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} + i\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right),$$

$$\mathbf{u}_1 = \operatorname{Re}\mathbf{u} = e^t\begin{pmatrix} \cos 2t \\ -\sin 2t \end{pmatrix}, \qquad \mathbf{u}_2 = \operatorname{Im}\mathbf{u} = e^t\begin{pmatrix} \sin 2t \\ \cos 2t \end{pmatrix}.$$

These are independent real solutions with $\mathbf{u}_1(0) = \left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right)$ and $\mathbf{u}_2(0) = \left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right)$, and the general real solution is $a\mathbf{u}_1 + b\mathbf{u}_2$ with a, b real. Both spiral clockwise around the origin while moving away from it exponentially (Figure 41).

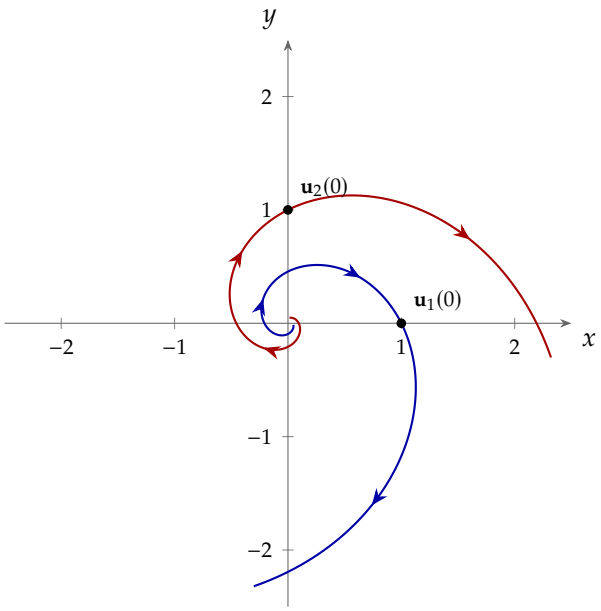


Figure 41: The real solutions $\mathbf{u}_1 = e^t(\cos 2t, -\sin 2t)$ (blue) and $\mathbf{u}_2 = e^t(\sin 2t, \cos 2t)$ (red) of $\mathbf{u}' = A\mathbf{u}$ with $A = \left(\begin{smallmatrix} 1 & 2 \\ -2 & 1 \end{smallmatrix}\right)$, drawn for $-3 \le t \le 0.85$. Both spiral clockwise and outward.

In general, if $\lambda = \alpha \pm i\beta$ with $\beta \neq 0$, the real solutions are $e^{\alpha t}$ times combinations of $\cos \beta t$ and $\sin \beta t$:

- nonreal eigenvalues give spiral trajectories;
- $\alpha > 0$: solutions go to infinity as $t \to \infty$ (unstable);
- $\alpha < 0$: solutions go to zero as $t \to \infty$ (stable);

- $\alpha = 0$: trajectories are ellipses.

So complex eigenvalues are no failure of the method: there are ray solutions, but they are complex and do not appear in the real phase plane.

34.3 Repeated eigenvalues

Let $A = \left(\begin{smallmatrix} -2 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. Then $p_A(\lambda) = \lambda^2 + 2\lambda + 1 = (\lambda + 1)^2$, with the single repeated root $\lambda_1 = \lambda_2 = -1$. From

$$A + I = \begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix}, \qquad \mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

every eigenvector is a multiple of $\mathbf{v}$. Up to multiples there is only one normal mode, $\mathbf{u}_1 = e^{-t}\mathbf{v}$, and the phase portrait has only one pair of opposed ray trajectories. A second solution comes from the following method.

Proposition 34.1 [Second solution for a repeated eigenvalue]. *Let λ_1 be an eigenvalue of A with eigenvector $\mathbf{v}$, and let $\mathbf{w}$ satisfy $(A - \lambda_1 I)\mathbf{w} = \mathbf{v}$. Then*

$$\mathbf{u}_2 = e^{\lambda_1 t}(t\mathbf{v} + \mathbf{w})$$

solves $\mathbf{u}' = A\mathbf{u}$.

Proof. Differentiate: $\mathbf{u}_2' = e^{\lambda_1 t}\big(\lambda_1 t\mathbf{v} + \lambda_1\mathbf{w} + \mathbf{v}\big)$. On the other hand, $A\mathbf{v} = \lambda_1\mathbf{v}$ and $A\mathbf{w} = \lambda_1\mathbf{w} + \mathbf{v}$, so $A\mathbf{u}_2 = e^{\lambda_1 t}\big(\lambda_1 t\mathbf{v} + \lambda_1\mathbf{w} + \mathbf{v}\big)$. □

In the example, $\left(\begin{smallmatrix} -1 & 1 \\ -1 & 1 \end{smallmatrix}\right)\mathbf{w} = \left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)$ has the solution $\mathbf{w} = \left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right)$, so

$$\mathbf{u}_2 = e^{-t}\left(t\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) = e^{-t}\begin{pmatrix} t \\ t+1 \end{pmatrix}.$$

Any $\mathbf{w} + c\mathbf{v}$ works equally well; it only adds a multiple of $\mathbf{u}_1$. With this choice $\mathbf{u}_1(0) = \left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)$ and $\mathbf{u}_2(0) = \left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right)$ are independent, and the general solution is $\mathbf{u} = a\mathbf{u}_1 + b\mathbf{u}_2$.

34.4 Defective and complete matrices

Definition 34.2 [Defective and complete]. A matrix with a repeated eigenvalue but only one line of eigenvectors is defective. A matrix whose repeated eigenvalue has a full plane of eigenvectors is complete.

The matrix above is defective. A repeated eigenvalue does not force this: $A = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 2 \end{smallmatrix}\right)$ has $p(\lambda) = \lambda^2 - 4\lambda + 4 = (\lambda - 2)^2$, so $\lambda_1 = \lambda_2 = 2$, but $A - 2I$ is the zero matrix and *every* vector is an eigenvector.

For every $\mathbf{v}$, $e^{2t}\mathbf{v}$ is a solution, so every solution is a normal mode.

In the 2×2 case, a repeated eigenvalue λ_1 is complete exactly when $A - \lambda_1 I$ kills every vector, that is, when $A = \left(\begin{smallmatrix} \lambda_1 & 0 \\ 0 & \lambda_1 \end{smallmatrix}\right)$. Every other matrix with a repeated eigenvalue is defective.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 34, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 35: Qualitative behavior of linear systems; phase plane

The phase portrait of $\mathbf{u}' = A\mathbf{u}$ is the (x, y)-plane, the phase plane, with trajectories of solutions drawn on it and the direction of time indicated. For these homogeneous linear systems the basic rule is: *the eigenvalues of A determine the phase portrait*, up to a few marginal cases where the matrix itself must be inspected.

35.1 Nodes: real eigenvalues of the same sign

Example: *Two rabbit populations.* Let $A = \left(\begin{smallmatrix} 0.3 & 0.1 \\ 0.2 & 0.4 \end{smallmatrix}\right)$. Then $p_A(\lambda) = \lambda^2 - 0.7\lambda + 0.1$, with roots $\lambda_1 = 0.5$ and $\lambda_2 = 0.2$. From

$$A - 0.5I = \begin{pmatrix} -0.2 & 0.1 \\ 0.2 & -0.1 \end{pmatrix}, \qquad A - 0.2I = \begin{pmatrix} 0.1 & 0.1 \\ 0.2 & 0.2 \end{pmatrix},$$

the eigenvectors are $\mathbf{v}_1 = \left(\begin{smallmatrix} 1 \\ 2 \end{smallmatrix}\right)$ and $\mathbf{v}_2 = \left(\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right)$, and the general solution is

$$\mathbf{u} = c_1 e^{0.5t} \begin{pmatrix} 1 \\ 2 \end{pmatrix} + c_2 e^{0.2t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The first normal mode says that if the second population is twice the first, it stays so forever. The second mode is not meaningful for populations, but it is a trajectory in the phase plane.

Both eigenvalues are positive, so all solutions flee from the origin. For large t the $\mathbf{v}_1$ component (larger eigenvalue) dominates, and trajectories become parallel to the $\mathbf{v}_1$ eigenline. For $t \to -\infty$ the $\mathbf{v}_2$ component dominates, so near the origin trajectories become tangent to the eigenline of the eigenvalue of smaller absolute value. This phase portrait is a node. It occurs whenever the eigenvalues are real, distinct, and of the same sign: an unstable node or source if they are positive, a stable node or sink if they are negative.

35.2 The trace-determinant plane

For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, $p_A(\lambda) = \lambda^2 - (\operatorname{tr} A)\lambda + \det A$. Comparing with $(\lambda - \lambda_1)(\lambda - \lambda_2)$,

$$\operatorname{tr} A = a + d = \lambda_1 + \lambda_2, \qquad \det A = ad - bc = \lambda_1 \lambda_2.$$

Completing the square,

$$p_A(\lambda) = \left(\lambda - \frac{\operatorname{tr} A}{2}\right)^2 + \left(\det A - \frac{(\operatorname{tr} A)^2}{4}\right), \qquad \lambda_{1,2} = \frac{\operatorname{tr} A}{2} \pm \sqrt{\frac{(\operatorname{tr} A)^2}{4} - \det A}.$$

So the eigenvalues are

- nonreal if $\det A > (\operatorname{tr} A)^2/4$, with real part $\operatorname{tr} A/2$;
- equal if $\det A = (\operatorname{tr} A)^2/4$, the CRITICAL PARABOLA;
- real and distinct if $\det A < (\operatorname{tr} A)^2/4$. They then have the same sign exactly when $\det A > 0$, and that sign is the sign of $\operatorname{tr} A$.

This divides the (tr, det)-plane into regions with the following phase portraits (Figure 42).

- NODES: below the parabola with $\det > 0$.
- SPIRALS: above the parabola. They move in when $\operatorname{Re}\lambda < 0$ (STABLE SPIRAL) and out when $\operatorname{Re}\lambda > 0$ (UNSTABLE SPIRAL). When $\operatorname{Re}\lambda = 0$, that is $\operatorname{tr} A = 0 < \det A$, the trajectories are ellipses and the portrait is a CENTER.
- SADDLES: $\det < 0$, real eigenvalues of opposite sign. There are two eigenlines: normal modes move out along the one with positive eigenvalue and in along the one with negative eigenvalue. Every other solution is a combination of the two.

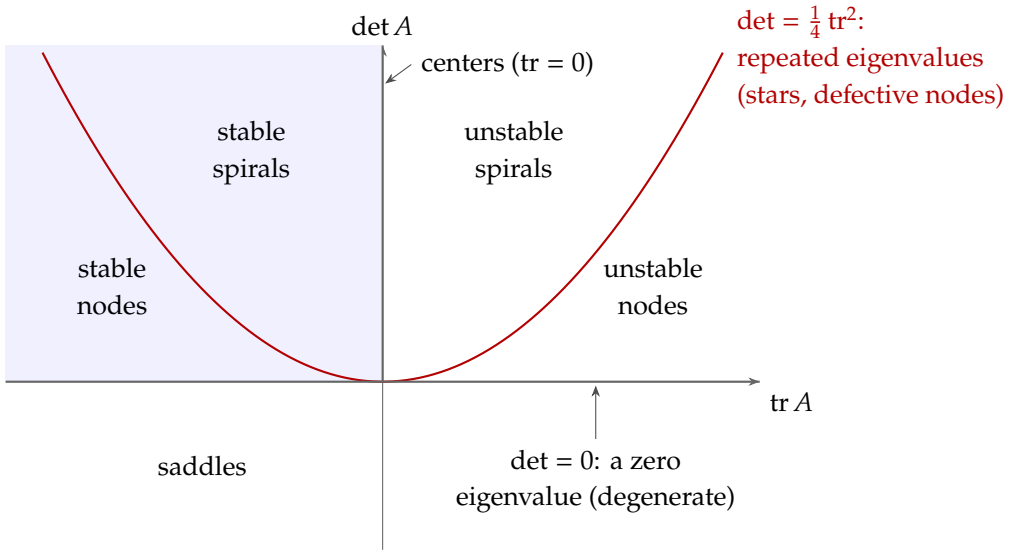


Figure 42: Phase portraits of $\mathbf{u}' = A\mathbf{u}$ by position in the trace-determinant plane. The shaded quadrant $\operatorname{tr} < 0 < \det$ is where all solutions decay.

Example: *A saddle.* $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & -2 \end{smallmatrix}\right)$ has $\operatorname{tr} A = -1$ and $\det A = -2$, so $\lambda_1 = 1$ and $\lambda_2 = -2$. The first column of A is $\left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right)$, so this is an eigenvector for $\lambda_1 = 1$. For λ_2, $A + 2I = \left(\begin{smallmatrix} 3 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ gives $\mathbf{v}_2 = \left(\begin{smallmatrix} -1 \\ 3 \end{smallmatrix}\right)$. Other trajectories are asymptotic to the eigenline for $\lambda_1 = 1$ as $t \to \infty$ and to the eigenline for $\lambda_2 = -2$ as $t \to -\infty$.

35.3 Marginal cases

On the boundaries between the regions, the trace and determinant do not determine the portrait, and the matrix must be inspected.

Repeated eigenvalues ($\det = \mathrm{tr}^2/4$). Lecture 34 treated $A = \left(\begin{smallmatrix} -2 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, with $\lambda_1 = \lambda_2 = -1$, eigenvector $\mathbf{v} = \left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)$, $\mathbf{w} = \left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right)$ solving $(A + I)\mathbf{w} = \mathbf{v}$, and

$$\mathbf{u}_1 = e^{-t}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \mathbf{u}_2 = e^{-t}(t\mathbf{v} + \mathbf{w}) = e^{-t}\begin{pmatrix} t \\ t+1 \end{pmatrix}.$$

Besides the two ray solutions, $\mathbf{u}_2(0) = \mathbf{w}$; as t increases the x-coordinate te^{-t} becomes positive, all solutions decay, and since $\mathbf{u}_2 = te^{-t}\left(\mathbf{v} + \mathbf{w}/t\right)$ the trajectory becomes tangent to the eigenline. It hooks around and enters the origin from the other side (Figure 43). This portrait is a DEFECTIVE NODAL SINK.

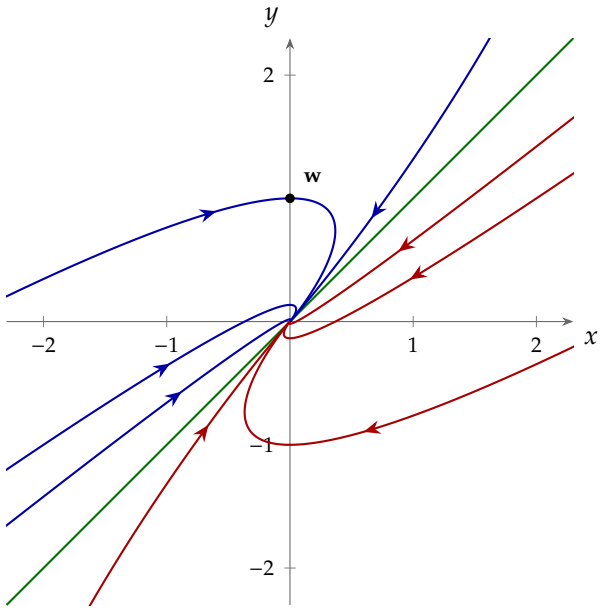


Figure 43: Defective nodal sink of $A = \left(\begin{smallmatrix} -2 & 1 \\ -1 & 0 \end{smallmatrix}\right)$: the solutions $e^{-t}\left(a\mathbf{v} + b(t\mathbf{v} + \mathbf{w})\right)$ for $b = 1$ (blue) and $b = -1$ (red), $a \in \{0, \pm 2, \pm 4\}$, and the eigenline $y = x$ (green). The blue curve through $\mathbf{w} = (0, 1)$ is $\mathbf{u}_2$.

For $A = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 2 \end{smallmatrix}\right)$ the eigenvalue 2 is also repeated, but $A - 2I = 0$, so every vector is an eigenvector (the complete case). Every ray from the origin is a trajectory; the portrait is a STAR SOURCE. In the 2×2 case with a repeated eigenvalue, either

- A is diagonal: A is complete and the portrait is a STAR; or
- A is not diagonal: A is defective, the second solution comes from Lecture 34's method, and the portrait is a DEFECTIVE NODE.

A zero eigenvalue (det = 0). If $\mathbf{v}$ is an eigenvector for the eigenvalue 0, then $\mathbf{u}(t) = c\mathbf{v}$ is a constant solution for every c, so there is (at least) a line of equilibria. Several patterns are possible.

Direction of rotation. Whether a spiral or center turns clockwise or counterclockwise is independent of whether it is a sink or a source, and must also be read off the matrix. At the point $\binom{1}{0}$, the velocity $\mathbf{u}' = A\mathbf{u}$ is the first column $\binom{a}{c}$ of A. If $c > 0$ the trajectory crosses the positive x-axis upward and the spiral turns counterclockwise; if $c < 0$ it turns clockwise.

35.4 Stability

Every linear system falls into one of three categories.

- Asymptotically stable: all solutions tend to $\mathbf{0}$ as $t \to \infty$. These are the systems in the open quadrant $\operatorname{tr} A < 0 < \det A$, where both eigenvalues have negative real part.
- Unstable: most solutions tend to infinity as $t \to \infty$. Saddles and unstable nodes and spirals are examples.
- Neutrally stable: all solutions stay bounded as $t \to \infty$, but most do not tend to $\mathbf{0}$. These lie on the boundary of the stable quadrant: the ray $\operatorname{tr} = 0$, $\det > 0$ (nonzero, purely imaginary eigenvalues; a center with elliptical trajectories), and the ray $\det = 0$, $\operatorname{tr} < 0$ (one zero and one negative eigenvalue; the portrait is a comb, with every solution approaching the line of equilibria along the other eigenline).

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 35, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 36: Normal modes and the matrix exponential

The scalar equation $x' = rx$ has solution $x = e^{rt}x(0)$, and e^{it} was defined as the solution of $x' = ix$, $x(0) = 1$, which led to $e^{it} = \cos t + i \sin t$. The same idea defines a matrix exponential that solves initial value problems for $\mathbf{u}' = A\mathbf{u}$.

36.1 Definition of e^{At}

We want e^{At} to satisfy

$$\text{the solution of } \mathbf{u}' = A\mathbf{u} \text{ with initial value } \mathbf{u}(0) \text{ is } \mathbf{u}(t) = e^{At}\mathbf{u}(0). \tag{36.1}$$

Since $\mathbf{u}(0)$ is a vector and $\mathbf{u}(t)$ a vector-valued function, e^{At} must be a matrix-valued function. The first column of any matrix B is $B\binom{1}{0}$, so (36.1) forces the following.

Definition 36.1 [Matrix exponential]. For a 2×2 matrix A, the matrix exponential e^{At} is the matrix-valued function whose first column is the solution of $\mathbf{u}' = A\mathbf{u}$ with $\mathbf{u}(0) = \binom{1}{0}$, and whose second column is the solution with $\mathbf{u}(0) = \binom{0}{1}$.

This makes (36.1) true for every $\mathbf{u}(0)$: $e^{At}\mathbf{u}(0)$ is a linear combination of the columns of e^{At}, hence a solution, and at $t = 0$ it equals $e^{A0}\mathbf{u}(0) = I\mathbf{u}(0) = \mathbf{u}(0)$.

36.2 Computing e^{At} from a fundamental matrix

Definition 36.2 [Fundamental matrix]. A fundamental matrix $\Phi(t)$ for A is a matrix-valued function whose columns are independent solutions of $\mathbf{u}' = A\mathbf{u}$. The general solution is then $\mathbf{u} = \Phi(t)\mathbf{c}$ for a constant vector $\mathbf{c}$.

The matrix exponential is the fundamental matrix normalized by $\Phi(0) = I$. A square matrix B has an inverse B^{-1}, with $BB^{-1} = I$ and $B^{-1}B = I$ (either implies the other), exactly when $\det B \neq 0$. In the 2×2 case

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \qquad \text{since} \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = (ad - bc)I.$$

Theorem 36.3 [Matrix exponential from any fundamental matrix]. *If $\Phi(t)$ is any fundamental matrix for A, then*

$$\boxed{e^{At} = \Phi(t)\,\Phi(0)^{-1}}.$$

Proof. The columns of $\Phi(0)$ are independent, so $\Phi(0)$ is invertible. Each column of $\Phi(t)\Phi(0)^{-1}$ is a linear combination of the columns of $\Phi(t)$, hence a solution, and at $t = 0$ the product is $\Phi(0)\Phi(0)^{-1} = I$. So its columns are the solutions with initial values $\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}0\\1\end{smallmatrix}\right)$, which by uniqueness are the columns of e^{At}. □

Example: Let $A = \left(\begin{smallmatrix}1 & 1\\0 & 2\end{smallmatrix}\right)$. It is upper triangular, so its eigenvalues are the diagonal entries 1 and 2; with $(\mathrm{tr}, \det) = (3, 2)$ below the critical parabola, the phase portrait is an unstable node. From

$$A - I = \begin{pmatrix}0 & 1\\0 & 1\end{pmatrix}, \quad \mathbf{v}_1 = \begin{pmatrix}1\\0\end{pmatrix}; \qquad A - 2I = \begin{pmatrix}-1 & 1\\0 & 0\end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix}1\\1\end{pmatrix},$$

the normal modes $\left(\begin{smallmatrix}e^t\\0\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}e^{2t}\\e^{2t}\end{smallmatrix}\right)$ give the general solution

$$\mathbf{u} = c_1\begin{pmatrix}e^t\\0\end{pmatrix} + c_2\begin{pmatrix}e^{2t}\\e^{2t}\end{pmatrix} = \Phi(t)\begin{pmatrix}c_1\\c_2\end{pmatrix}, \qquad \Phi(t) = \begin{pmatrix}e^t & e^{2t}\\0 & e^{2t}\end{pmatrix}.$$

Here $\Phi(0) = \left(\begin{smallmatrix}1 & 1\\0 & 1\end{smallmatrix}\right)$ and $\Phi(0)^{-1} = \left(\begin{smallmatrix}1 & -1\\0 & 1\end{smallmatrix}\right)$, so

$$e^{At} = \begin{pmatrix}e^t & e^{2t}\\0 & e^{2t}\end{pmatrix}\begin{pmatrix}1 & -1\\0 & 1\end{pmatrix} = \begin{pmatrix}e^t & e^{2t} - e^t\\0 & e^{2t}\end{pmatrix}.$$

36.3 Uncoupled and defective examples

Uncoupled systems. For $A = \left(\begin{smallmatrix}a & 0\\0 & d\end{smallmatrix}\right)$ the eigenvalues are $\lambda_1 = a$ and $\lambda_2 = d$, with eigenvectors $\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}0\\1\end{smallmatrix}\right)$. The normal modes give

$$\Phi(t) = \begin{pmatrix}e^{\lambda_1 t} & 0\\0 & e^{\lambda_2 t}\end{pmatrix},$$

which already satisfies $\Phi(0) = I$, so it is e^{At}.

A defective example. Let $A = \left(\begin{smallmatrix}0 & 1\\0 & 0\end{smallmatrix}\right)$. Then $\mathrm{tr}\, A = \det A = 0$, so 0 is an eigenvalue of multiplicity two; A is not diagonal, hence defective. It is also the companion matrix of $x'' = 0$: with $\mathbf{u} = \left(\begin{smallmatrix}x\\x'\end{smallmatrix}\right)$, the solutions $x_1 = 1$ and $x_2 = t$ give

$$\mathbf{u}_1 = \begin{pmatrix}1\\0\end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix}t\\1\end{pmatrix}, \qquad \Phi(t) = \begin{pmatrix}1 & t\\0 & 1\end{pmatrix}.$$

Since $\Phi(0) = I$, $e^{At} = \left(\begin{smallmatrix}1 & t\\0 & 1\end{smallmatrix}\right)$.

36.4 The exponential law

Setting $t = 1$ defines e^A for any square matrix A; for instance

$$\exp\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

As expected, $e^0 = I$. The exponential law holds only for commuting matrices:

$$e^A e^B = e^{A+B} \qquad \text{provided that } AB = BA.$$

For example, A commutes with itself, so $(e^A)^n = e^{nA}$.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 36, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 37: Nonlinear systems

37.1 Autonomous systems: deer and wolves

An ODE is autonomous if x' depends only on x and not on t: $x' = f(x)$. A deer population on an island grows like $x' = kx$ when small, but the island limits it. Taking the growth rate to be $k(1 - x/a)$, where a is the maximal sustainable population, gives the logistic equation

$$x' = k\Big(1 - \frac{x}{a}\Big)x.$$

With $k = a = 3$, this is $x' = (3 - x)x$. Its critical points are $x = 0$ and $x = 3$; $x' > 0$ for $0 < x < 3$, and $x' < 0$ for $x > 3$ (and, unrealistically, for $x < 0$). So on the phase line 0 is unstable and 3 is stable.

Now add wolves, with population y. Wolves eat deer, which depresses the deer growth rate: $x' = (3 - x - y)x$. Wolves alone follow a logistic equation $y' = (1 - y)y$, and deer increase their growth rate: $y' = (1 - y + x)y$. The result is a nonlinear autonomous system

$$x' = (3 - x - y)x, \qquad y' = (1 - y + x)y. \tag{37.1}$$

37.2 Vector fields

The general autonomous system in the plane is

$$x' = f(x, y), \qquad y' = g(x, y).$$

A solution traces a curve $\mathbf{u}(t) = \left(\begin{smallmatrix} x(t) \\ y(t) \end{smallmatrix}\right)$ whose velocity is determined by its current position:

$$\mathbf{u}'(t) = \mathbf{F}(\mathbf{u}(t)), \qquad \mathbf{F}(x, y) = \begin{pmatrix} f(x, y) \\ g(x, y) \end{pmatrix}.$$

$\mathbf{F}$ assigns a vector to each point of the plane: it is a vector field. An autonomous system is the same thing as a vector field, and a solution is a curve whose velocity at each of its points is the value of the field there. The trajectories drawn in the plane form the phase portrait of the system; the phase line of a single autonomous equation is the one-dimensional version.

Linear systems have vector fields too. For $A = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ the field is $A\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = \left(\begin{smallmatrix} -y \\ x \end{smallmatrix}\right)$, a vector of the same length as $\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$ at right angles to it. Here $\operatorname{tr} A = 0$ and $\det A = 1$, a center, and in fact the trajectories are circles.

In (37.1), the x-axis is invariant ($y = 0$ gives $y' = 0$) and carries the deer-only phase line; the y-axis

carries the wolf-only phase line (Figure 44).

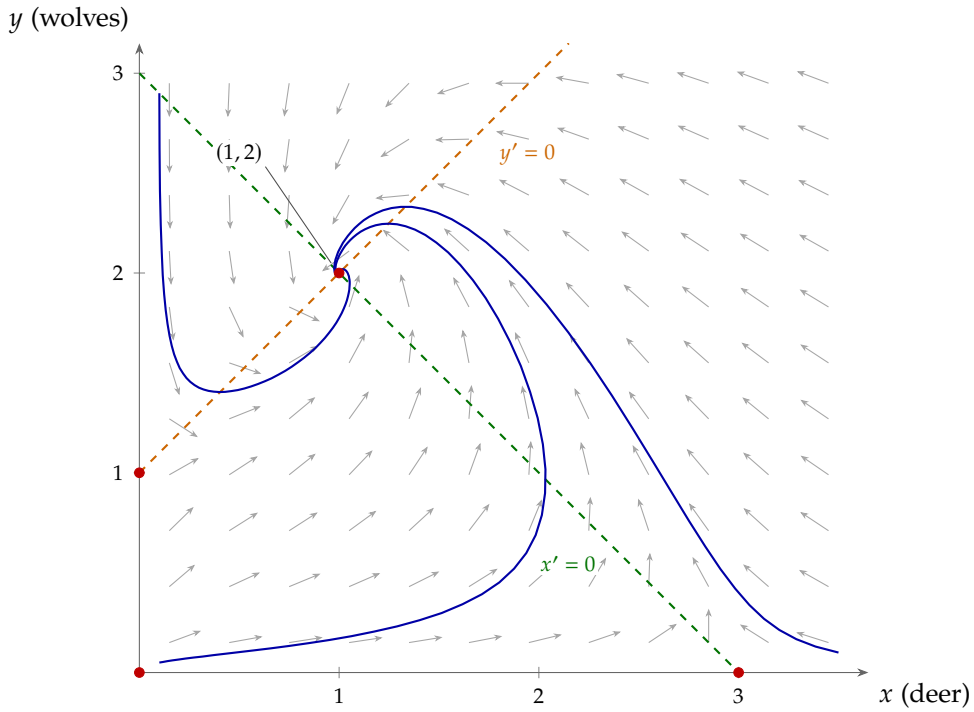


Figure 44: Direction field of the deer-wolf system (37.1) (arrows normalized to equal length), the nullclines $y = 3 - x$ where the field is vertical and $y = 1 + x$ where it is horizontal (dashed; the axes are nullclines too), the four equilibria, and three numerically integrated trajectories converging to $(1, 2)$.

37.3 Equilibria

An EQUILIBRIUM or CRITICAL POINT is a point where the vector field vanishes, giving a constant solution. In (37.1), the horizontal component vanishes, so the field is vertical, where $(3 - x - y)x = 0$: on the lines $x = 0$ and $x + y = 3$. The vertical component vanishes, so the field is horizontal, where $(1 - y + x)y = 0$: on the lines $y = 0$ and $y = 1 + x$. The field vanishes where a line of the first kind meets one of the second:

$$(0, 0), \qquad (3, 0), \qquad (0, 1), \qquad (1, 2),$$

the last from $y = 3 - x$ and $y = 1 + x$. The computed phase portrait looks like a node near $(0, 0)$, saddles at $(3, 0)$ and $(0, 1)$, and a spiral at $(1, 2)$. This illustrates a basic fact: near an equilibrium, an autonomous system is usually well approximated by a linear system, which can be analyzed with eigenvalues and eigenvectors. The eigenvectors are visible in the nonlinear phase portrait.

37.4 Linearization

Near the origin, expand

$$x' = 3x - x^2 - xy, \qquad y' = y + xy - y^2.$$

The quadratic terms are insignificant, so to first order $x' = 3x$ and $y' = y$: both populations grow naturally, uncoupled. The matrix $\left(\begin{smallmatrix} 3 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ has eigenvalues 3 and 1, so the origin is a node (a source). Near the origin all solutions except the normal mode on the x-axis become tangent to the y-axis, the eigenline of the smaller eigenvalue.

At another point $P = (a, b)$, use the tangent plane approximation

$$f(x, y) \approx f(a, b) + \frac{\partial f}{\partial x}\Big|_P (x - a) + \frac{\partial f}{\partial y}\Big|_P (y - b),$$

and similarly for g. If P is an equilibrium then $f(a, b) = g(a, b) = 0$. In the shifted coordinates $\mathbf{u} = \left(\begin{smallmatrix} x-a \\ y-b \end{smallmatrix}\right)$,

$$\mathbf{u}' \approx J(a, b)\,\mathbf{u}, \qquad J(x, y) = \begin{pmatrix} \partial f/\partial x & \partial f/\partial y \\ \partial g/\partial x & \partial g/\partial y \end{pmatrix},$$

where J is the JACOBIAN MATRIX. Near P the system behaves very much like the linear system with matrix $J(a, b)$, its LINEARIZATION at P.

For (37.1),

$$J(x, y) = \begin{pmatrix} 3 - 2x - y & -x \\ y & 1 + x - 2y \end{pmatrix}.$$

At the axis equilibria, $J(3, 0) = \left(\begin{smallmatrix} -3 & -3 \\ 0 & 4 \end{smallmatrix}\right)$ and $J(0, 1) = \left(\begin{smallmatrix} 2 & 0 \\ 1 & -1 \end{smallmatrix}\right)$ have $\det < 0$, confirming the saddles. At the interior equilibrium,

$$A = J(1, 2) = \begin{pmatrix} -1 & -1 \\ 2 & -2 \end{pmatrix}, \qquad p_A(\lambda) = \lambda^2 + 3\lambda + 4 = \left(\lambda + \frac{3}{2}\right)^2 + \frac{7}{4},$$

so $\lambda = -\frac{3}{2} \pm i\frac{\sqrt{7}}{2}$. This is a stable spiral, turning counterclockwise because the lower left entry is positive, matching the nonlinear portrait near $(1, 2)$.

So the populations approach the stable levels $(1, 2)$. As they approach, they oscillate about them, decaying like $e^{-3t/2}$, with period

$$\frac{2\pi}{\sqrt{7}/2} = \frac{4\pi}{\sqrt{7}} \approx 4.75.$$

The phase portrait shows that this happens for all positive initial populations: the open first quadrant is the BASIN of the equilibrium $(1, 2)$, which is a SINK or ATTRACTOR.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 37, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 38: Linearization near equilibria; the nonlinear pendulum

Linearization at equilibria explains the phase portrait of the damped pendulum, gives the phugoid oscillation of an airplane, and shows how wind can turn damping into anti-damping on a bridge.

38.1 The nonlinear pendulum

A bob of mass m on a rod of length L swings in a plane, possibly all the way around the pivot, under gravity g. Let θ be the angle from straight down, measured to the right, and $s = L\theta$ the arc length from the bottom (Figure 45). The component of gravity along the direction of motion is $-mg\sin\theta$. Model friction by $-cs' = -cL\theta'$. Newton's law $ms'' = F$ gives

$$mL\theta'' = -mg\sin\theta - cL\theta',$$

and dividing by mL,

$$\boxed{\theta'' + b\theta' + k\sin\theta = 0}, \qquad k = \frac{g}{L}, \quad b = \frac{c}{m}.$$

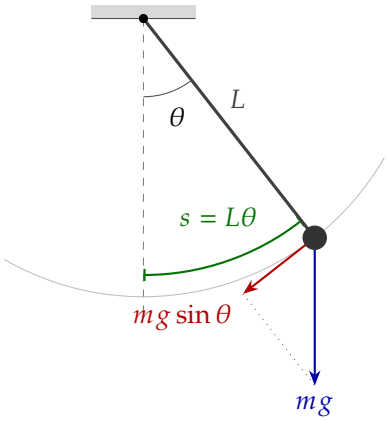


Figure 45: The pendulum. Gravity mg has component $mg\sin\theta$ along the circle, pointing back toward $\theta = 0$.

This second-order equation is nonlinear, but it still has a companion first-order system. With $x = \theta$ and $y = x'$,

$$x' = y, \qquad y' = -k\sin x - by.$$

This is an autonomous system. Its equilibria have $y = 0$ and $\sin x = 0$: the points $(n\pi, 0)$, $n \in \mathbb{Z}$. The Jacobian is

$$J(x,y) = \begin{pmatrix} 0 & 1 \\ -k\cos x & -b \end{pmatrix}.$$

- At $x = 0, \pm 2\pi, \pm 4\pi, \ldots$ (hanging straight down), $\cos x = 1$ and $J = \left(\begin{smallmatrix} 0 & 1 \\ -k & -b \end{smallmatrix}\right)$, with det $= k$ and tr $= -b$. For small damping ($b^2 < 4k$) these are stable spirals.
- At $x = \pm\pi, \pm 3\pi, \ldots$ (straight up), $\cos x = -1$ and $J = \left(\begin{smallmatrix} 0 & 1 \\ k & -b \end{smallmatrix}\right)$, with det $= -k < 0$: saddles.

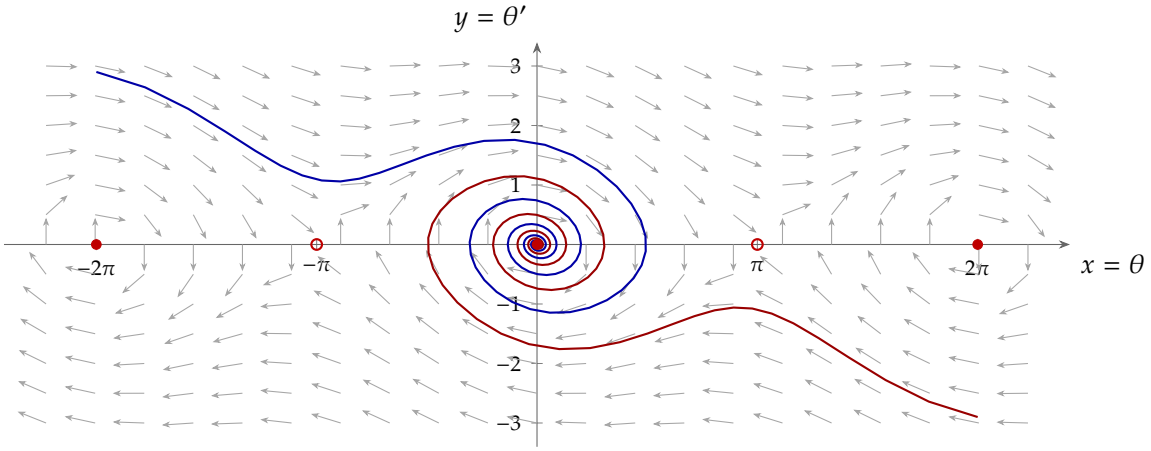


Figure 46: Phase portrait of the damped pendulum for the illustrative values $k = 1$, $b = 0.25$: normalized direction field, spiral sinks at $x = 0, \pm 2\pi$ (filled), saddles at $x = \pm\pi$ (open), and two numerically integrated trajectories that whirl over the top once before spiraling into $(0, 0)$.

The full phase portrait (Figure 46) shows a feature common to all companion systems: since $x' = y$, trajectories above the x-axis move right and those below move left. Trajectories coming down from the upper left are the pendulum whirling counterclockwise in complete circles; trajectories coming up from the lower right are clockwise circles. Each dip of such a trajectory is a passage through the bottom position. In very exceptional cases a trajectory heads straight into a saddle and converges to it exponentially (the pendulum comes to rest pointing straight up); most trajectories miss it and move away from it exponentially. Every other trajectory is eventually caught in the basin of one of the sinks and spirals in to it: the pendulum comes to rest hanging straight down. (It was in that basin all along.)

38.2 Phugoid oscillation

An airplane in flight feels thrust F along its direction of motion, its weight mg downward, and an air force with two components: drag, opposite the thrust, and lift, perpendicular to the motion. At equilibrium it moves horizontally at speed v_0. After a disturbance (a downdraft, say) the horizontal velocity is $v_0 + u$ and the vertical velocity is w, with u, w small relative to v_0. The linearized equations near equilibrium are

$$\frac{d}{dt}\begin{pmatrix} u \\ w \end{pmatrix} = \begin{pmatrix} -2F/(mv_0) & -g/v_0 \\ 2g/v_0 & 0 \end{pmatrix}\begin{pmatrix} u \\ w \end{pmatrix}, \qquad p_A(\lambda) = \lambda^2 + \frac{2F}{mv_0}\lambda + \frac{2g^2}{v_0^2}.$$

Writing $p_A(\lambda) = \lambda^2 + 2\zeta\omega_n\lambda + \omega_n^2$ in terms of natural frequency and damping ratio,

$$\omega_n = \frac{\sqrt{2}\,g}{v_0}, \qquad 2\zeta\omega_n = \frac{2F}{mv_0} \implies \zeta = \frac{1}{\sqrt{2}}\,\frac{F}{mg}.$$

Two conclusions follow.

1. The natural period $2\pi/\omega_n = \sqrt{2}\,\pi v_0/g$ of this PHUGOID OSCILLATION depends *only on the equilibrium speed*, not on thrust or mass. In meters and seconds it is about $0.45\,v_0$: about 118 seconds for a Boeing 747 at 260 m/s, and about 380 seconds for an F15 at 838 m/s.
2. The damping ratio depends only on the dimensionless *thrust-to-weight ratio* F/mg. The system is underdamped when $\zeta < 1$, that is $F/mg < \sqrt{2}$. Even an F15 is far from $F/mg > \sqrt{2}$, so in practice the phugoid is underdamped: for a 747, $F/mg \approx 0.27$ and $\zeta \approx 0.19$; for an F15, $F/mg \approx 0.67$ and $\zeta \approx 0.47$.

38.3 Anti-damping: a twisting bridge

The Tacoma Narrows suspension bridge (1940) oscillated vertically from the start, and in a November storm it began to twist with growing amplitude. The cause was not resonance with gusts at the bridge's natural frequency. Let θ be the angle of the deck from horizontal at some point along the bridge, so that

$$\theta'' + b\theta' + \omega_n^2\theta = F.$$

The wind's torsional force F changes as the deck tips, so model it as $F = a(v)\theta + c(v)\theta'$, depending on the wind speed v. The dominant contribution is the θ' term; dropping the θ term,

$$\theta'' + \big(b - c(v)\big)\theta' + \omega_n^2\theta = 0.$$

The characteristic polynomial is $s^2 + (b - c(v))s + \omega_n^2 = \left(s + \frac{b-c(v)}{2}\right)^2 + \omega_d^2$, with nonreal roots (as observed)

$$r = \frac{c(v) - b}{2} \pm i\omega_d.$$

The graph of $c(v)$ is shaped like a W. For v near zero, $c(v) < 0$, which increases the damping and stabilizes the bridge. At some speed $c(v) = 0$, which is still harmless. But once $c(v)$ exceeds the damping constant b, the effective damping $b - c(v)$ is negative: the real part of the roots is positive, and the solutions oscillate inside growing exponentials.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 38, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 39: Limitations of the linear: limit cycles and chaos

39.1 Stability

Definition 39.1 [Stable equilibrium]. An equilibrium of an autonomous system is stable if all nearby trajectories stay near it and converge to it as $t \to \infty$, and unstable otherwise.

For linear systems $\mathbf{u}' = A\mathbf{u}$, the stable region of the (tr, det)-plane is the open upper left quadrant $\mathrm{tr} < 0 < \det$. It is bounded by two half-lines: $\mathrm{tr} = 0$, $\det > 0$, where the eigenvalues are purely imaginary, and $\det = 0$, $\mathrm{tr} < 0$, where one eigenvalue is zero and the other negative.

39.2 Limits of linearization

Linearizing at an equilibrium works *generically*, that is, almost all the time: it predicts the phase portrait near the equilibrium and whether the equilibrium is stable. Precisely:

1. If the linearization does not lie on the boundary of the stable quadrant, it correctly predicts the stability of the equilibrium.
2. If the eigenvalues of the Jacobian have nonzero real parts, so that (tr, det) lies neither on the tr-axis nor on the half-line $\mathrm{tr} = 0 < \det$, then near the equilibrium the phase portrait is a small deformation of the linear one.

On the borderlines, linearization can fail.

Example: *A linear center that is a nonlinear spiral.* For $x' = -y - y^2$, $y' = x + y^2$, the Jacobian at the origin is $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, so the linearization $x' = -y$, $y' = x$ has $\mathrm{tr} = 0$ and $\det = 1$: a center, which is not stable. A computed phase portrait instead shows stable spirals, which decay to the origin more slowly than any exponential. Other examples with the same linearization give unstable spirals, or centers; the linearization alone cannot decide.

Example: *Six ray solutions.* For $x' = 4xy$, $y' = x^2 - y^2$, the Jacobian at the origin is the zero matrix. The linearization $\mathbf{u}' = \mathbf{0}$ is completely degenerate: both eigenvalues are zero, every vector is an eigenvector, and every point is a constant solution. The actual phase portrait (Figure 47) has six ray solutions, unlike any linear portrait; second-order effects dominate. The field is radial where $x\,y' - y\,x' = x(x^2 - 5y^2) = 0$, that is, on the lines $x = 0$ and $x = \pm\sqrt{5}\,y$.

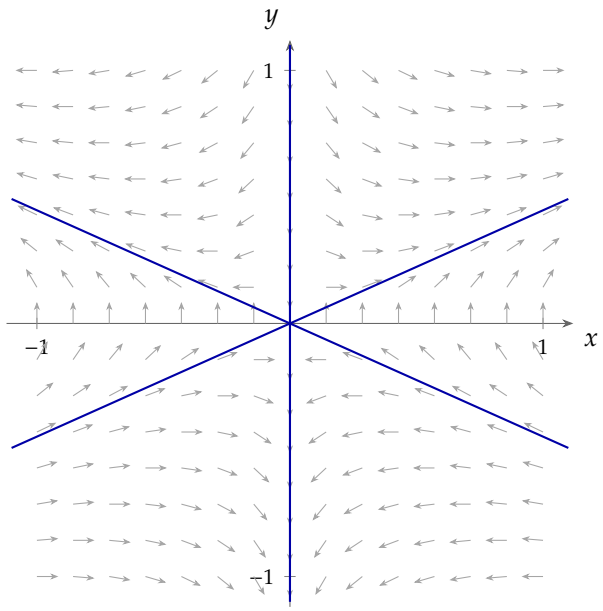


Figure 47: Normalized direction field of $x' = 4xy$, $y' = x^2 - y^2$ near the origin, with the three lines $x = 0$, $x = \pm\sqrt{5}\,y$ that carry the six ray solutions.

39.3 Limit cycles

After equilibria, the most important feature of a nonlinear autonomous system is the possible presence of periodic solutions. The VAN DER POL EQUATION is

$$x'' + a(x^2 - 1)x' + x = 0$$

for a fixed constant a. For $a = 0$ it is the harmonic oscillator with $\omega_n = 1$, and all nonzero solutions are periodic with period 2π. The companion system is

$$x' = y, \qquad y' = -x - a(x^2 - 1)y.$$

It continues to have periodic solutions, and for $a > 0$ the situation is even better: there is exactly one periodic trajectory, and all other nonzero solutions converge to it (Figure 48). Such an isolated periodic trajectory is a LIMIT CYCLE. It is an ATTRACTOR: every nearby trajectory stays nearby and converges to it. The heart is controlled by an equation like this, which is why it reliably returns to its normal periodic pattern after a disturbance.

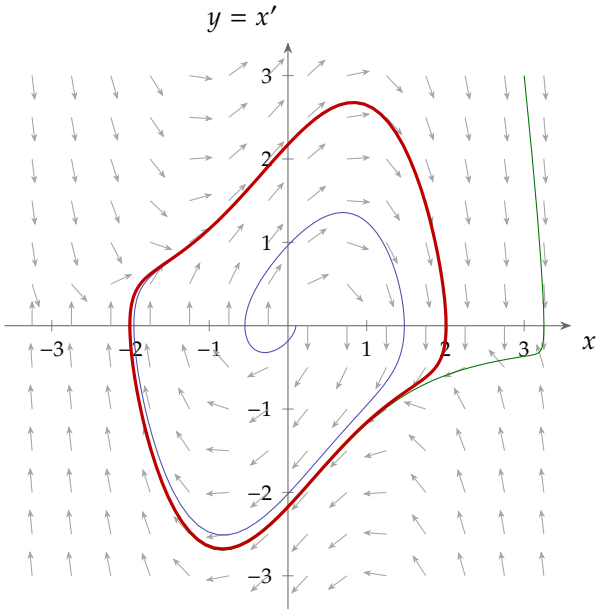


Figure 48: Van der Pol system with $a = 1$: normalized direction field, the limit cycle (red), and two numerically integrated trajectories approaching it from inside (from $(0.1, 0)$) and outside (from $(3, 3)$).

39.4 Strange attractors and chaos

Limit cycles are typical of two-dimensional systems; in three dimensions much more complicated behavior is possible. Edward Lorenz's 1963 model of convection rolls in the atmosphere is the nonlinear autonomous system

$$x' = -ax + by, \qquad y' = -xz + rx - y, \qquad z' = xy - bz$$

for constants a, b, r. Besides the origin it has two equilibria exactly when $r > a/b$: the first equation gives $y = ax/b$, the third $z = xy/b = ax^2/b^2$, and the second $x(r - z) = y$, so $z = r - a/b$, which needs to be positive. For suitable parameter values these two equilibria are unstable, and solutions neither settle into a periodic orbit nor run off to infinity. They wrap erratically around the two equilibria forever. The attracting set is much stranger than a loop and is called a STRANGE ATTRACTOR; this is the fundamental characteristic of CHAOS.

Source. Haynes Miller and Arthur Mattuck, MIT 18.03, *Differential Equations*, Lecture 39, Spring 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.